Chapter 3 - Derivatives

Calculus I

September 27, 2024

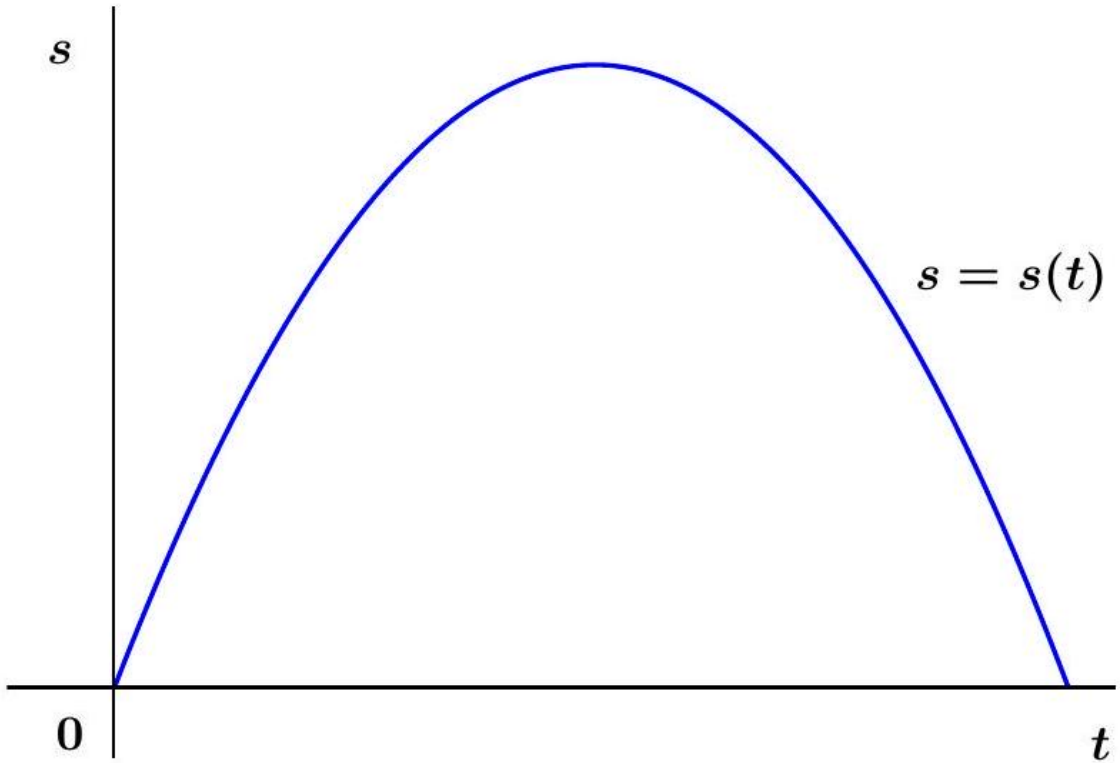
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Section 1

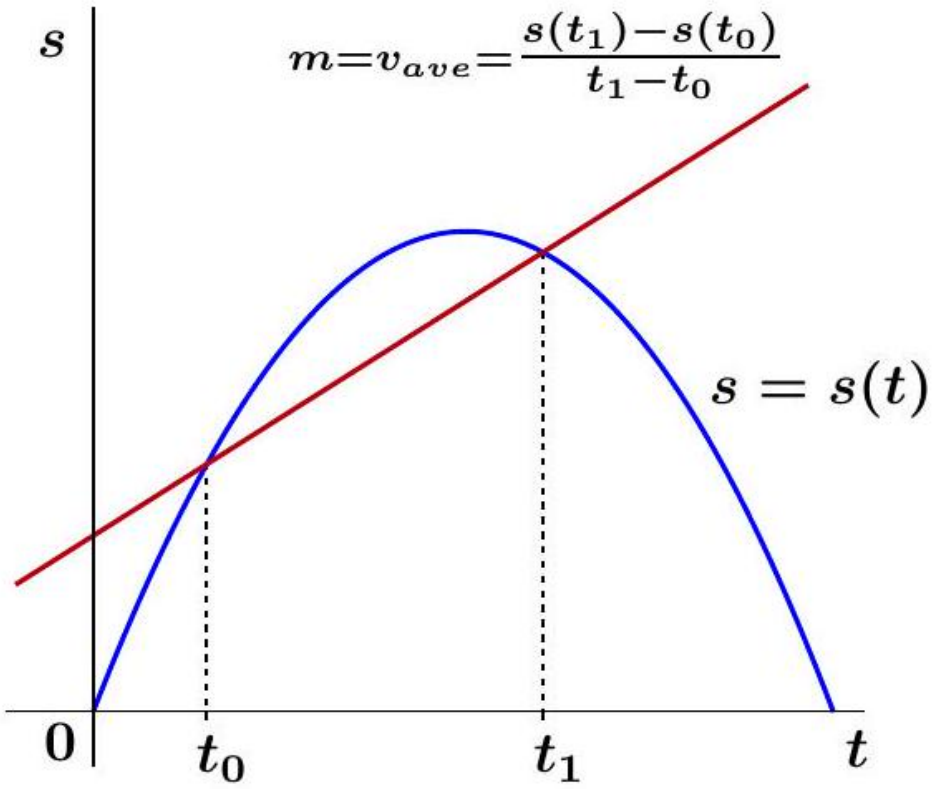
Derivatives

Instantaneous Velocity



Problem: A stone is thrown straight upward in the air and eventually falls back to the ground. How can we define the instantaneous velocity of the stone at any given time?

Instantaneous Velocity

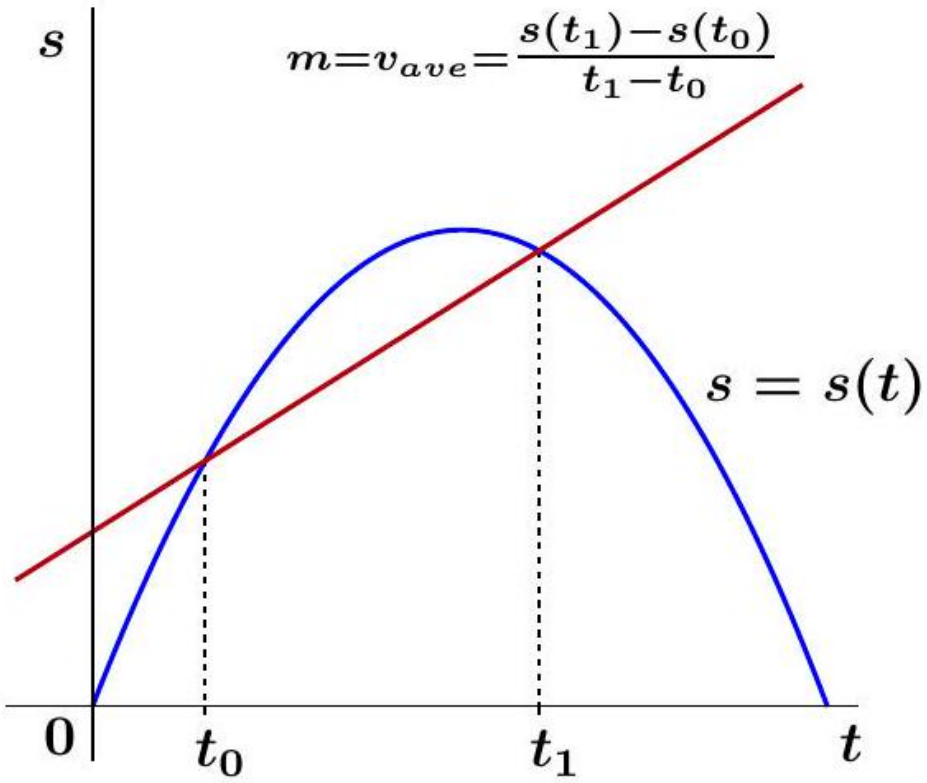


Recall: The average velocity of the stone relative to the ground over the period from time $t = t_0$ to $t = t_1$ is given by the formula where

$$V_{\text{ave}} = \frac{\text{displacement (change in position)}}{\text{elapsed time}}$$

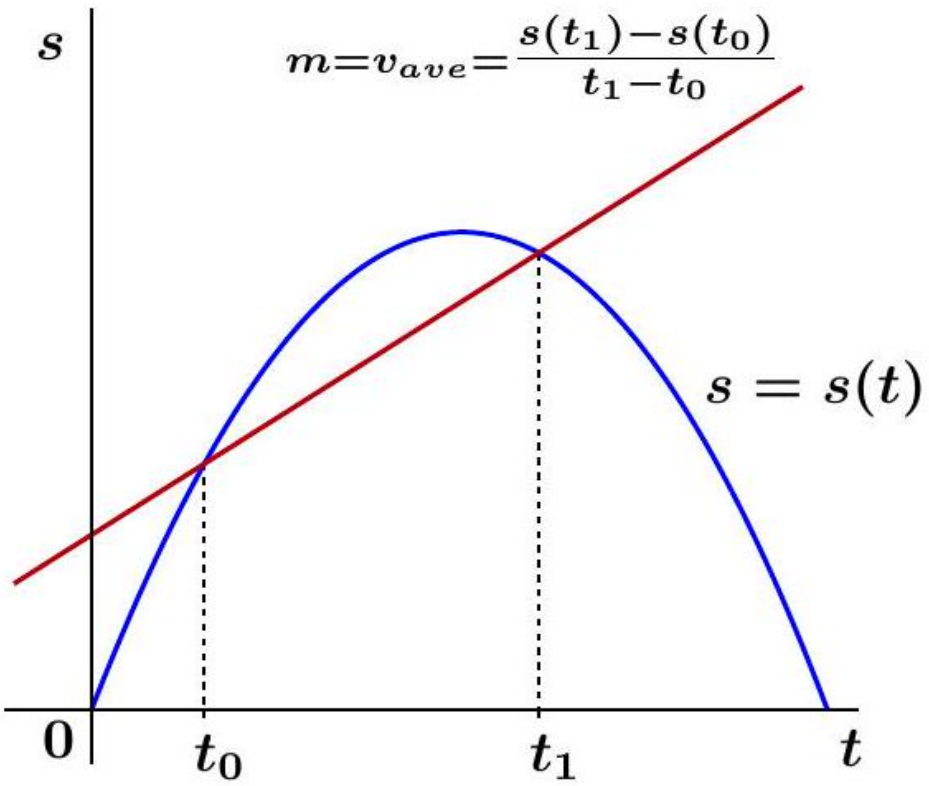
$$= \frac{s(t_1) - s(t_0)}{t_1 - t_0} = \frac{\triangle s}{\triangle t}$$

Instantaneous Velocity



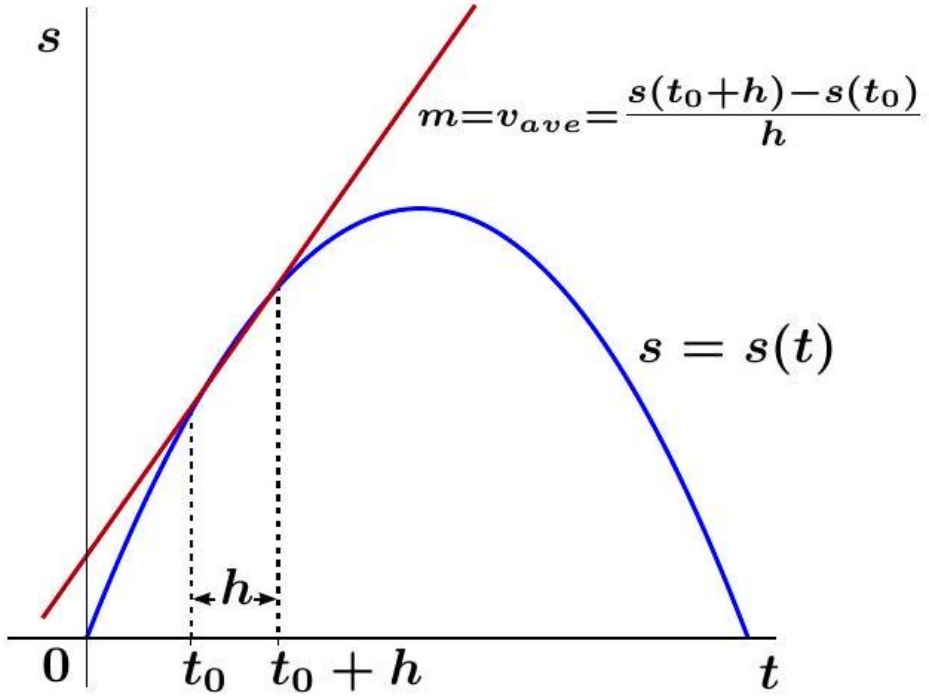
Geometric Interpretation: V_{ave} is the slope m of the "secant line" to the graph of $s(t)$ through the points $(t_0, s(t_0))$ and $(t_1, s(t_1))$.

Instantaneous Velocity



Question: How do we define instantaneous velocity at a point t_0?

Instantaneous Velocity



Key Assumption: The velocity of the stone should not vary too much over very small intervals of time. Therefore, if h is small

$$\begin{aligned} v(t_0) &\approx v_{\text{ave}} \\ &= \frac{s(t_0+h) - s(t_0)}{(t_0+h) - t_0} \\ &= \frac{s(t_0+h) - s(t_0)}{h} \end{aligned}$$

Instantaneous Velocity

Definition 3.1.1 (Instantaneous Velocity)

The instantaneous velocity of an object at time t_0 is given by

$$v(t_0) = \lim_{h \to 0} \frac{s(t_0 + h) - s(t_0)}{h}$$

provided this limit exists.

Average Rate of Change

Definition 3.1.2 (Average Rate of Change)

Given a function $f(t)$, we can define the average rate of change of $f(t)$ as t goes from t_0 to t_1 to be the ratio

$$\frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

Note: If we fix a point t_0 and let $h = t_1 - t_0$ be small, then

$$\frac{f(t_0 + h) - f(t_0)}{h}$$

is an estimate of the instantaneous rate of change of $f(t)$ at t_0.

Definition 3.1.3 (Instantaneous Rate of Change)

Given a function $f(t)$, we can define the instantaneous rate of change of $f(t)$ at t_0 to be

$$\lim_{h \to 0} \frac{f(t_0 + h) - f(t_0)}{h}$$

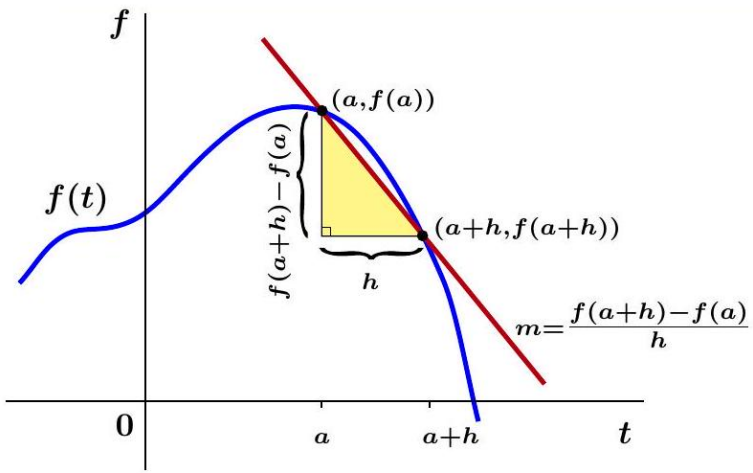
The Derivative

Definition 3.1.4 (Newton Quotient)

Given a function $f(t)$, a point $t = a$ and $h \neq 0$, the ratio

$$\frac{f(a+h) - f(a)}{h}$$

is called a Newton Quotient for $f(t)$ centered at $t = a$.



Note: Geometrically, the Newton Quotient represents the slope of the secant line to the graph of $f(t)$ through the points $(a, f(a))$ and

$(a+h, f(a+h))$.

The Derivative

Definition 3.1.5 (The Derivative at $t = a$)

We say that the function $f(t)$ is differentiable at $t = a$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

In this case, we write

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

and $f'(a)$ is the derivative of $f(t)$ at $t = a$.

Note: If $t = a + h$, then as $h \to 0$ we have $t \to a$. Furthermore, since $h = t - a$, we get

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{t \to a} \frac{f(t) - f(a)}{t - a} \end{aligned}$$

provided the limits exist.

Derivative as Instantaneous Rate of Change

Observation: If $y = f(t), \triangle y = f(t) - f(a)$ and $\Delta t = t - a$, then the Newton Quotient

$$\frac{\triangle y}{\triangle t} = \frac{f(t) - f(a)}{t - a}$$

is the average change in y.

Therefore,

$$f'(a) = \lim_{\Delta t \to 0} \frac{\Delta y}{\Delta t}$$

is the limit of average rates of change over smaller and smaller intervals and so this represents the instantaneous rate of change of y with respect to t.

Derivative as Instantaneous Rate of Change

Example 3.1.6

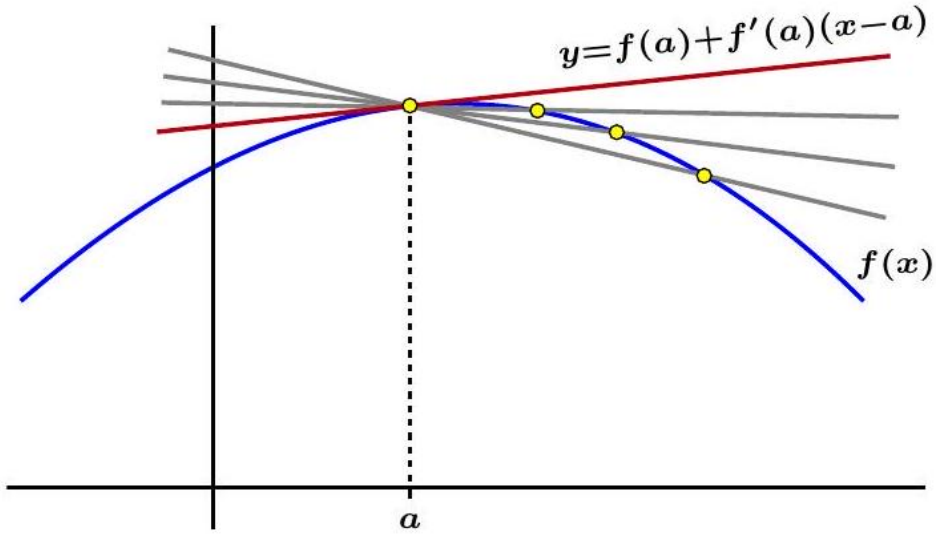
Let $s = s(t)$ represent the displacement of an object. Then

$$\begin{aligned} s'(t_0) &= \lim_{\Delta t \to 0} \frac{\triangle s}{\Delta t} \\ &= \lim_{t \to t_0} \frac{s(t) - s(t_0)}{t - t_0} \\ &= \lim_{h \to 0} \frac{s(t_0 + h) - s(t_0)}{h} \\ &= v(t_0) \end{aligned}$$

where $v(t_0)$ is the instantaneous velocity of the object at time t_0.

Fact: Velocity is the derivative of displacement.

The Tangent Line



Question: Does there exist a geometric interpretation of the derivative?

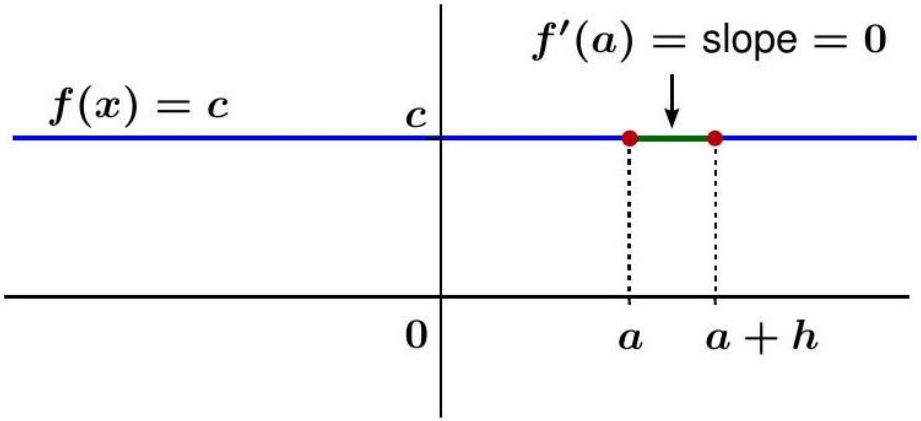
Observation: If $f(x)$ is differentiable at $x = a$, the slopes of the secant lines through $(a, f(a))$ and $(a + h, f(a + h))$ converge to $f'(a)$ as $h \to 0$.

Definition 3.1.7 (Tangent Line)

The tangent line to the graph of $f(x)$ at $x = a$ is the line passing through $(a, f(a))$ with slope equal to $f'(a)$. That is,

$$y = f(a) + f'(a)(x - a)$$

Derivative of a Constant Function



Example 3.1.8 (Derivative of a Constant Function)

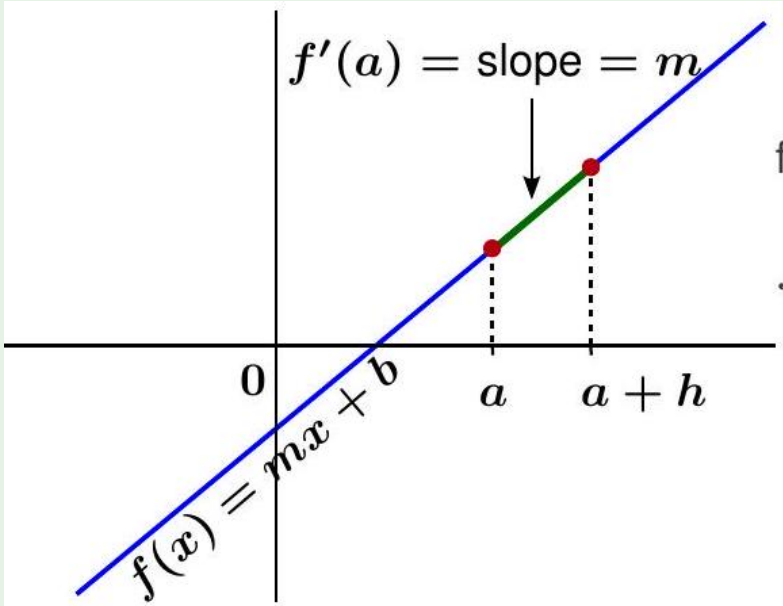
Assume that $f(x) = c$ for all $x \in \mathbb{R}$ and let $a \in \mathbb{R}$. Then

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= 0 \end{aligned}$$

Derivative of a Linear Function

Example 3.1.9 (Derivative of a Linear Function)

Assume that



$$f(x) = mx + b$$

for all $x \in \mathbb{R}$. Then

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h\to 0} \frac{(m(a+h)+b) - (ma+b)}{h} \\ &= \lim_{h\to 0} \frac{ma + mh + b - ma - b}{h} \\ &= \lim_{h\to 0} \frac{mh}{h} \\ &= m. \end{aligned}$$

Derivatives and Continuity

Definition 3.1.10 (Derivative)

We say that the function $f(t)$ is differentiable at $t = a$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

Equivalently:

$$f'(a) = \lim_{t \to a} \frac{f(t) - f(a)}{t - a}$$

if this limit exists.

Derivatives and Continuity

Question: Is there a relationship between differentiability and continuity?

Key Observation: Assume that $f(t)$ is differentiable at $t = a$. Then

$$f'(a) = \lim_{t \to a} \frac{f(t) - f(a)}{t - a}$$

So

$$0 = f'(a) \cdot \lim_{t \to a}(t - a) = \lim_{t \to a} f(t) - f(a)$$

Hence

$$\lim_{t \to a} f(t) = f(a)$$

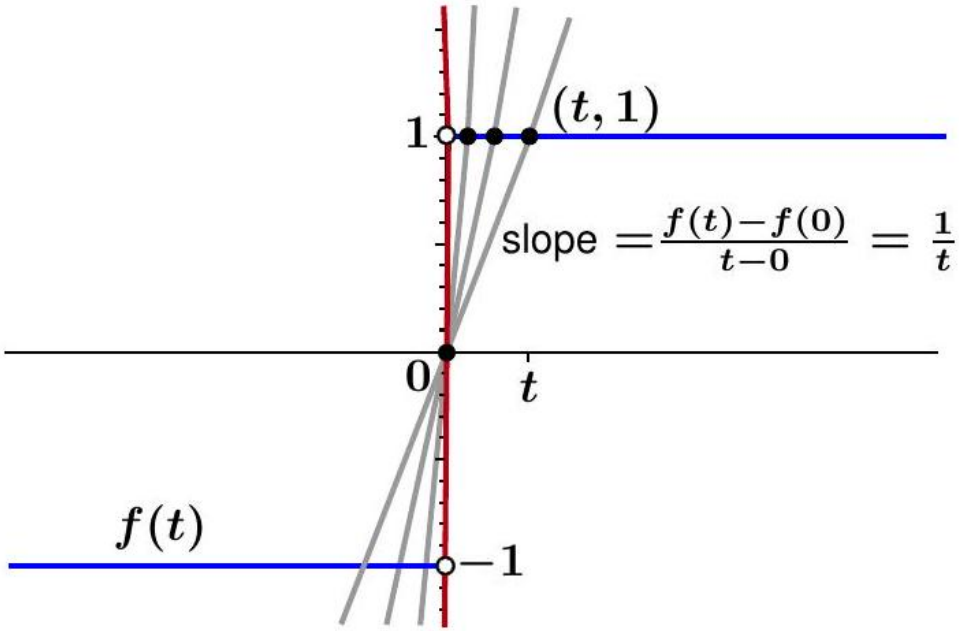
and $f(t)$ is continuous at $t = a$.

Derivatives and Continuity

Theorem 3.1.11 (Differentiation Implies Continuity)

Assume that $f(t)$ is differentiable at $t = a$. Then $f(t)$ is continuous at $t = a$.

Derivatives and Continuity



Example: Consider

$$f(t) = \begin{cases} \frac{|t|}{t} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases}.$$

Then we know that

$$f(t) = \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{if } t = 0 \\ -1 & \text{if } t < 0 \end{cases}.$$

Then

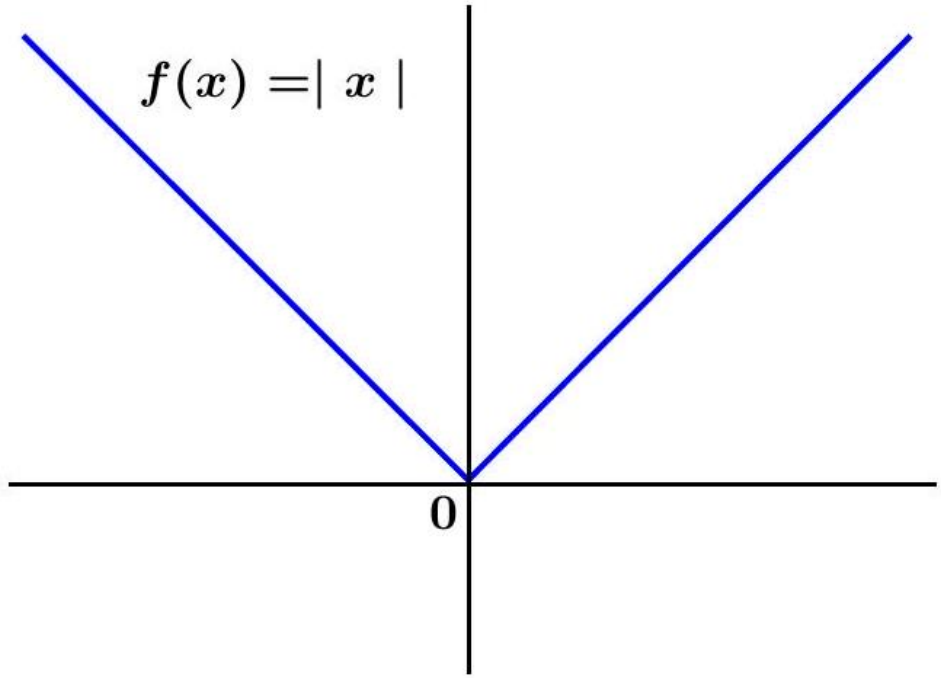
$$\lim_{t \to 0^+} \frac{f(t) - f(0)}{t - 0} = \lim_{t \to 0^+} \frac{1}{t} = \infty$$

since $f(0) = 0$ and $f(t) = 1$ if $t > 0$.

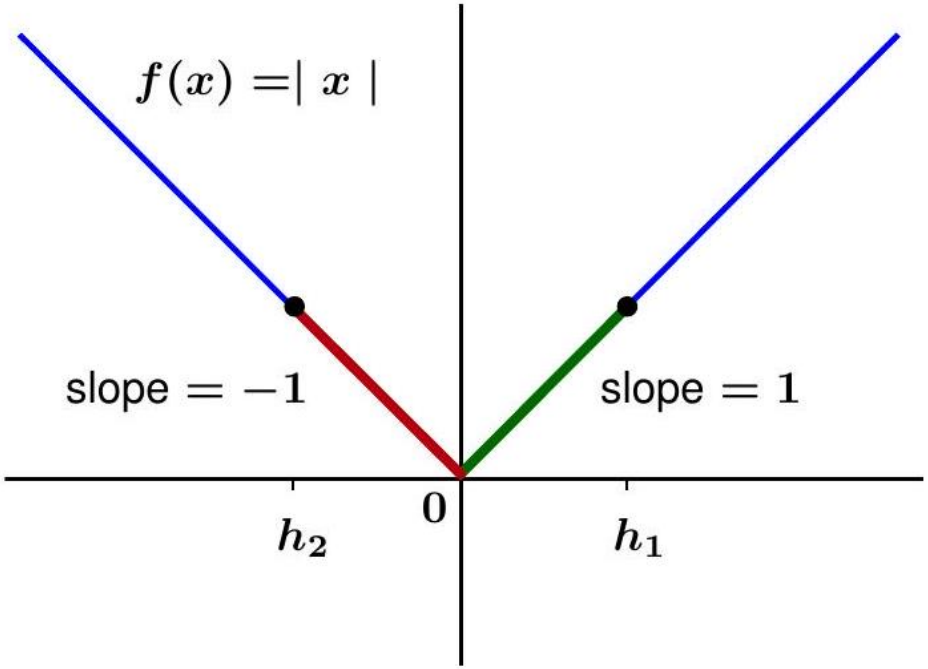
Derivatives and Continuity

Example: Let $f(x) = |x|$ and let $a = 0$. Then $f(x)$ is continuous at 0. However, since $f(0) = 0$, we get that

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{|x|}{x}$$



Derivatives and Continuity

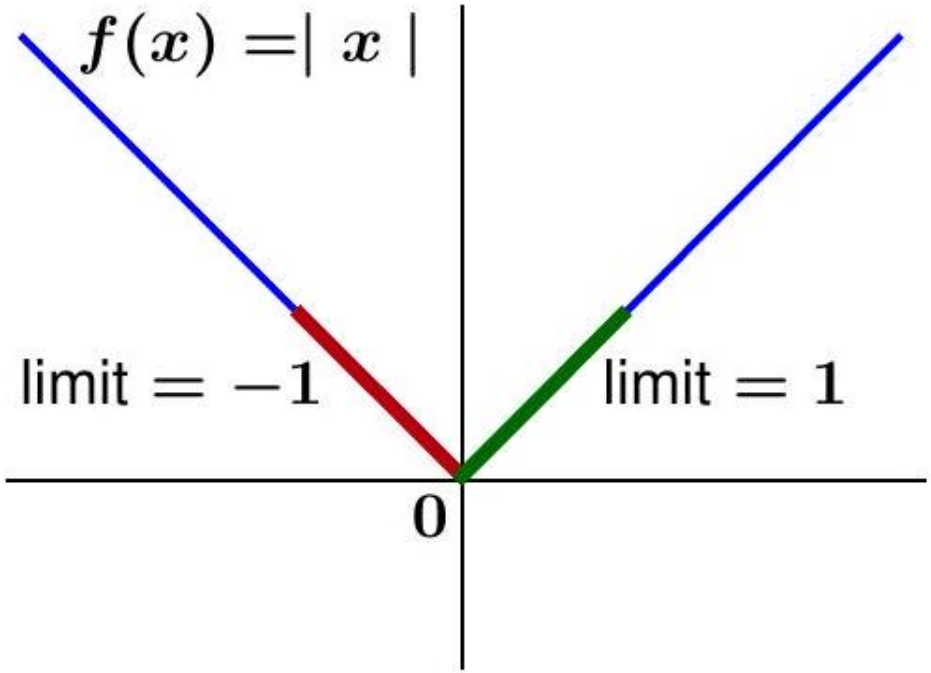


Example (continued):

If $h_1 > 0$, then the slope of the secant line through $(0, f(0)) = (0, 0)$ and $(h_1, f(h_1)) = (h_1, h_1)$ is 1 .

If $h_2 < 0$, then the slope of the secant line through $(0, f(0)) = (0, 0)$ and $(h_2, f(h_2)) = (h_2, -h_2)$ is -1 .

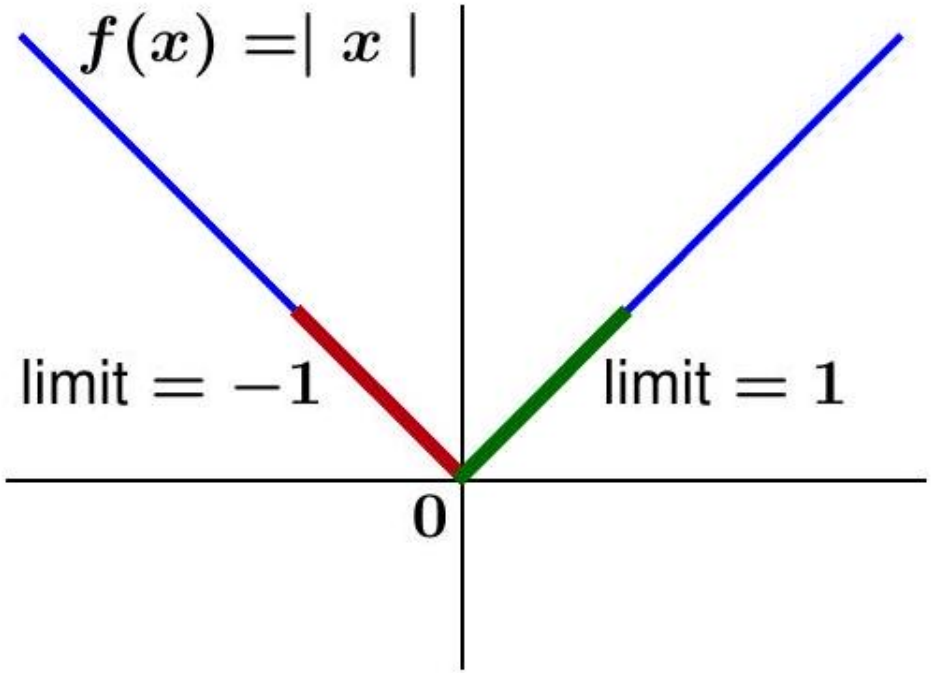
Derivatives and Continuity



Example (continued): Therefore

$$\begin{aligned}\lim_{x\to 0^+}\frac{f(x)-f(0)}{x-0} &= \lim_{x\to 0^+}\frac{|x|}{x}\\ &= \lim_{x\to 0^+}\frac{x}{x}\\ &= 1\end{aligned}$$

Derivatives and Continuity



while

$$\begin{aligned}\lim_{x\to 0^-}\frac{f(x)-f(0)}{x-0} &= \lim_{x\to 0^-}\frac{|x|}{x}\\ &= \lim_{x\to 0^-}\frac{-x}{x}\\ &= -1\end{aligned}$$

Hence $f(x) = |x|$ is continuous, but not differentiable at $x = 0$.

Derivatives and Continuity

Historically Important Question: Does there exist a function $f(x)$ that is continuous at each $x \in \mathbb{R}$, but not differentiable at even one point?

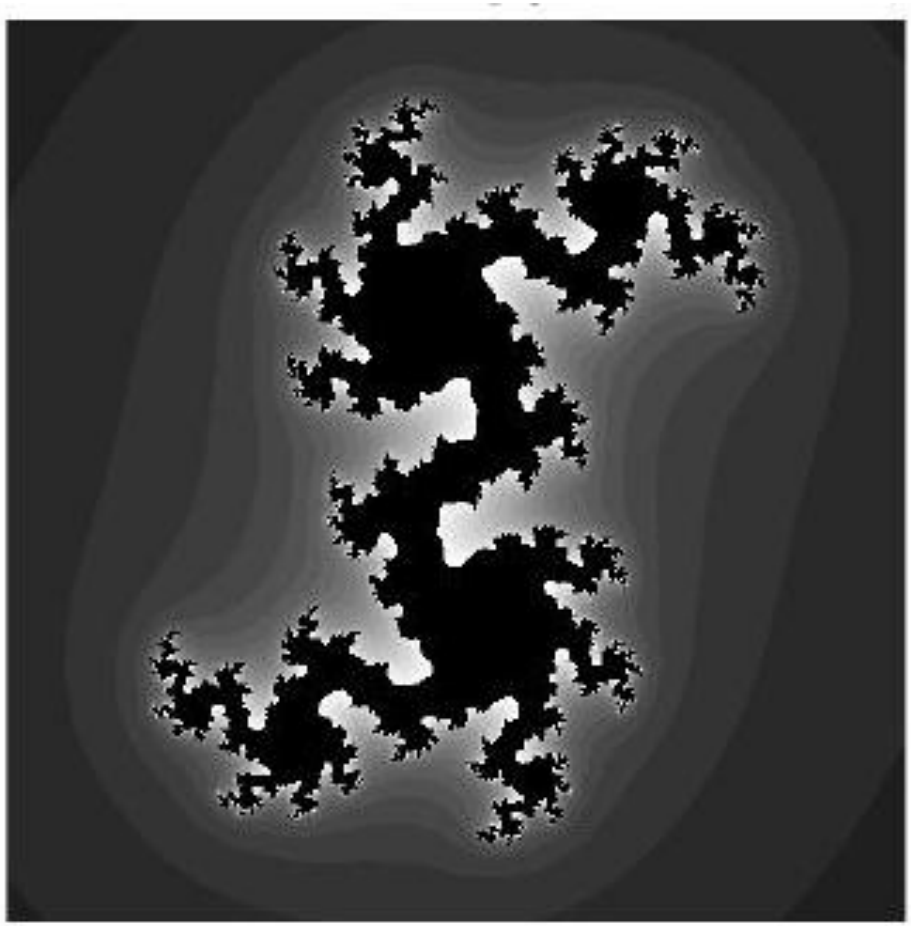
Answer: Yes! [Karl Weierstrass (1872)]

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{2^n} \sin\left(2^n x\right)$$

Derivatives and Continuity

Note:

1. The most famous application of these types of functions are fractals.



2. It can be shown that most continuous functions are nowhere differentiable.

The Derivative at a Point

Recall: We say that the function $f(x)$ is differentiable at $x = a$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

Equivalently:

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

if this limit exists.

The Derivative Function

Definition 3.1.12 (Derivative Function)

We say that a function $f(x)$ is differentiable on an interval I if $f'(a)$ exists for every $a \in I$.

In this case, we define the derivative function on I, denoted by f', where

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

That is, the value of the derivative function at x is simply the derivative of f at x for each $x \in I$.

Leibniz Notation

Leibniz Notation: Given a function $y = f(t)$, Leibniz wrote

$$\frac{dy}{dt} \text{ or } \frac{df}{dt}$$

to represent the derivative of y (or equivalently, of f) with respect to t.

An alternate form of Leibniz's notation is to write

$$\frac{d}{dt}(f(t))$$

to indicate that $f(t)$ is to be differentiated with respect to the variable t.

The symbol

$$\frac{d}{dt}$$

is called a differential operator.

In Leibniz's notation, we denote $f'(a)$, the derivative at $t = a$, by

$$\left.\frac{dy}{dt}\right|_a \quad \text{or} \quad \left.\frac{df}{dt}\right|_a$$

Higher Derivatives

Definition 3.1.13 (Higher Derivatives)

Let $f(x)$ be a differentiable function with derivative $f'(x)$. If $f'(x)$ is also differentiable, then its derivative

$$\frac{d}{dx}\left(f'(x)\right)$$

is called the second derivative of $f(x)$ and it is usually denoted by

$$f''(x) \text{ or } f^{(2)}(x) \text{ or } \frac{d^2}{dx^2}(f(x))$$

Higher Derivatives

Definition 3.1.13 (Higher Derivatives) (Continue)

If $f''(x)$ is also differentiable, then its derivative is called the third derivative of $f(x)$ and it is denoted by

$$f'''(x) \text{ or by } f^{(3)}(x)$$

In general, for any $n \geq 1$,

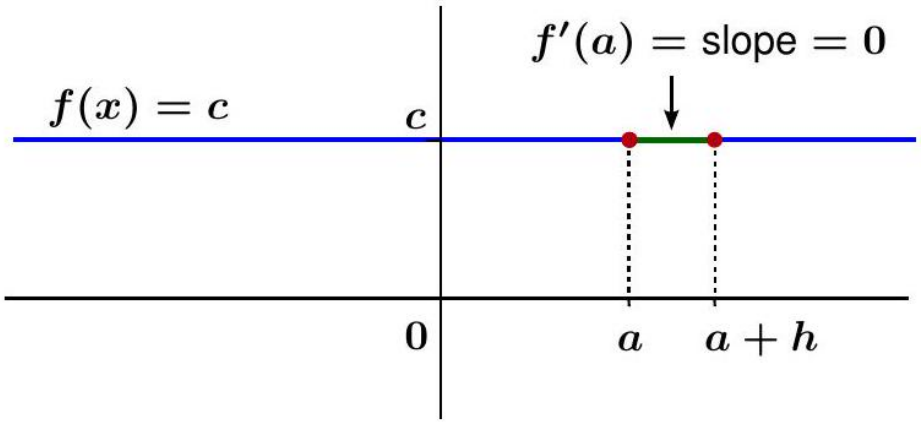
$$f^{(n+1)}(x) = \frac{d}{dx}\left(f^{(n)}(x)\right)$$

and $f^{(n)}(x)$ is called the n-th derivative of $f(x)$.

Higher Derivatives

Note: $f''(x)$ impacts the geometry of the graph of $f(x)$. In particular, the larger the magnitude of $f''(x)$, the more curved the graph of $f(x)$.

Derivative of a Constant Function



Example 3.1.14 (Derivative of a Constant Function)

Assume that $f(x) = c$ for all $x \in \mathbb{R}$ and let $a \in \mathbb{R}$. Then

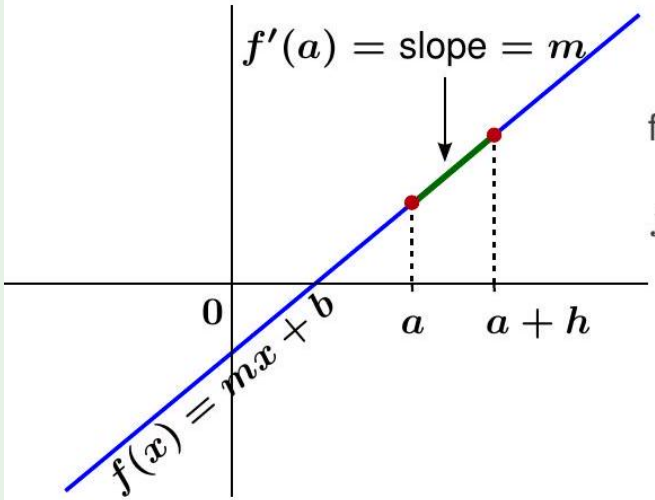
$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= 0 \end{aligned}$$

Hence $f'(x) = 0$ for all $x \in \mathbb{R}$.

Derivative of a Linear Function

Example 3.1.15 (Derivative of a Linear Function)

Assume that



$$f(x) = mx + b$$

for all $x \in \mathbb{R}$. Then

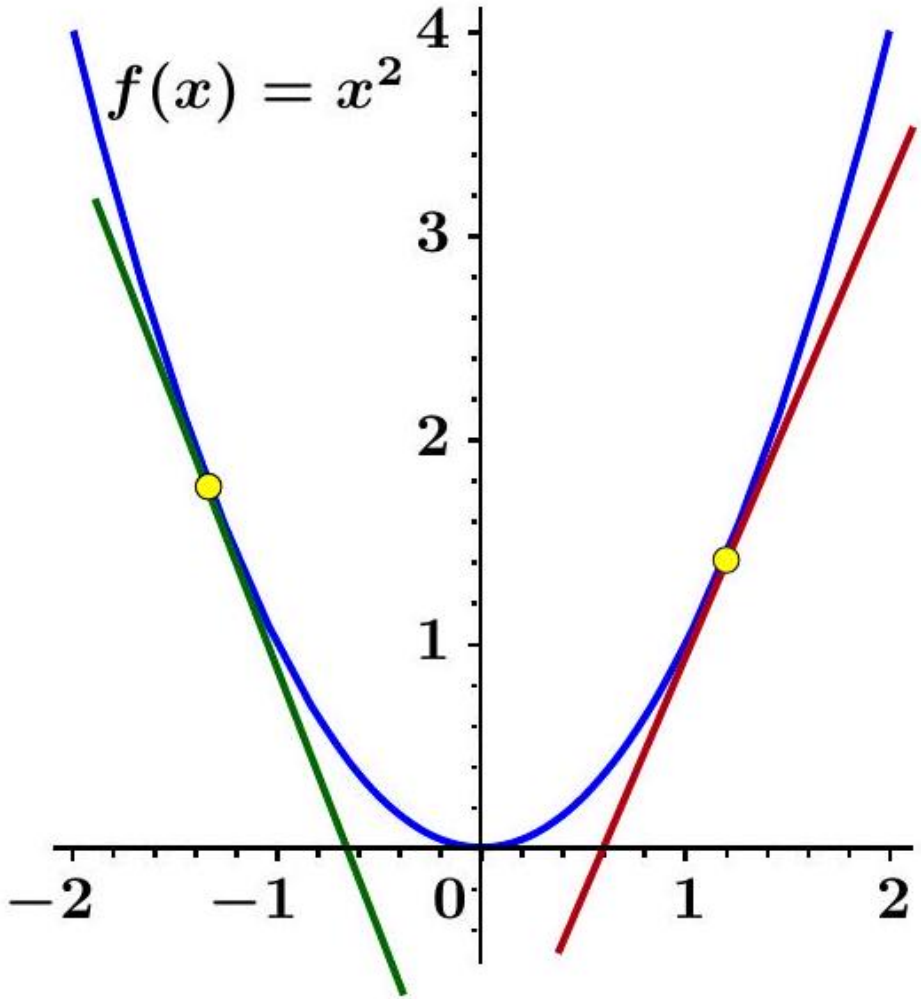
$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h\to 0} \frac{(m(a+h)+b) - (ma+b)}{h} \\ &= \lim_{h\to 0} \frac{ma + mh + b - ma - b}{h} \\ &= \lim_{h\to 0} \frac{mh}{h} \\ &= m. \end{aligned}$$

Hence

$$f'(x) = m$$

for all $x \in \mathbb{R}$.

Derivative of $f(x) = x^2$

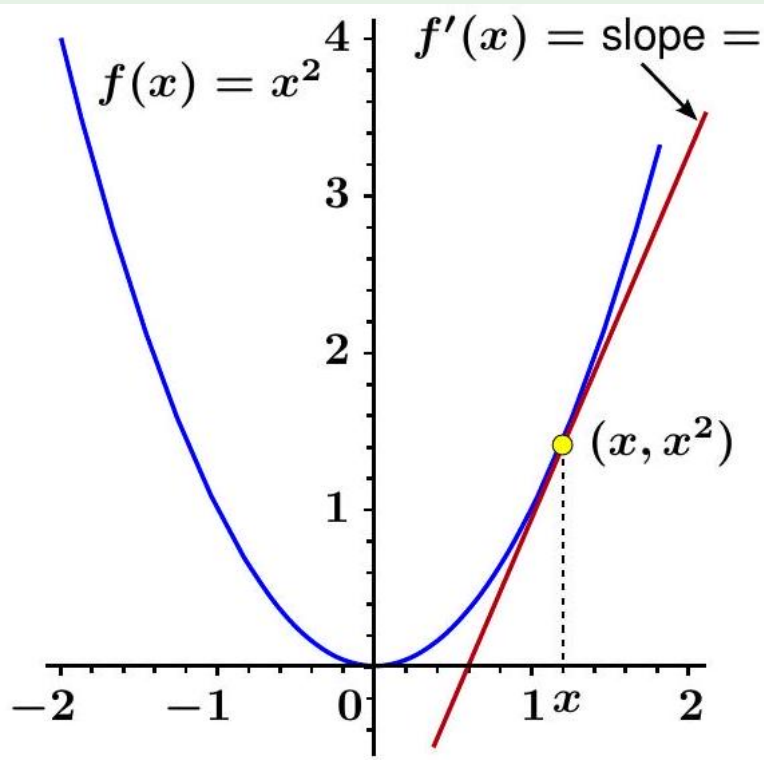


Example: Calculate the derivative of $f(x) = x^2$.

Note: Unlike the previous examples, the derivative appears to vary with the choice of x.

Derivative of $f(x) = x^2$

Example 3.1.16 (Derivative of a Quadratic Function)

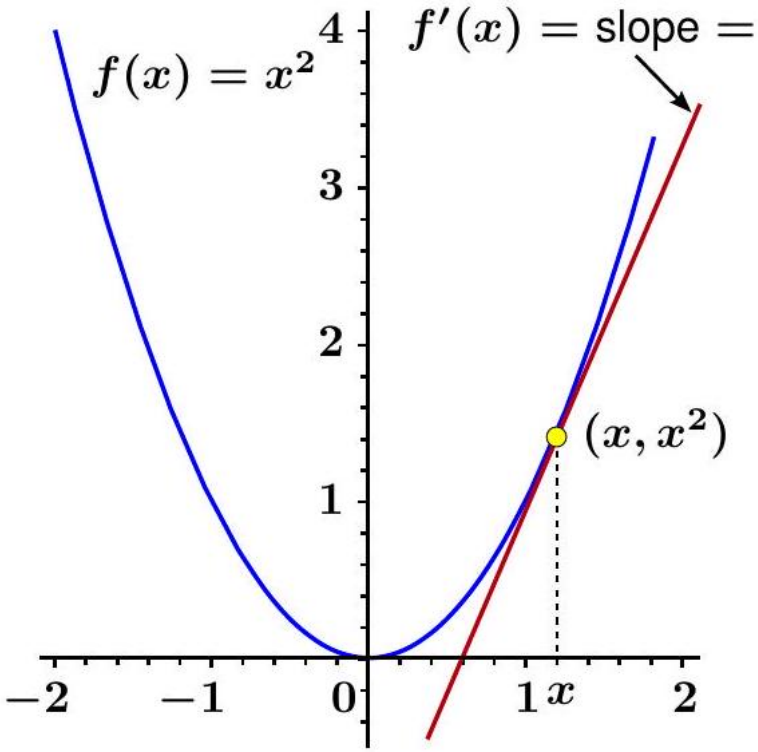


Calculate the derivative of $f(x) = x^2$.

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{2xh + h^2}{h} \\
&= \lim_{h \to 0} 2x + h \\
&= 2x.
\end{aligned}
$$

Derivative of $f(x) = x^2$

Example 3.1.16 (Derivative of a Quadratic Function) (Continue)

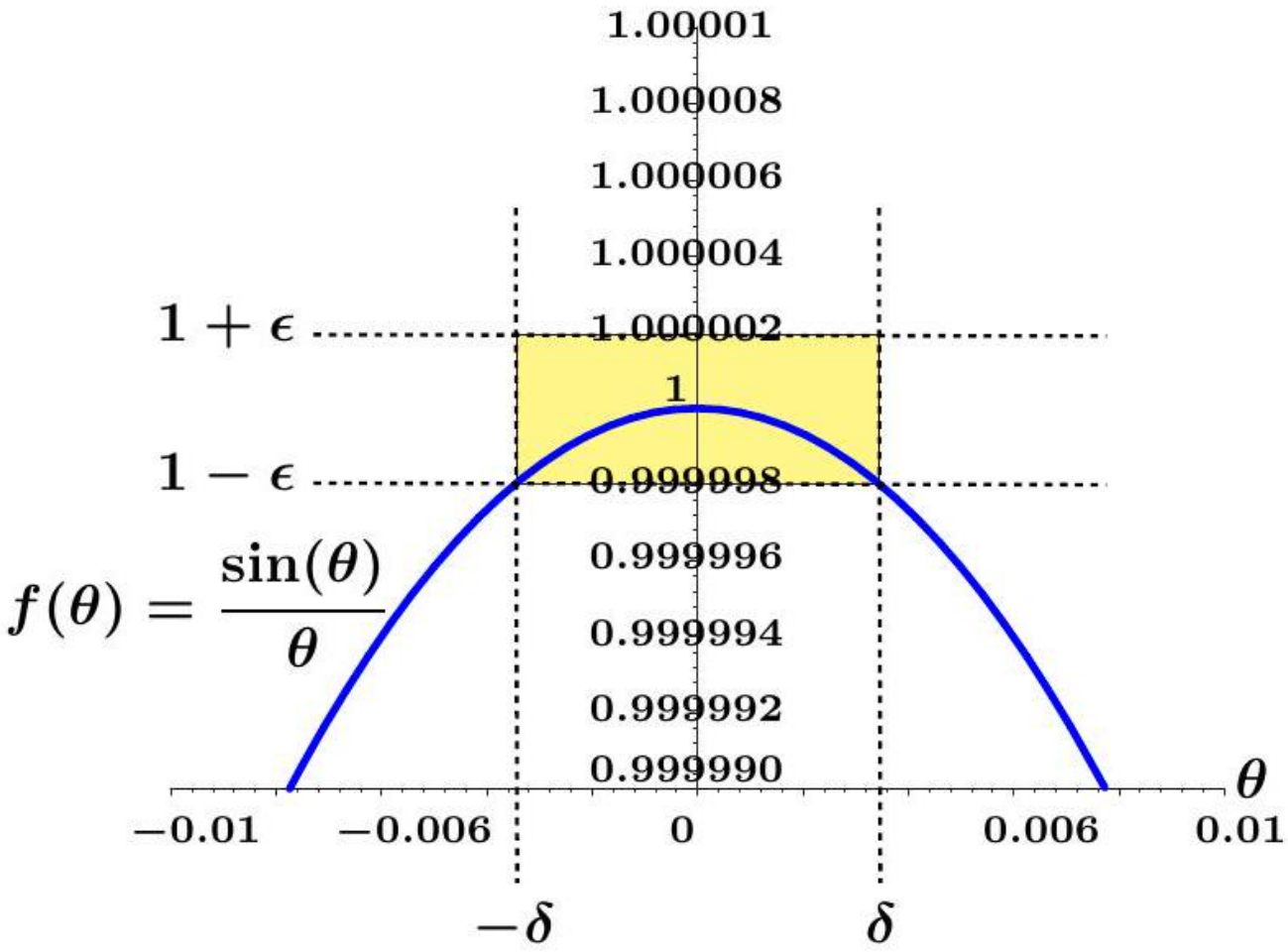


Calculate the derivative of $f(x) = x^2$.

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{2xh + h^2}{h} \\
&= \lim_{h \to 0} 2x + h \\
&= 2x.
\end{aligned}
$$

Note: If $f(x) = x^2$, then $f'(x) = 2x$, so $f''(x) = 2$.

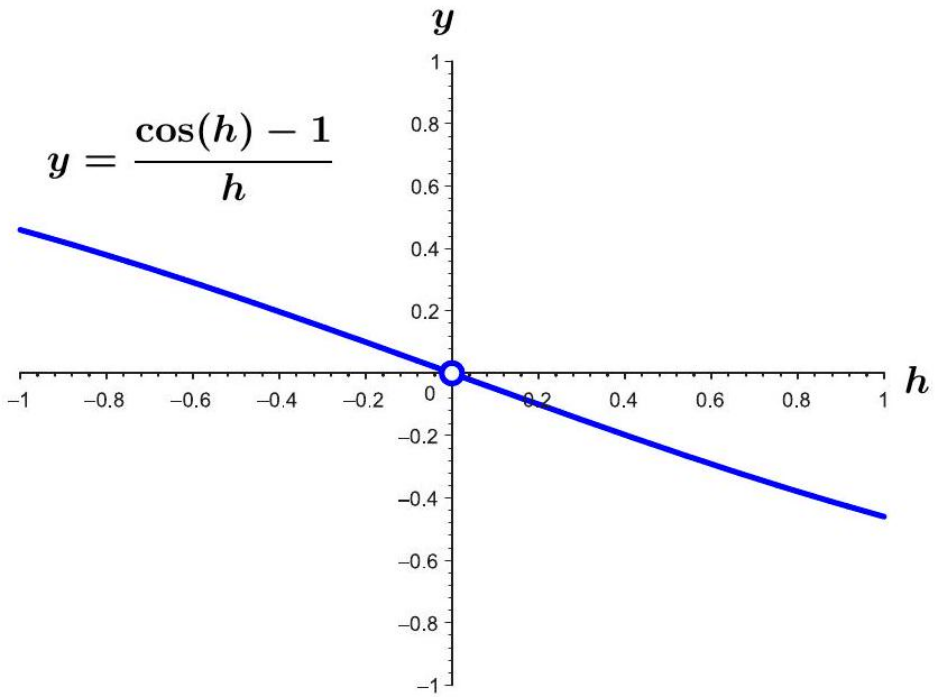
Derivative of $\sin(x)$



Theorem 3.1.17 (Fundamental Trig Limit (FTL))

$$\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta} = 1$$

Derivative of $\sin(x)$



Theorem 3.1.18

$$\lim_{h \to 0} \frac{\cos(h) - 1}{h} = 0$$

Derivative of $\sin(x)$

Theorem 3.1.18

$$\lim_{h\to 0}\frac{\cos(h)-1}{h}=0$$

Proof: We have

$$\begin{aligned}
\lim_{h\to 0}\frac{\cos(h)-1}{h} &= \lim_{h\to 0}\left(\frac{\cos(h)-1}{h}\right)\left(\frac{\cos(h)+1}{\cos(h)+1}\right)\\
&= \lim_{h\to 0}\frac{\cos^2(h)-1}{h(\cos(h)+1)}\\
&= \lim_{h\to 0}\frac{-\sin^2(h)}{h(\cos(h)+1)}\\
&= \lim_{h\to 0}\frac{\sin(h)}{h}\cdot\lim_{h\to 0}\frac{-\sin(h)}{(\cos(h)+1)}\\
&= 1\cdot 0\\
&= 0
\end{aligned}$$

since $\lim_{h\to 0}\frac{\sin(h)}{h}=1, \lim_{h\to 0}(\sin(h))=0$ and $\lim_{h\to 0}(\cos(h)+1)=2$.

Derivative of $\sin(x)$

Problem: Find $\frac{d}{dx}(\sin(x))$.

Solution:

$$\begin{aligned}
&\lim_{h\to 0} \frac{\sin(x+h)-\sin(x)}{h} \\
&= \lim_{h\to 0} \frac{(\sin(x)\cos(h)+\cos(x)\sin(h))-\sin(x)}{h} \\
&\quad = \lim_{h\to 0} \left(\sin(x)\frac{\cos(h)-1}{h}+\cos(x)\frac{\sin(h)}{h}\right) \\
&= \sin(x)\lim_{h\to 0}\frac{\cos(h)-1}{h}+\cos(x)\lim_{h\to 0}\frac{\sin(h)}{h} \\
&= \sin(x)\cdot 0+\cos(x)\cdot 1 \\
&= \cos(x).
\end{aligned}$$

Derivative of $\sin(x)$

Theorem 3.1.19 (The Derivative of $\sin(x)$)

Assume that $f(x) = \sin(x)$. *Then*

$$f'(x) = \cos(x)$$

for all $x \in \mathbb{R}$.

Derivative of $\cos(x)$

Theorem 3.1.20 (The Derivative of $\cos(x)$)

Assume that $f(x) = \cos(x)$. *Then*

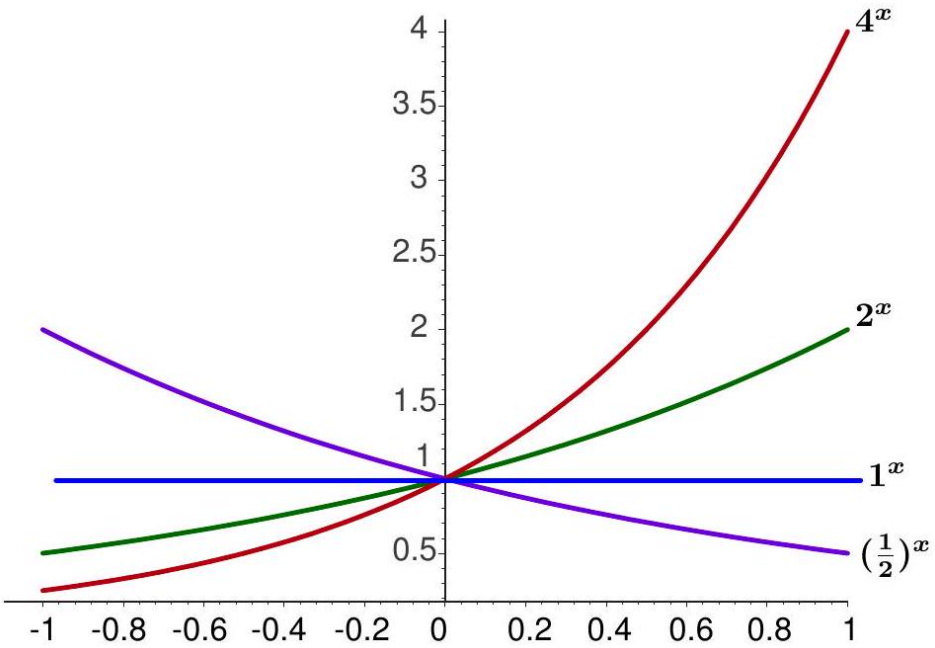
$$f'(x) = -\sin(x)$$

for all $x \in \mathbb{R}$.

Proof: We have

$$\begin{aligned}
&\lim_{h\to 0} \frac{\cos(x+h) - \cos(x)}{h} \\
&= \lim_{h\to 0} \frac{(\cos(x)\cos(h) - \sin(x)\sin(h)) - \cos(x)}{h} \\
&= \lim_{h\to 0} \left(\cos(x)\frac{\cos(h)-1}{h} - \sin(x)\frac{\sin(h)}{h} \right) \\
&= \cos(x) \lim_{h\to 0} \frac{\cos(h)-1}{h} - \sin(x) \lim_{h\to 0} \frac{\sin(h)}{h} \\
&= \cos(x)\cdot 0 - \sin(x)\cdot 1 \\
&= -\sin(x)
\end{aligned}$$

Derivative of a^x



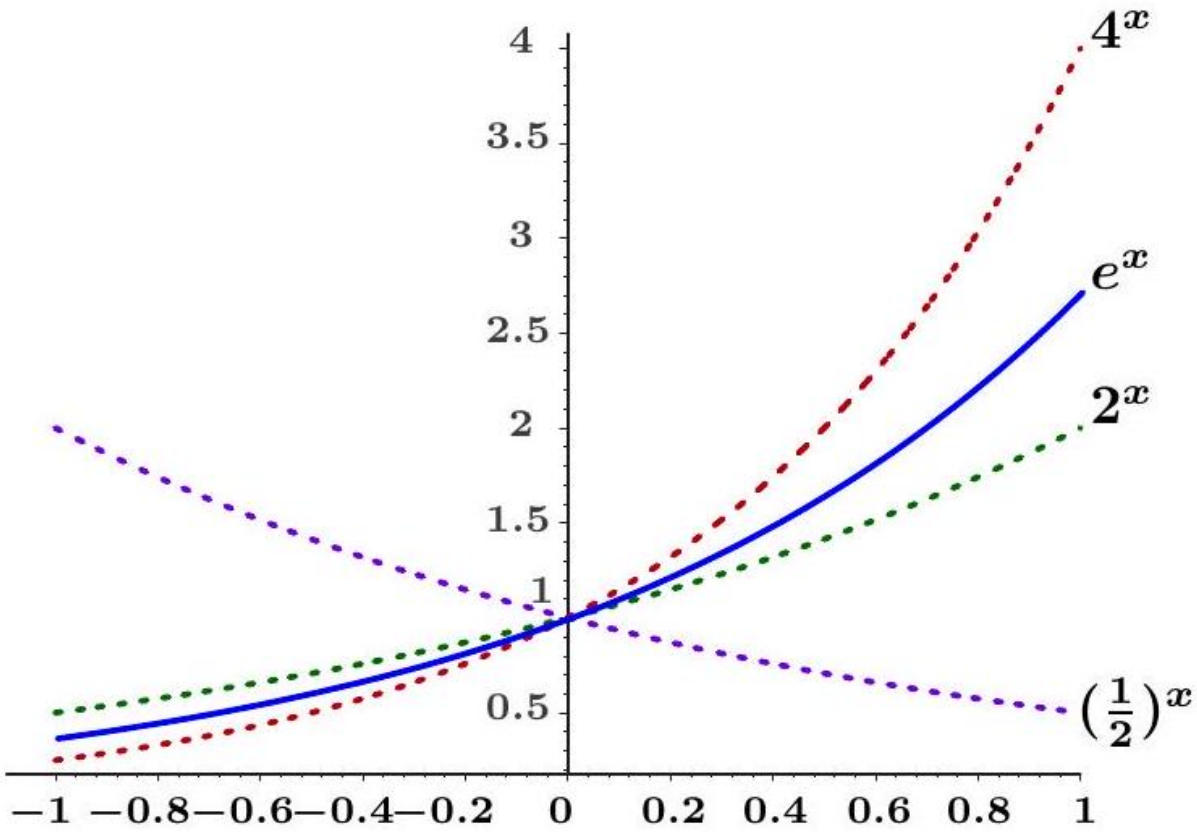
Problem: If $f(x) = a^x$ where $a > 0$, what is $f'(x)$?

Derivative of a^x

Problem: If $f(x) = a^x$ where $a > 0$, what is $f'(x)$?

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h \to 0} \frac{a^x \left(a^h - 1\right)}{h} \\ &= a^x \cdot \left(\lim_{h \to 0} \frac{a^h - 1}{h} \right) \\ &= a^x \cdot f'(0) \end{aligned}$$

Derivative of a^x



Definition 3.1.21 (Euler's Constant e)

Let $f(x) = a^x$. Of all the possible choices for $a > 0$, e is the unique base such that if $f(x) = e^x$, then $f'(0) = 1$.

$$2 < e < 4 \Rightarrow e \approx 2.718281828$$

Derivative of a^x

Summary: Let $a > 0$. Then:

1. If $f(x) = a^x$, then $f'(x) = f'(0) \cdot a^x$.
2. If $f(x) = e^x$, then $f'(x) = e^x$.

Problem: If $f(x) = a^x$, and $f'(x) = f'(0) \cdot a^x$, what is $f'(0)$?

Answer: $\ln(a)$

Derivative of a^x

Theorem 3.1.22 (Derivative of a^x)

Let $a > 0$. If $f(x) = a^x$, then

$$f'(x) = \ln(a) \cdot a^x$$

for all $x \in \mathbb{R}$.

Derivative of e^x

Special Case: Since $\ln(e) = 1$, we have:

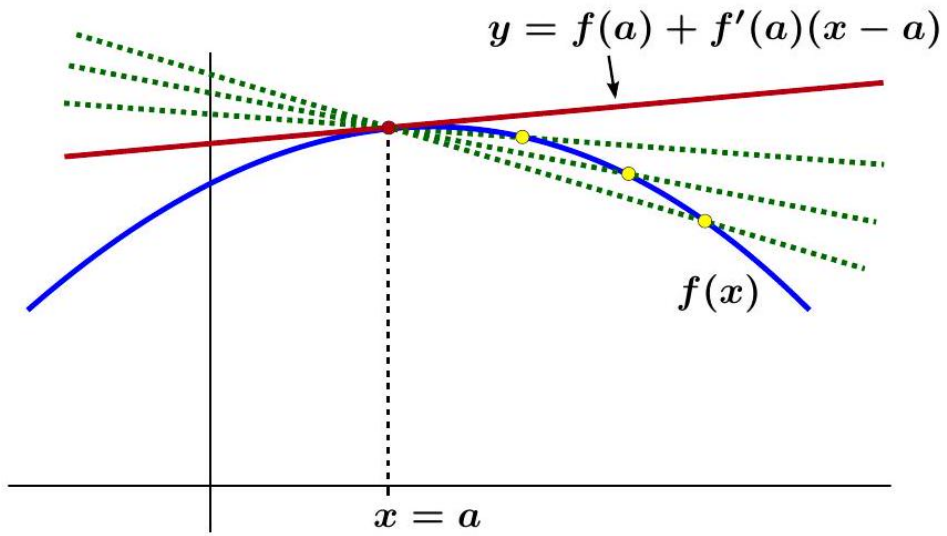
Theorem 3.1.23 (Derivative of e^x)

Let $f(x) = e^x$. Then

$$f'(x) = e^x$$

for all $x \in \mathbb{R}$.

Tangent Line



Question: Does there exist a geometric interpretation of the derivative?

Definition 3.1.24 (Tangent Line)

The tangent line to the graph of $f(x)$ at $x = a$ is the line passing through $(a, f(a))$ with slope equal to $f'(a)$. That is

$$y = f(a) + f'(a)(x - a)$$

Linear Approximation

Fundamental Observation: Suppose that $f(x)$ is differentiable at $x = a$ with derivative $f'(a)$. Then

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = f'(a)$$

Hence for values of x that are close to a we have

$$\frac{f(x) - f(a)}{x - a} \approx f'(a)$$

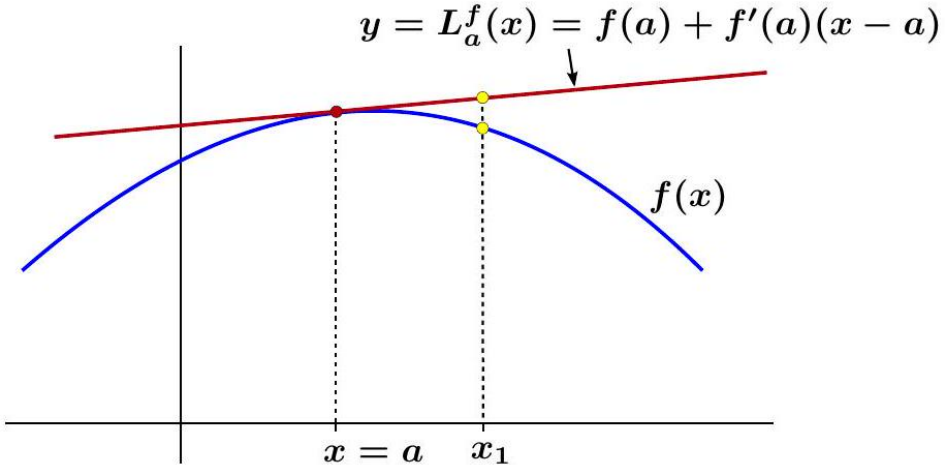
Rearranging $(*)$, we get

$$f(x) - f(a) \approx f'(a)(x - a)$$

and finally that

$$f(x) \approx f(a) + f'(a)(x - a). \quad (**)$$

Linear Approximation



Definition 3.1.25 (Linear Approximation)

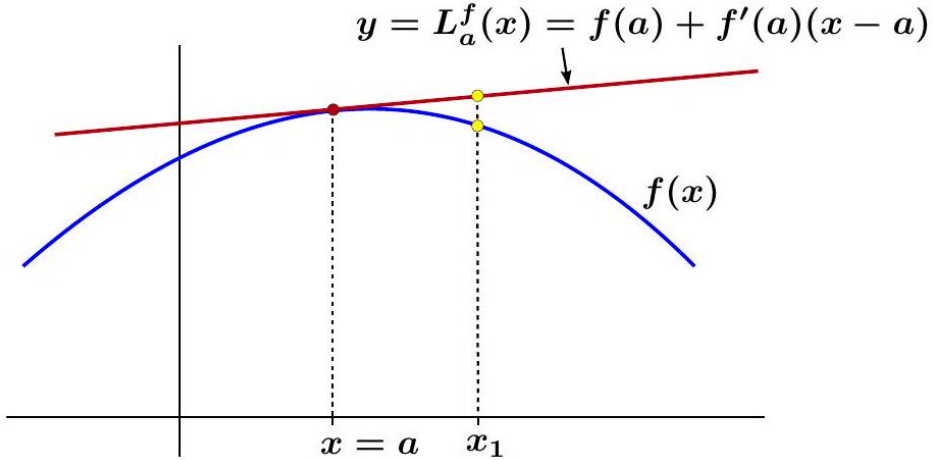
Let $y = f(x)$ be differentiable at $x = a$. The linear approximation to $f(x)$ at $x = a$ is the function

$$L_a^f(x) = f(a) + f'(a)(x - a)$$

$L_a^f(x)$ is also called the linearization of $f(x)$ or the tangent line approximation to $f(x)$ at $x = a$.

Note: If $f(x)$ is clear from the context, then we will simply write $L_a(x)$.

Linear Approximation



Summary: If

$$L_a^f(x) = f(a) + f'(a)(x - a)$$

then if $x \approx a$,

$$L_a^f(x) \approx f(x)$$

Observation: The graph of $L_a^f(x)$ is the tangent line to the graph of $f(x)$ through $(a, f(a))$.

Fundamental Properties of $L_a^f(x)$

Three Fundamental Properties of $L_a^f(x)$:

Assume that $f(x)$ is differentiable at $x = a$ with

$$L_a^f(x) = f(a) + f'(a)(x - a)$$

Then:

1. $L_a^f(a) = f(a)$.
2. $L_a^f(x)$ is differentiable at $x = a$ and $L_a^{f\prime}(a) = f'(a)$.
3. $L_a^f(x)$ is the only first degree polynomial with Property (1) and Property (2).

Section 2

Arithmetic Rules for Differentiation

Arithmetic Rules for Differentiation

Theorem 3.2.1 (The Arithmetic Rules for Differentiation)

Assume that $f(x)$ and $g(x)$ are both differentiable at $x = a$.

1. *The Constant Multiple Rule:*

Let $h(x) = cf(x)$. Then $h(x)$ is differentiable at $x = a$ and

$$h'(a) = c \cdot f'(a)$$

2. *The Sum Rule:*

Let $h(x) = f(x) + g(x)$. Then $h(x)$ is differentiable at $x = a$ and

$$h'(a) = f'(a) + g'(a)$$

3. *The Product Rule:*

Let $h(x) = f(x)g(x)$. Then $h(x)$ is differentiable at $x = a$ and

$$h'(a) = f'(a)g(a) + f(a)g'(a)$$

Arithmetic Rules for Differentiation

Theorem 3.2.1 (The Arithmetic Rules for Differentiation) (continued)

Assume that $f(x)$ and $g(x)$ are both differentiable at $x = a$.

4. The Reciprocal Rule:

Let $h(x) = \frac{1}{f(x)}$. If $f(a) \neq 0$, then $h(x)$ is differentiable at $x = a$ and

$$h'(a) = \frac{-f'(a)}{(f(a))^2}$$

5. The Quotient Rule:

Let $h(x) = \frac{f(x)}{g(x)}$. If $g(a) \neq 0$, then $h(x)$ is differentiable at $x = a$ and

$$h'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}$$

Arithmetic Rules for Differentiation

1. Proof of the Constant Multiple Rule:

Assume that $c \in \mathbb{R}$ and that $f(x)$ is differentiable at $x = a$. Then

$$\begin{aligned}
(cf)'(a) &= \lim_{h \to 0} \frac{(cf)(a+h) - (cf)(a)}{h} \\
&= \lim_{h \to 0} \frac{c \cdot f(a+h) - c \cdot f(a)}{h} \\
&= c \cdot \left(\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \right) \\
&= c \cdot f'(a)
\end{aligned}$$

Arithmetic Rules for Differentiation

2. Proof of the Sum Rule:

Assume that $f(x)$ and $g(x)$ are differentiable at $x = a$. Then

$$\begin{aligned}(f+g)'(a) &= \lim_{h\to 0} \frac{(f+g)(a+h)-(f+g)(a)}{h} \\ &= \lim_{h\to 0} \frac{f(a+h)+g(a+h)-f(a)-g(a)}{h} \\ &= \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} + \lim_{h\to 0} \frac{g(a+h)-g(a)}{h} \\ &= f'(a)+g'(a)\end{aligned}$$

Arithmetic Rules for Differentiation

3. Proof of the Product Rule:

Observe: We have

$$(fg)(a+h) - (fg)(a) = [f(a+h)g(a+h) - f(a+h)g(a)]$$
$$+ [f(a+h)g(a) - f(a)g(a)]$$
$$(fg)'(a) = \lim_{h\to 0} \frac{(fg)(a+h) - (fg)(a)}{h}$$
$$= \lim_{h\to 0} \frac{f(a+h)(g(a+h) - g(a))}{h}$$
$$+ \lim_{h\to 0} \frac{g(a)(f(a+h) - f(a))}{h}$$
$$= \lim_{h\to 0} f(a+h) \cdot \lim_{h\to 0} \frac{(g(a+h) - g(a))}{h}$$
$$+ g(a) \cdot \lim_{h\to 0} \frac{(f(a+h) - f(a))}{h}$$
$$= f(a)g'(a) + f'(a)g(a)$$

Arithmetic Rules for Differentiation

4. Proof of the Reciprocal Rule:

Assume that $f(x)$ is differentiable at $\boldsymbol{x = a}$. Then

$$\begin{aligned}
\left(\frac{1}{f}\right)'(a) &= \lim_{h\to 0} \frac{\frac{1}{f(a+h)} - \frac{1}{f(a)}}{h} \\
&= \lim_{h\to 0} \frac{f(a) - f(a+h)}{f(a+h)f(a)h} \\
&= -\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h\to 0} \frac{1}{f(a+h)f(a)} \\
&= -f'(a) \cdot \frac{1}{(f(a))^2} \qquad \text{(by continuity at } x = a\text{)} \\
&= \frac{-f'(a)}{(f(a))^2}
\end{aligned}$$

5. Proof of the Quotient Rule:

The proof of the Quotient Rule is a combination of the Product Rule and the Reciprocal Rule.

Power Rule for Differentiation

Note: We have seen that

$$\frac{d}{dx}(x) = 1 \quad \text{and} \quad \frac{d}{dx}\left(x^2\right) = 2x$$

Using the Binomial Theorem we can show that if $\boldsymbol{n} \in \mathbb{N}$, then

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

Theorem 3.2.2 (The Power Rule for Differentiation)

Assume that $\alpha \in \mathbb{R}, \alpha \neq 0$, and $f(x) = x^\alpha$. Then $f(x)$ is differentiable and

$$f'(x) = \alpha x^{\alpha-1}$$

wherever $x^{\alpha-1}$ is defined.

Differentiating Polynomials and Rational Functions

Example 3.2.3 (Differentiating Polynomials and Rational Functions)

1. Let $P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ be a polynomial.

Then $\boldsymbol{P(x)}$ is always differentiable and

$$P'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + n a_n x^{n-1}$$

2. Using the Quotient Rule, we see that a rational function

$$R(x) = \frac{P(x)}{Q(x)}$$

is differentiable at any point where $Q(x) \neq 0$.

Differentiating Polynomials and Rational Functions

Example 3.2.4

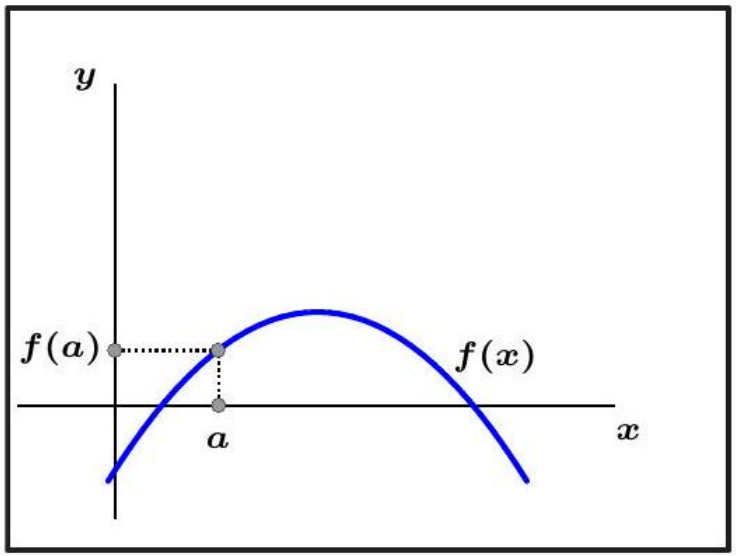
If

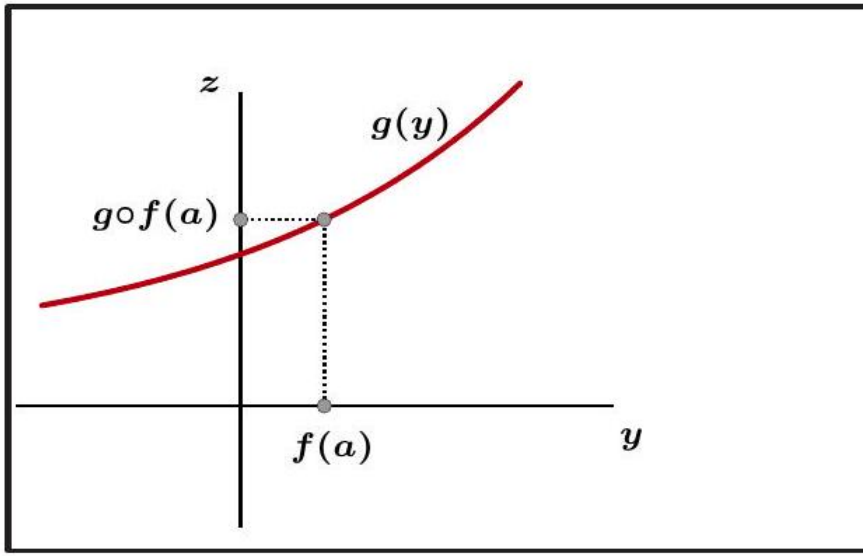
$$R(x) = \frac{x+2}{x^2-1}$$

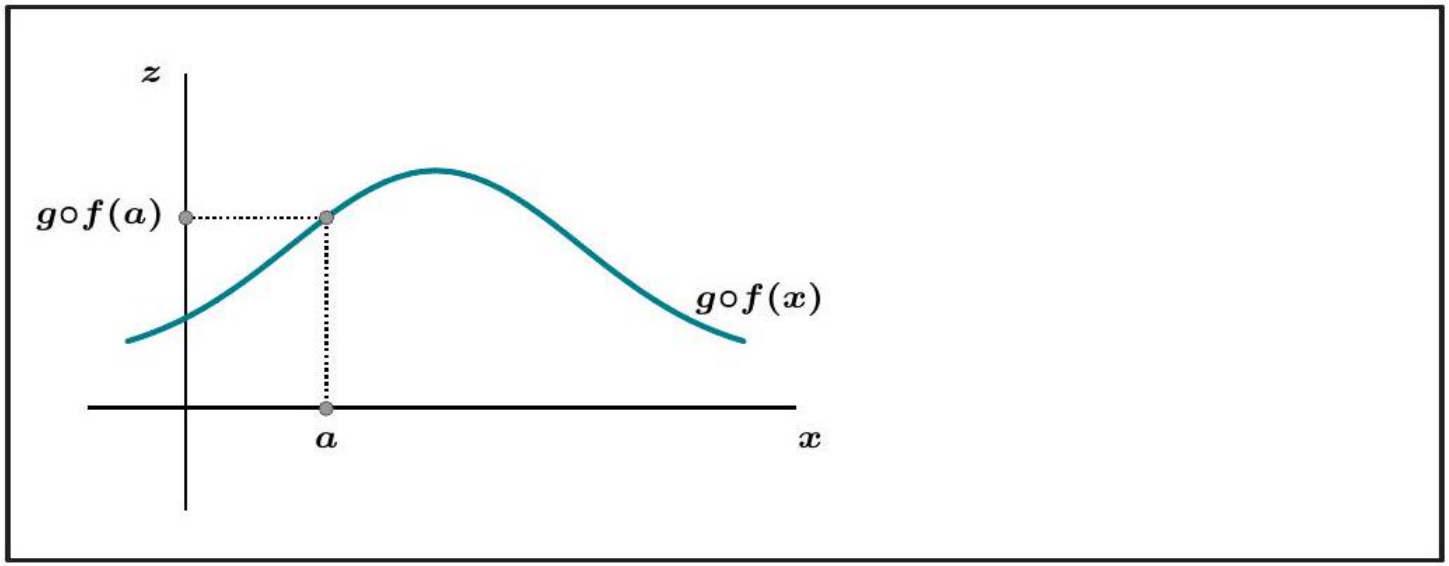
then $R(x)$ is differentiable provided that $x^2 - 1 \neq 0$. That is, when $x \neq \pm 1$. Moreover,

$$\begin{aligned} R'(x) &= \frac{\left(\frac{d}{dx}(x+2)\right)\left(x^2-1\right) - (x+2)\left(\frac{d}{dx}\left(x^2-1\right)\right)}{\left(x^2-1\right)^2} \\ &= \frac{1 \cdot \left(x^2-1\right) - (x+2)(2x)}{\left(x^2-1\right)^2} \\ &= \frac{\left(x^2-1\right) - 2x^2 - 4x}{\left(x^2-1\right)^2} \\ &= \frac{-x^2-4x-1}{\left(x^2-1\right)^2} \end{aligned}$$

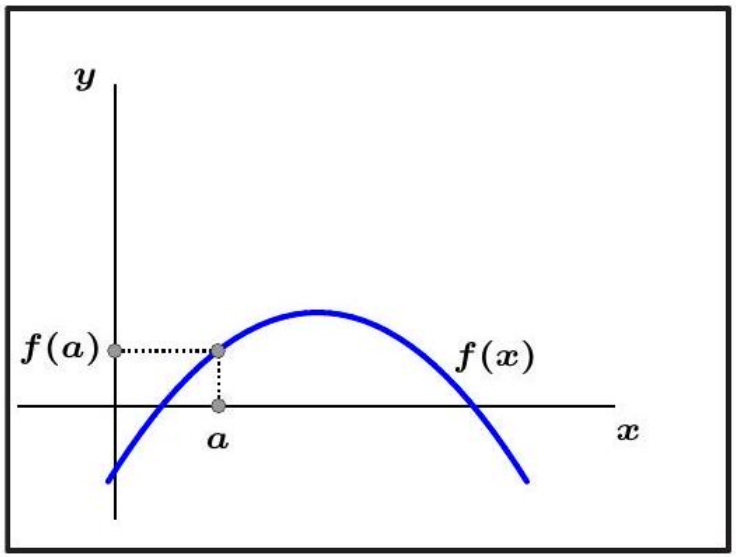
Chain Rule

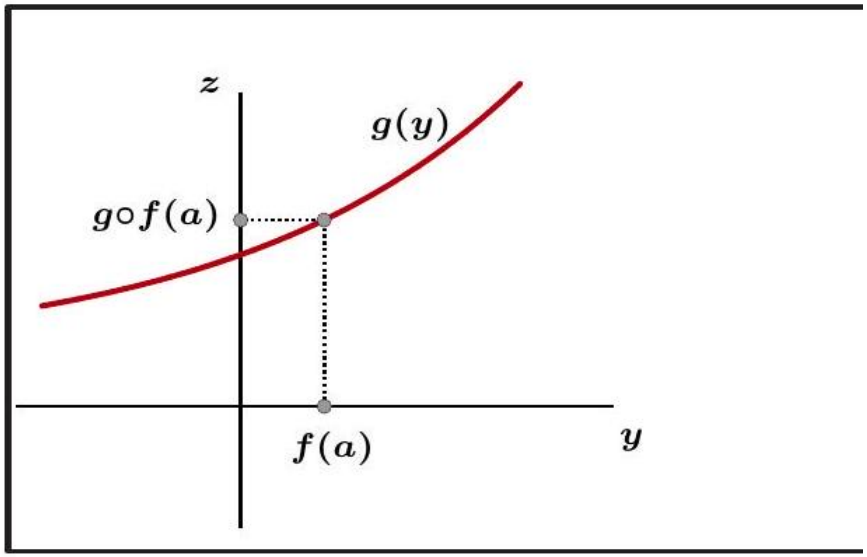


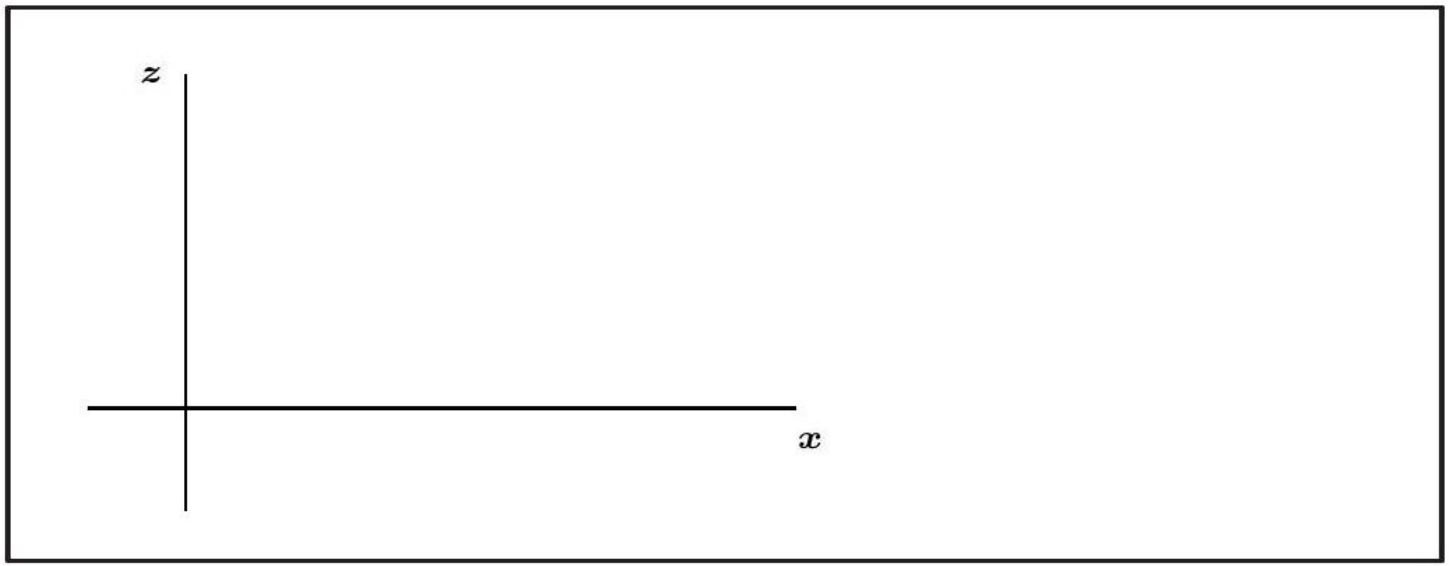




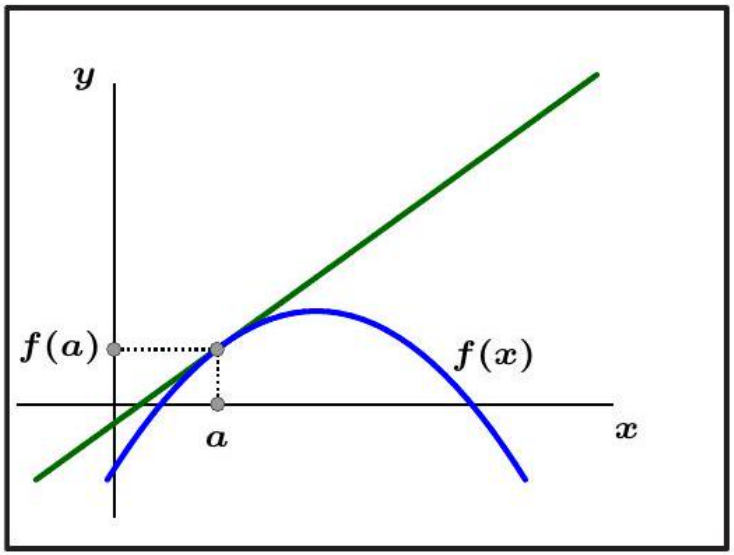
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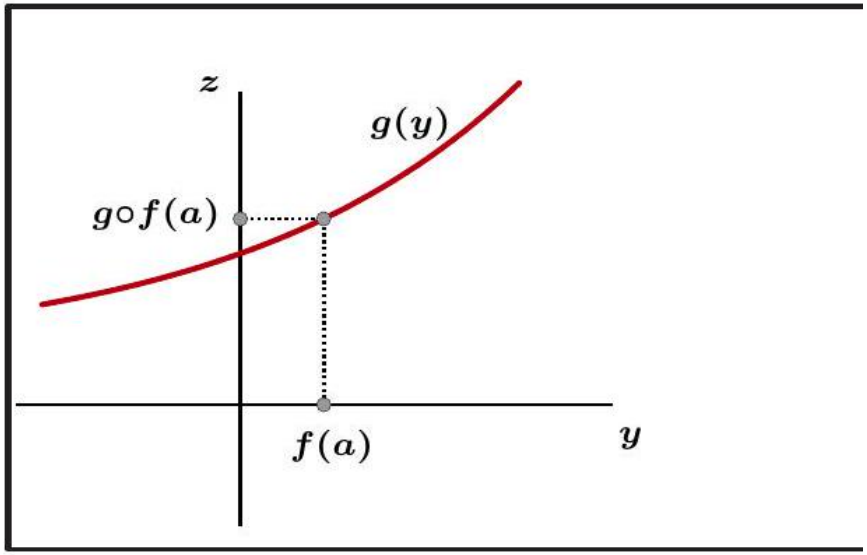


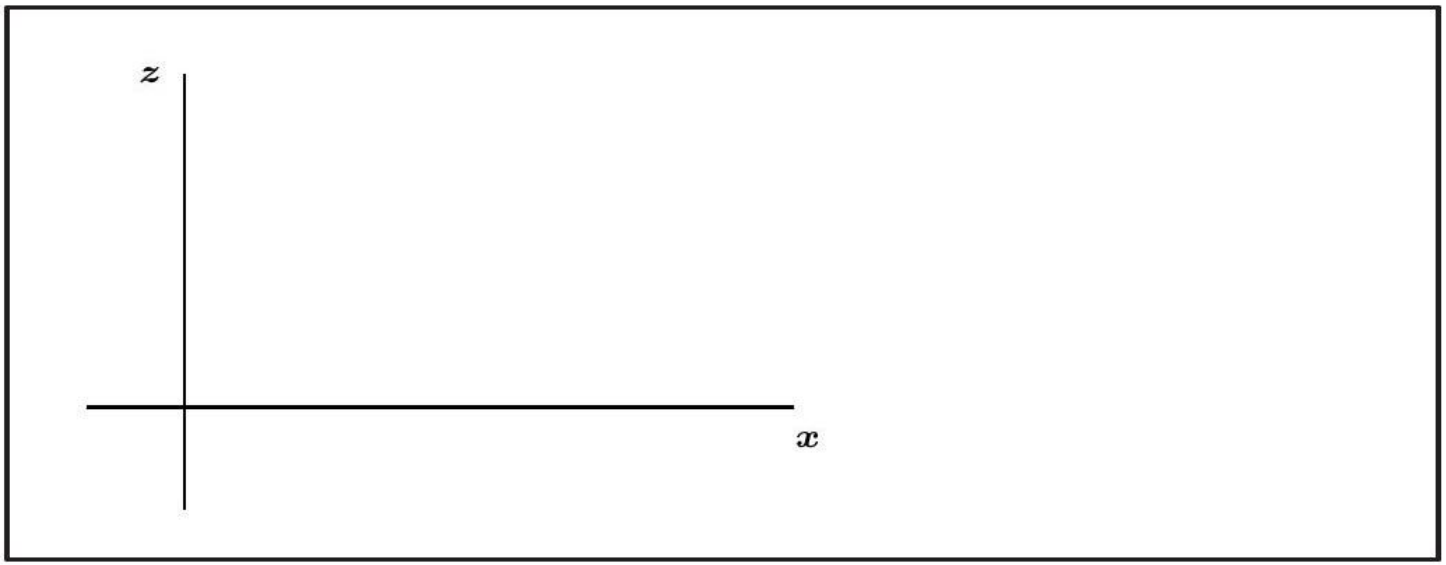




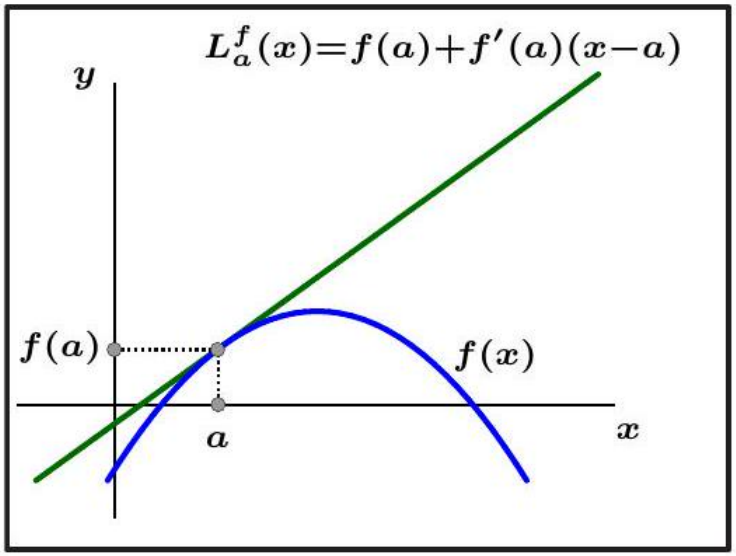
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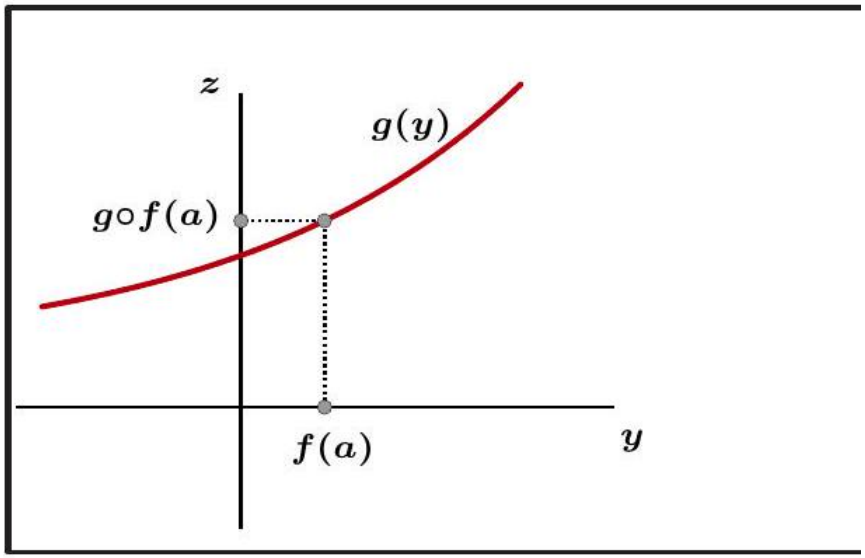


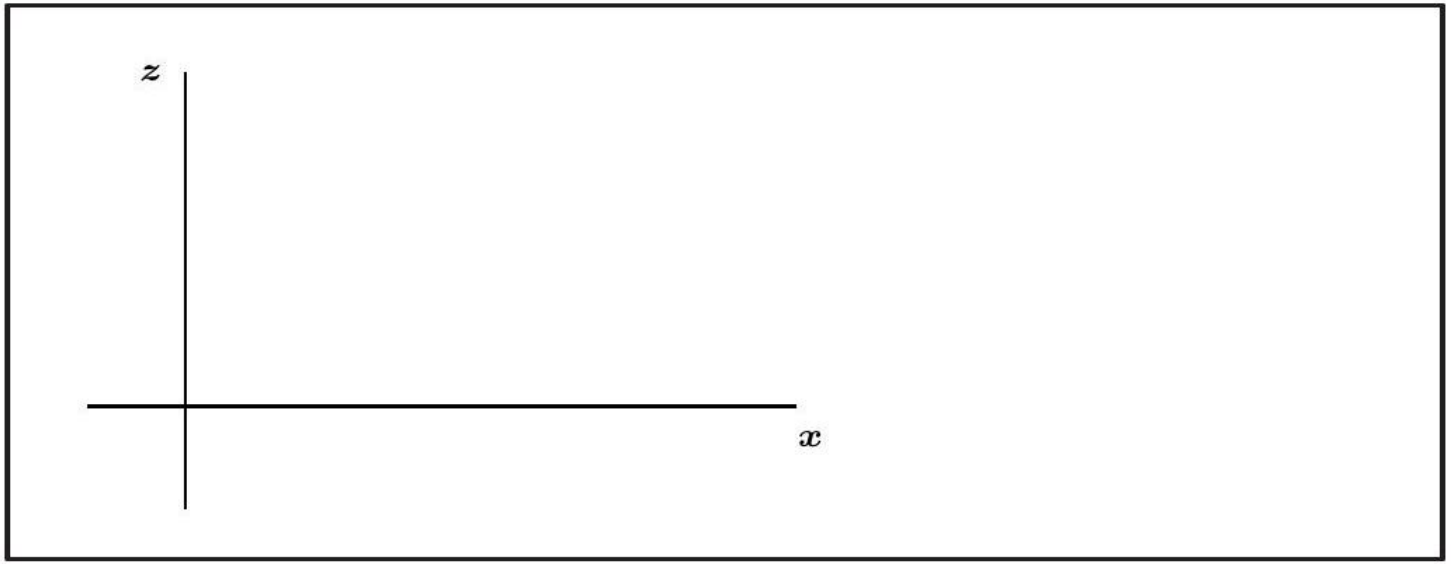




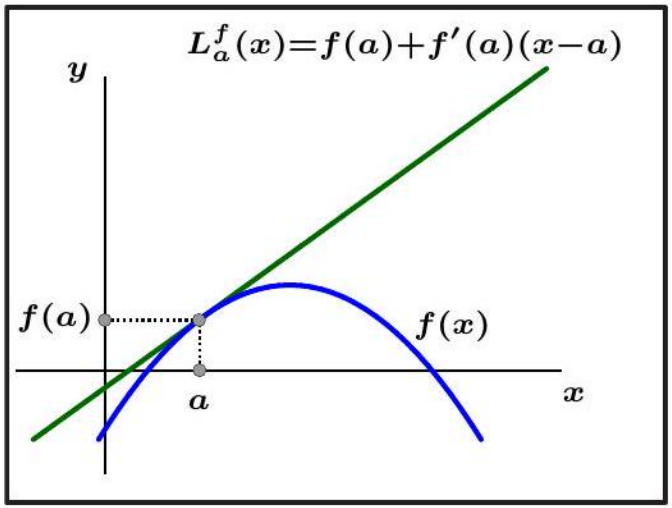
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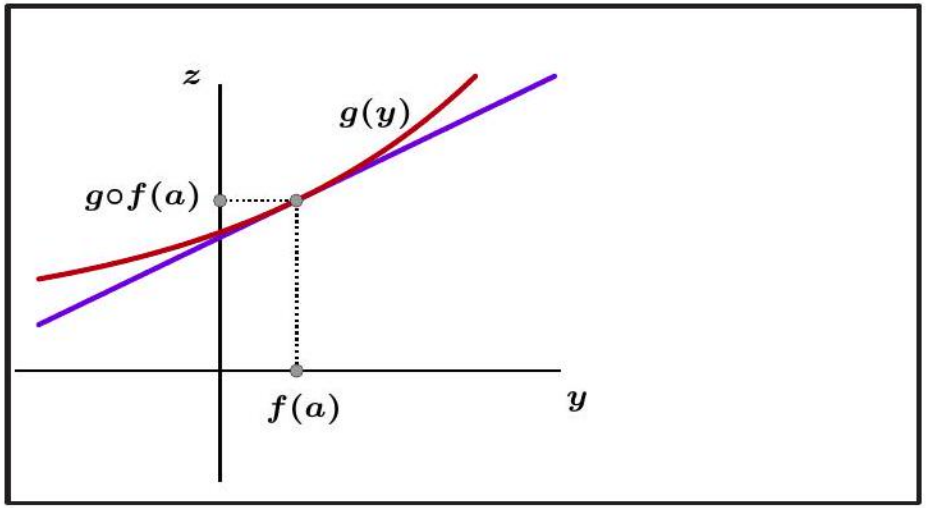


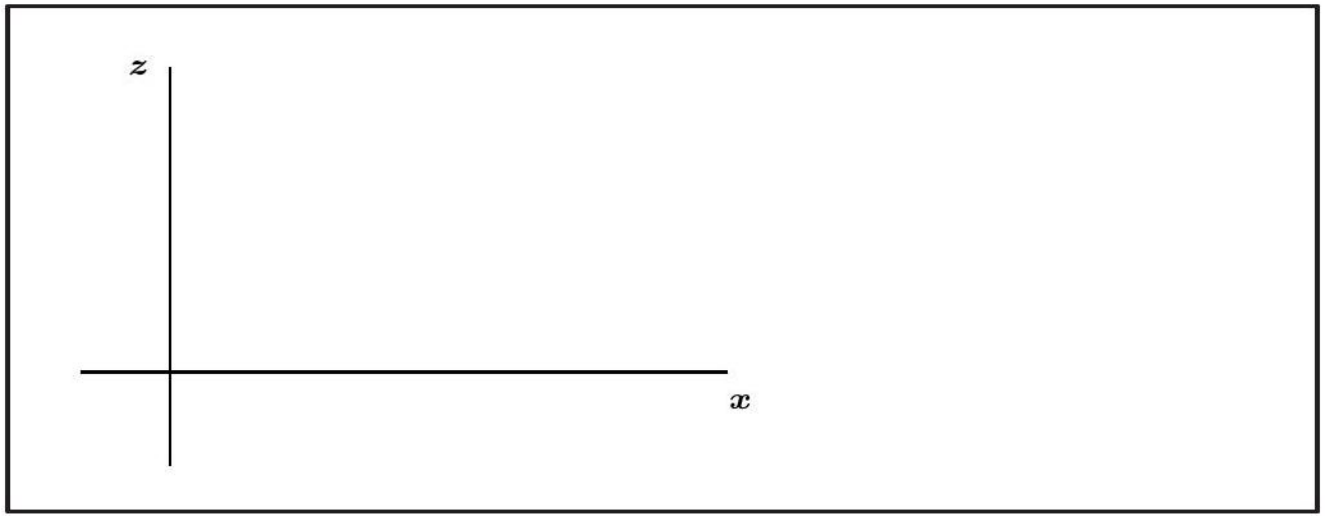




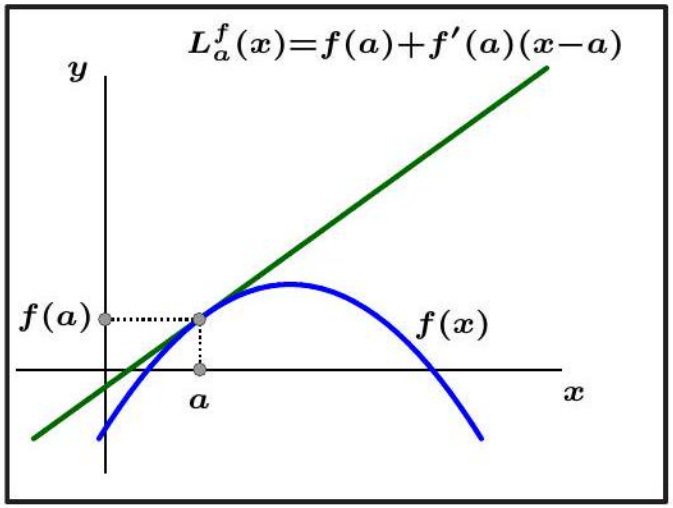
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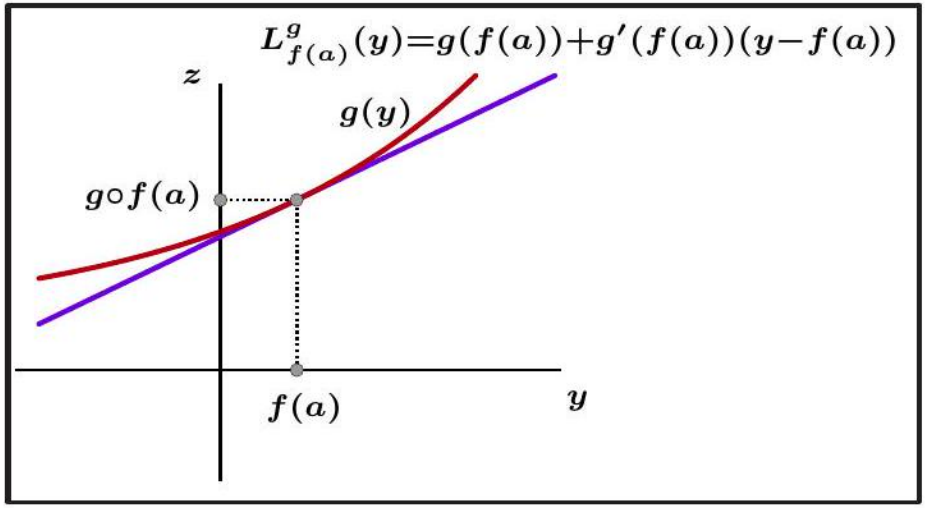


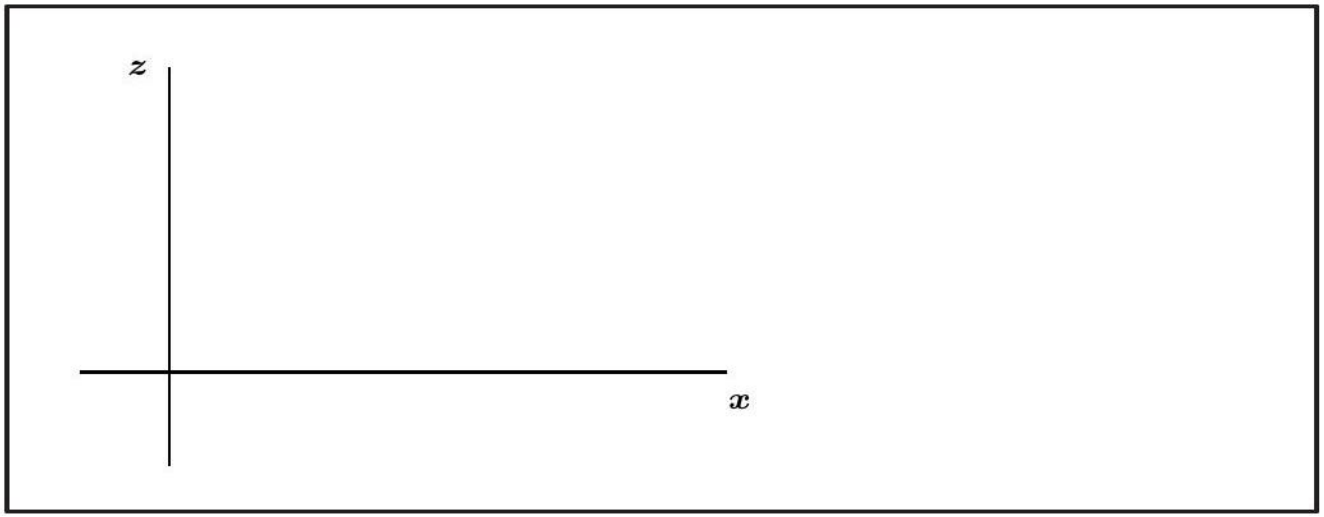




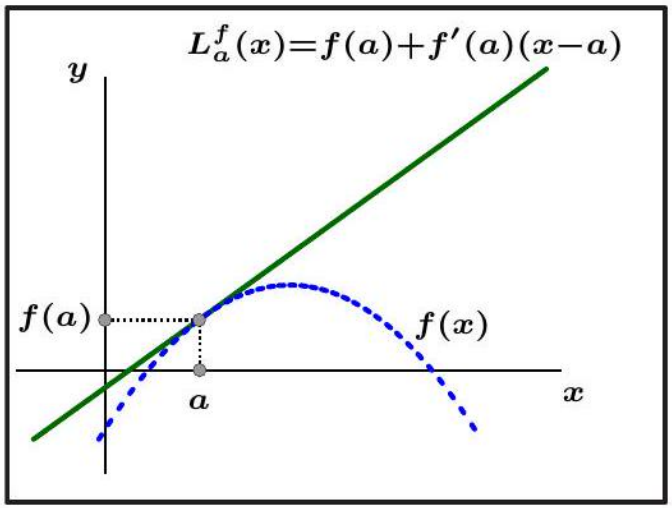
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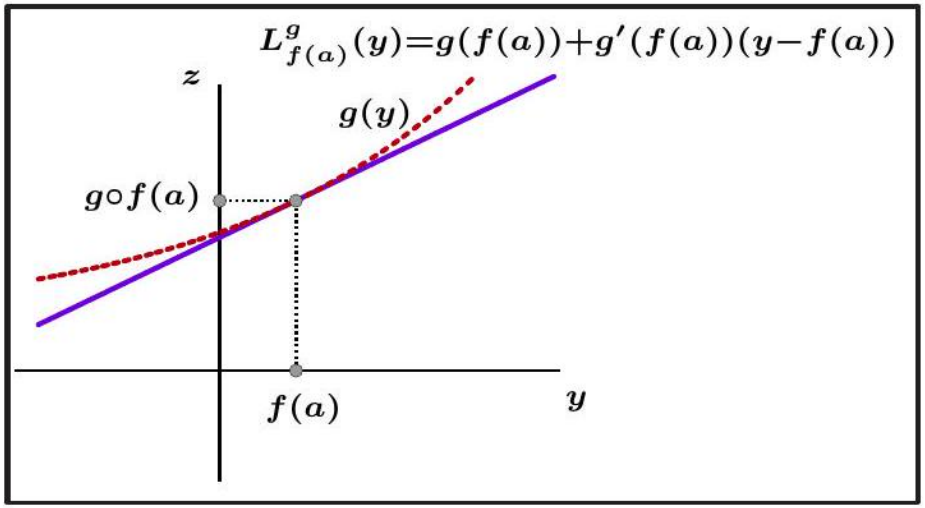


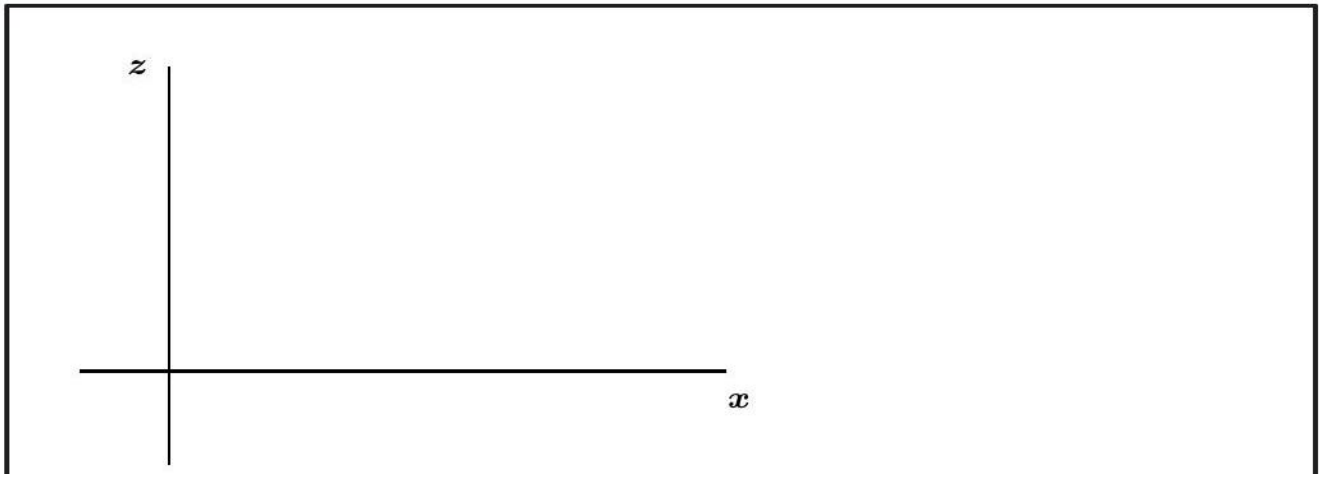




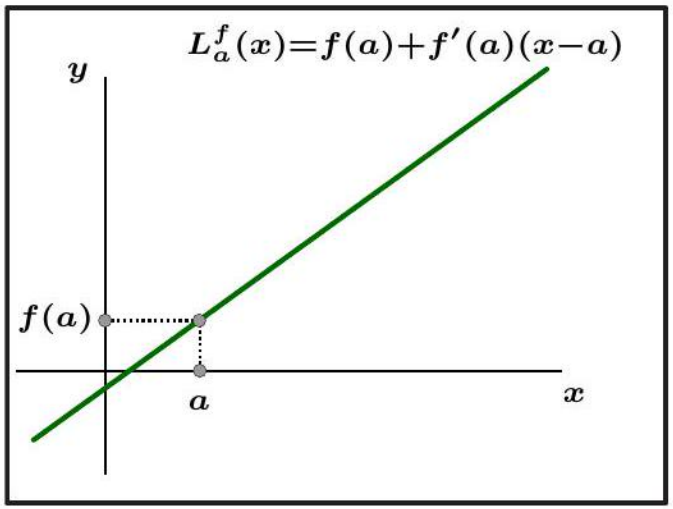
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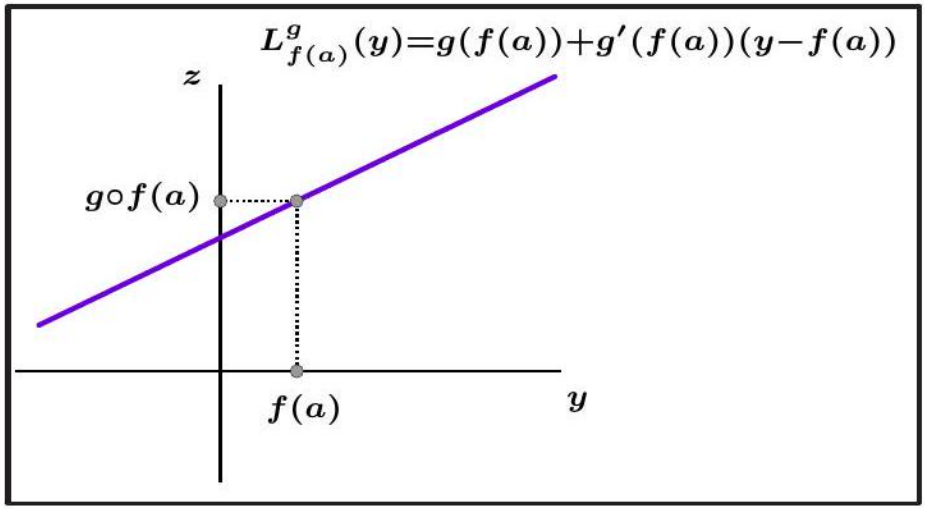


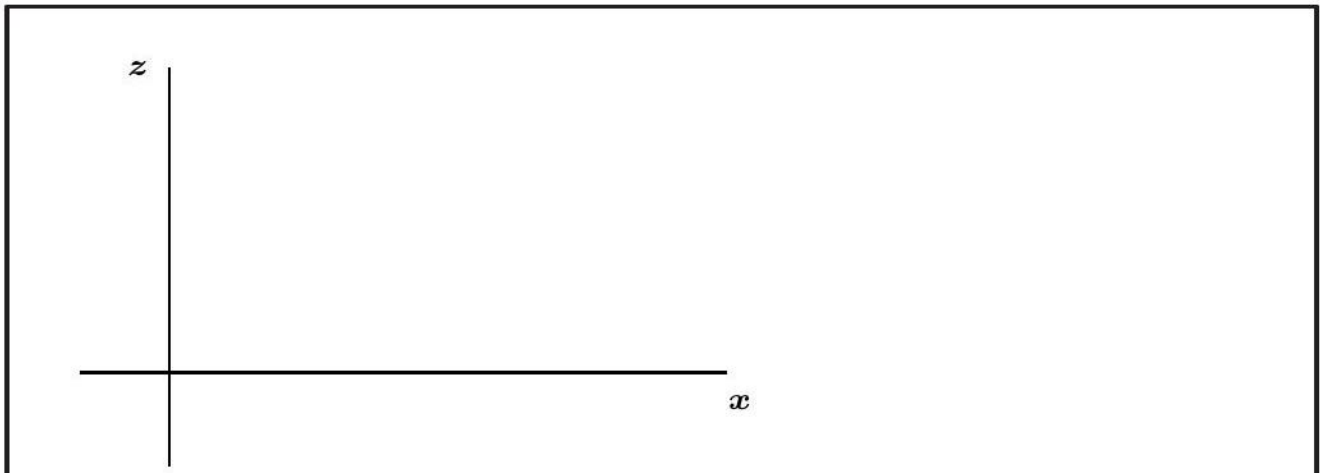




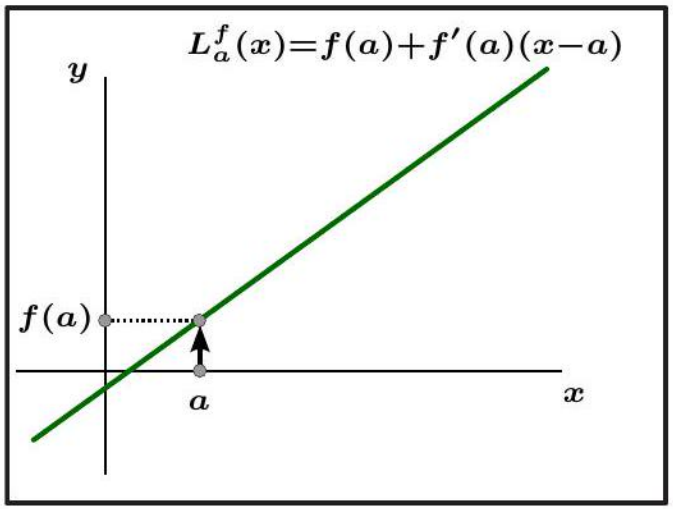
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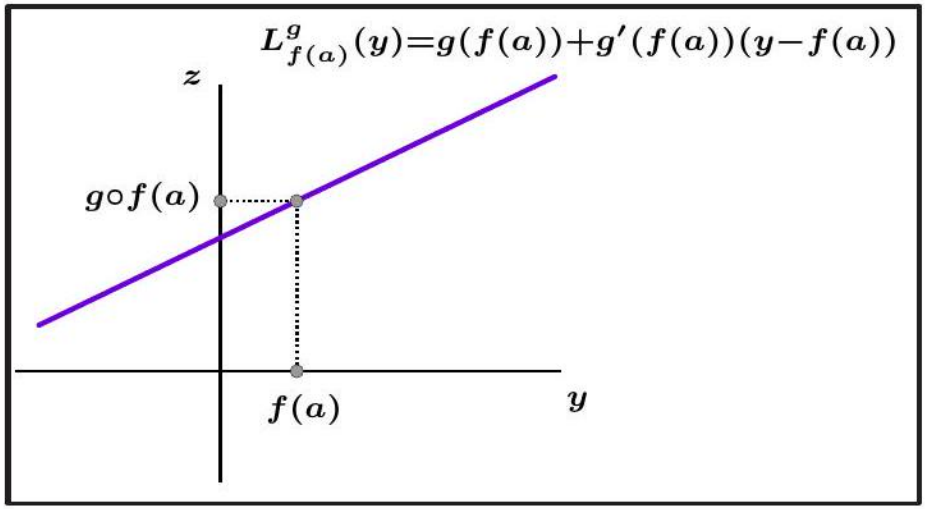


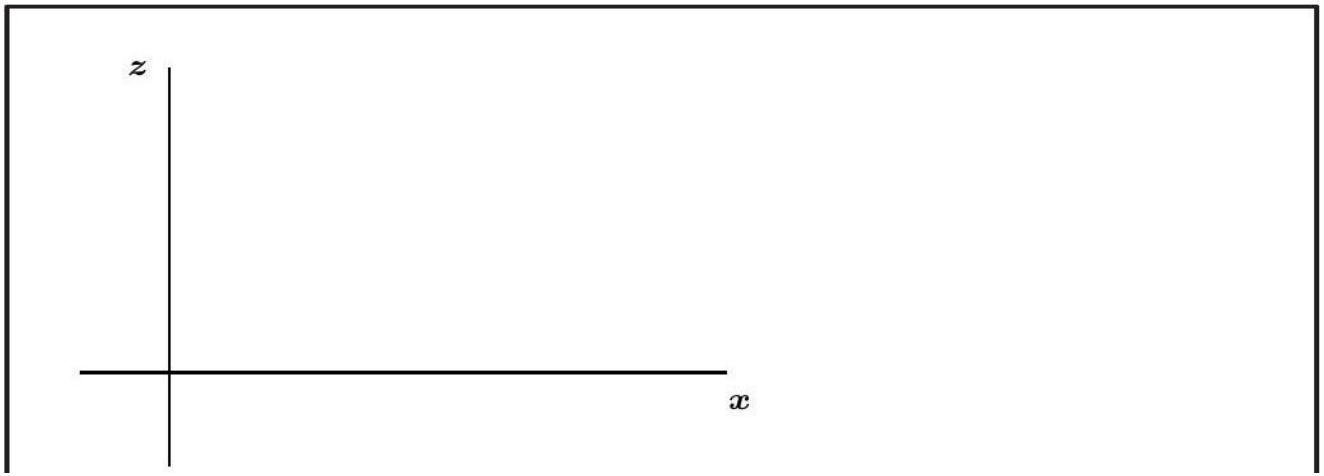




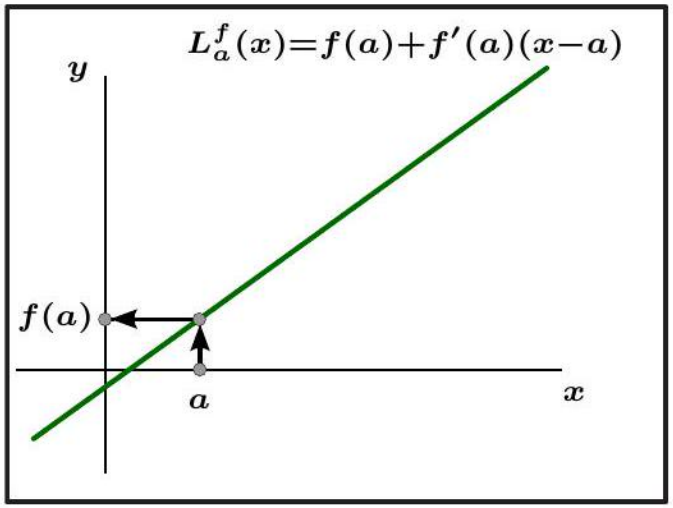
Chain Rule

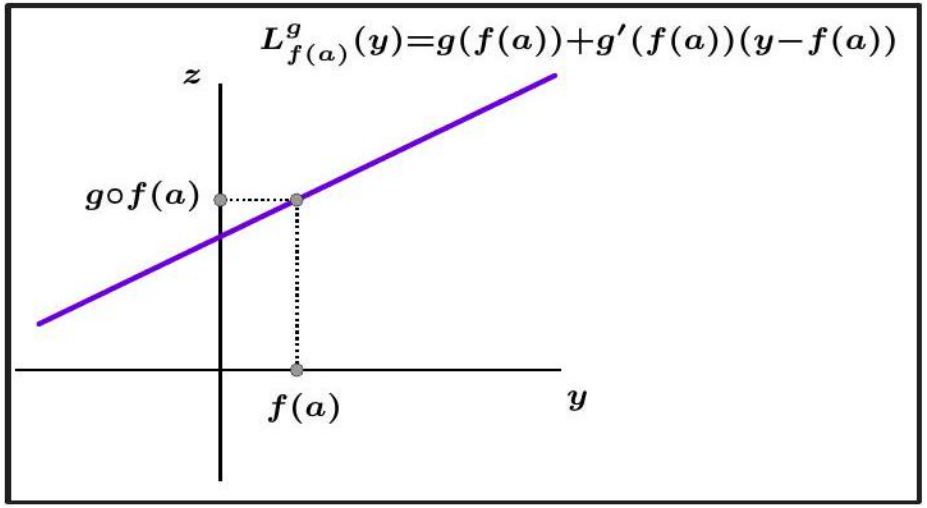


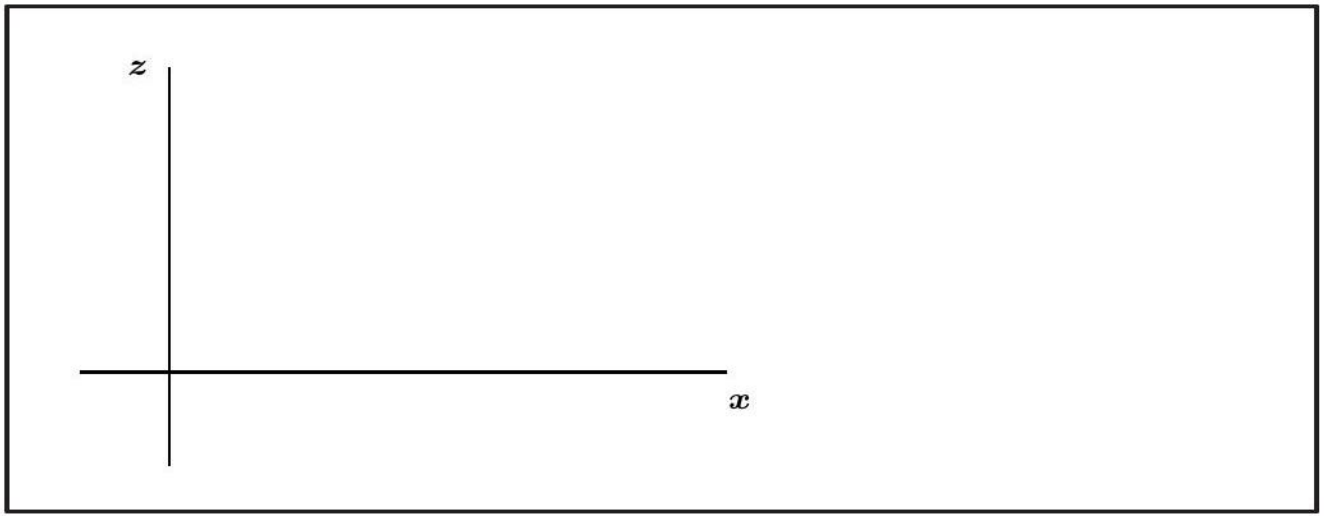




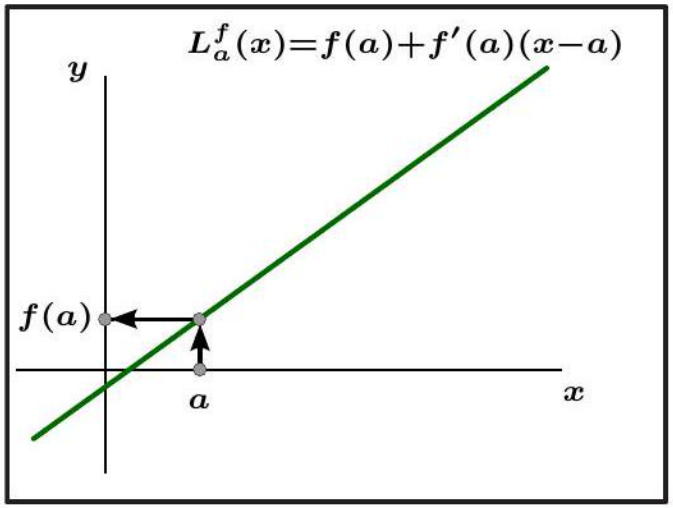
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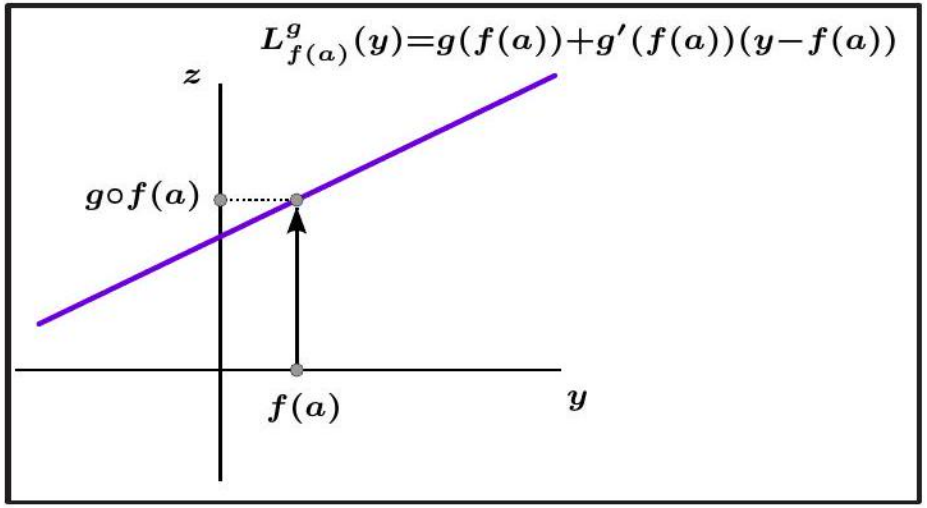


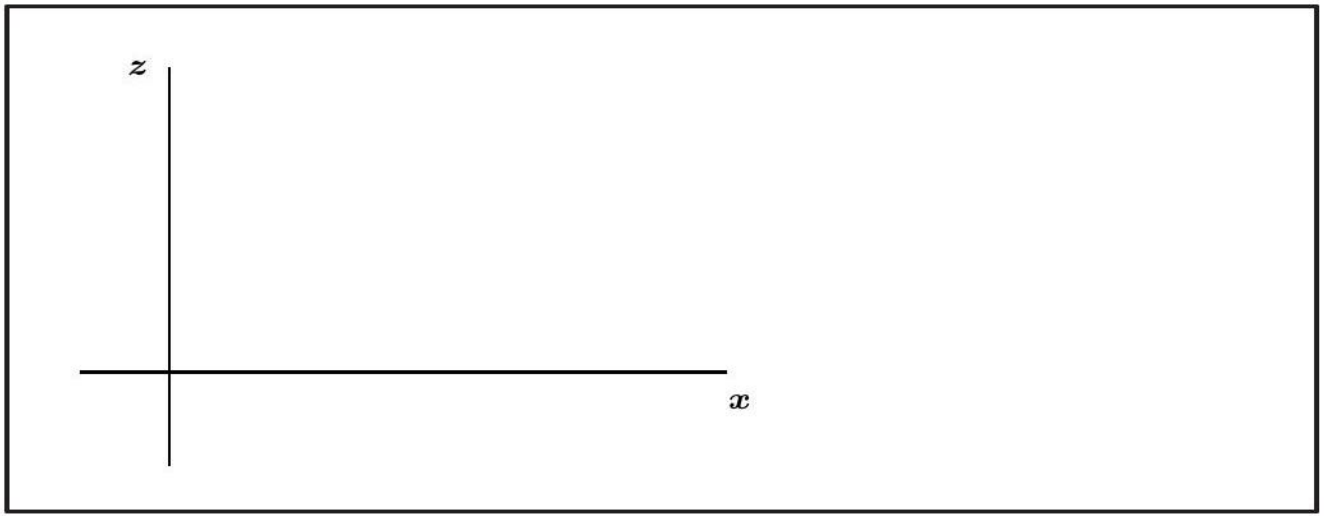




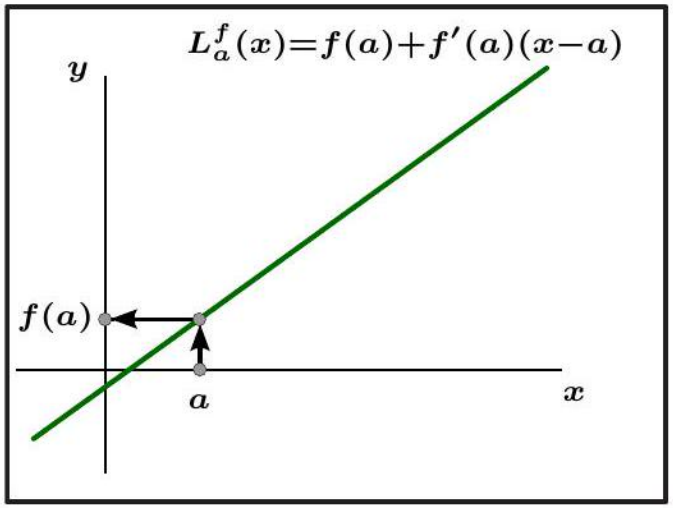
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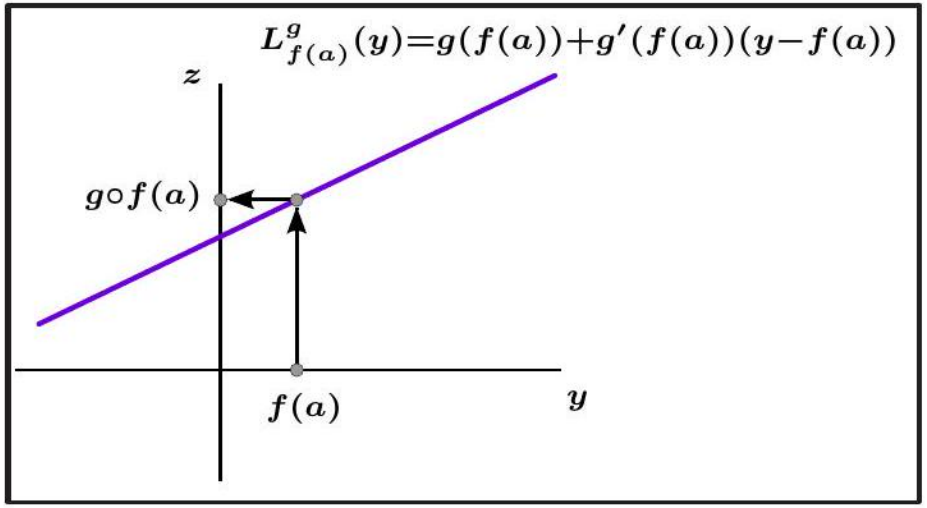


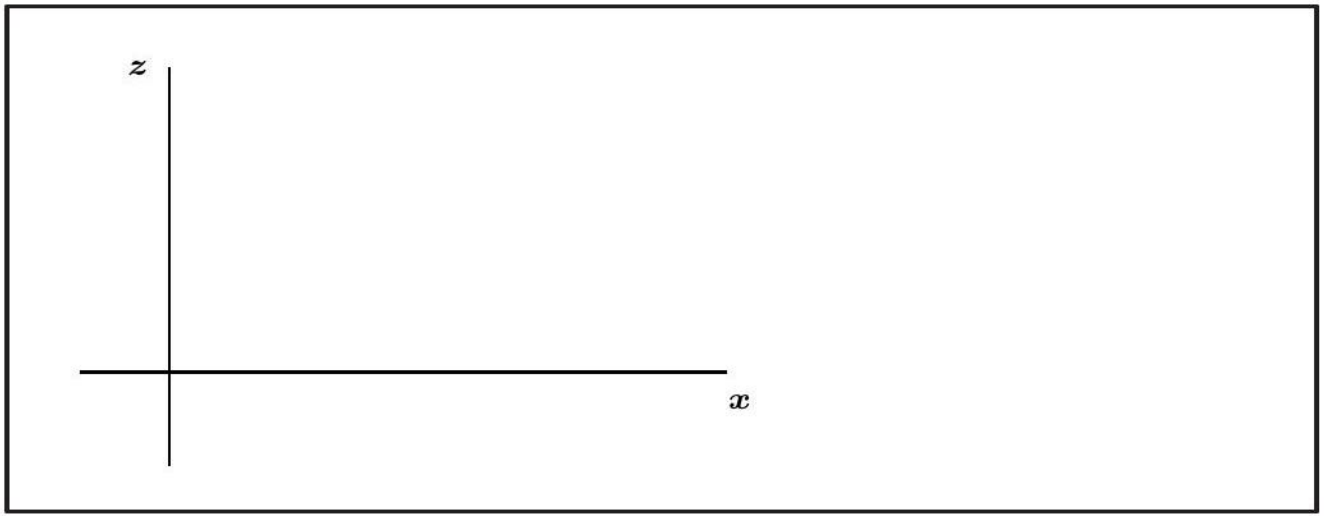




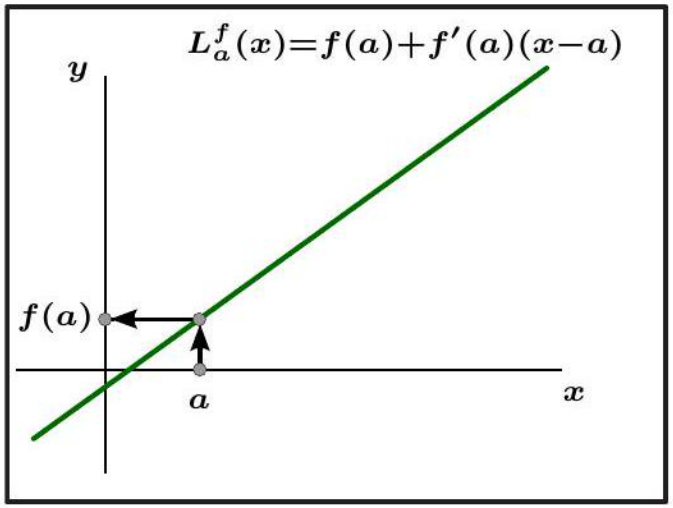
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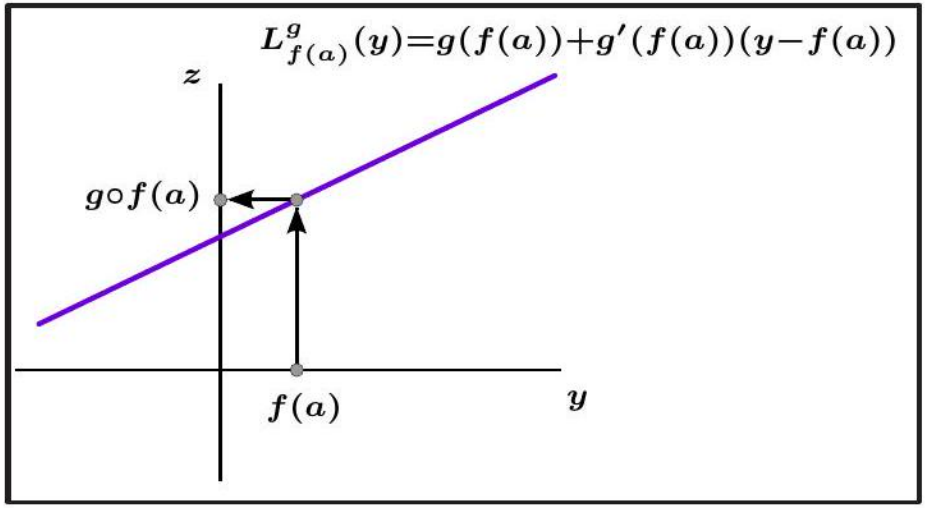


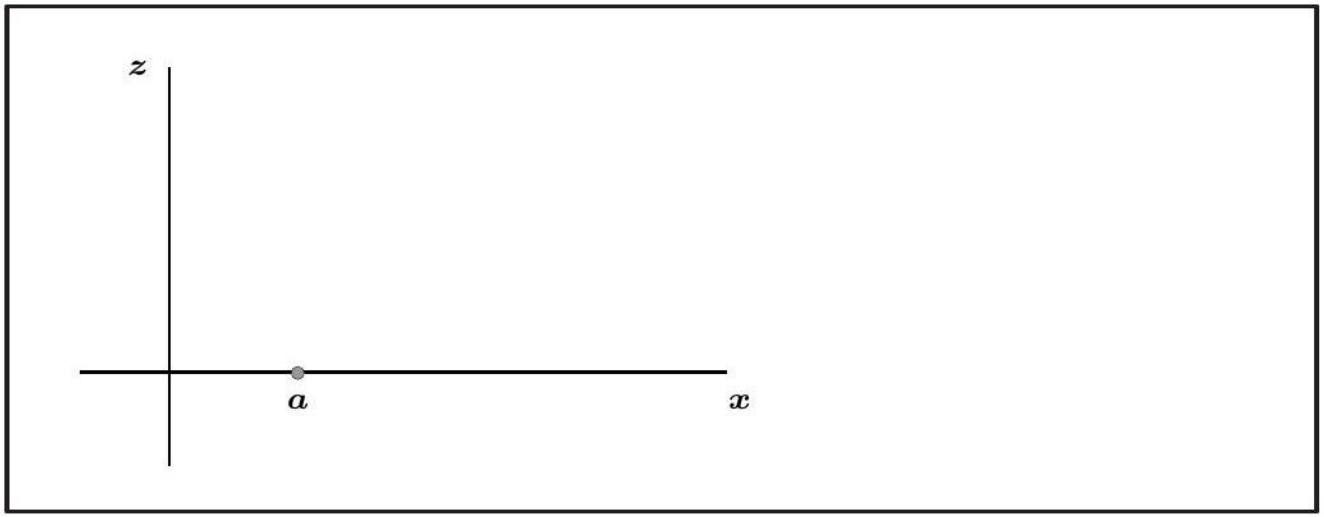




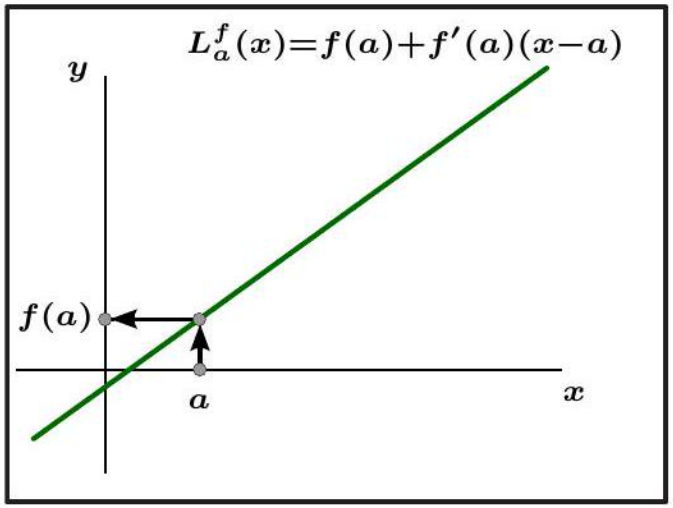
Chain Rule

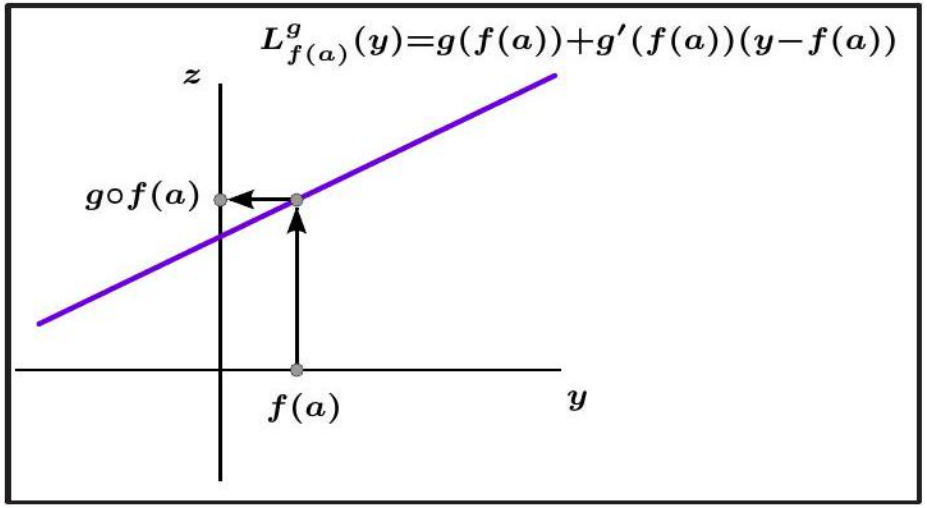


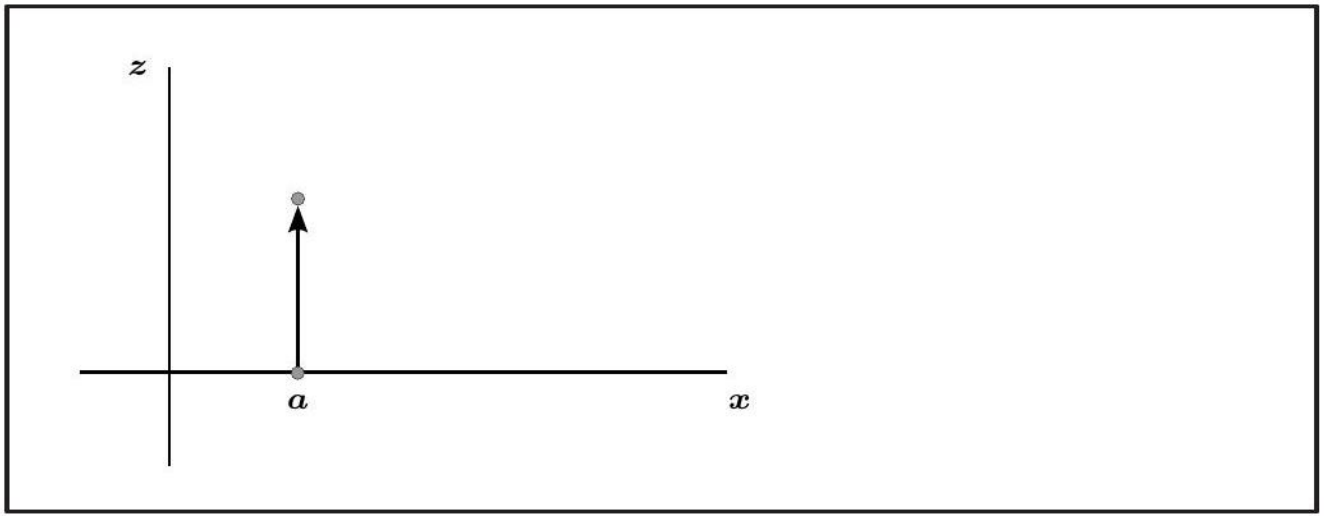




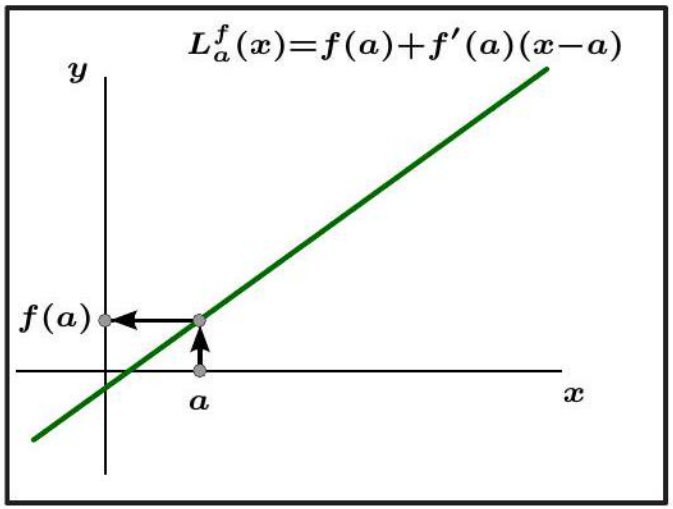
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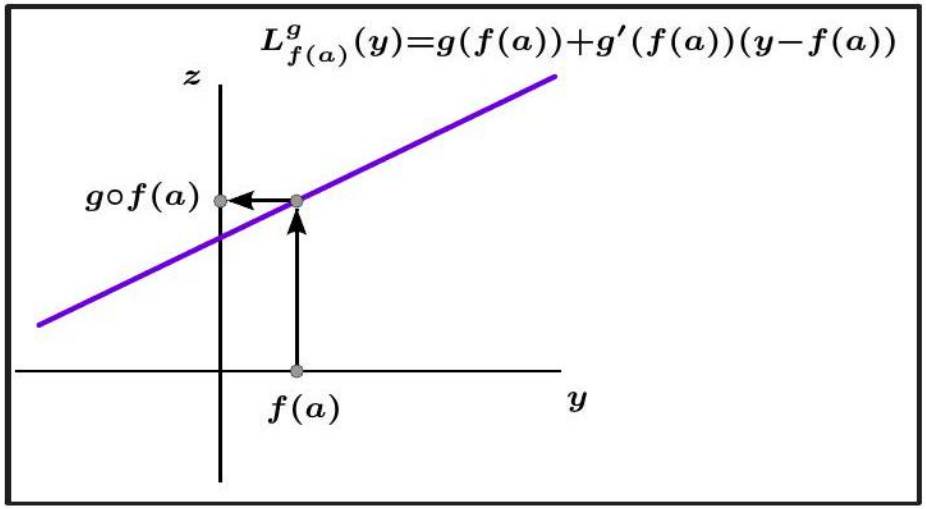


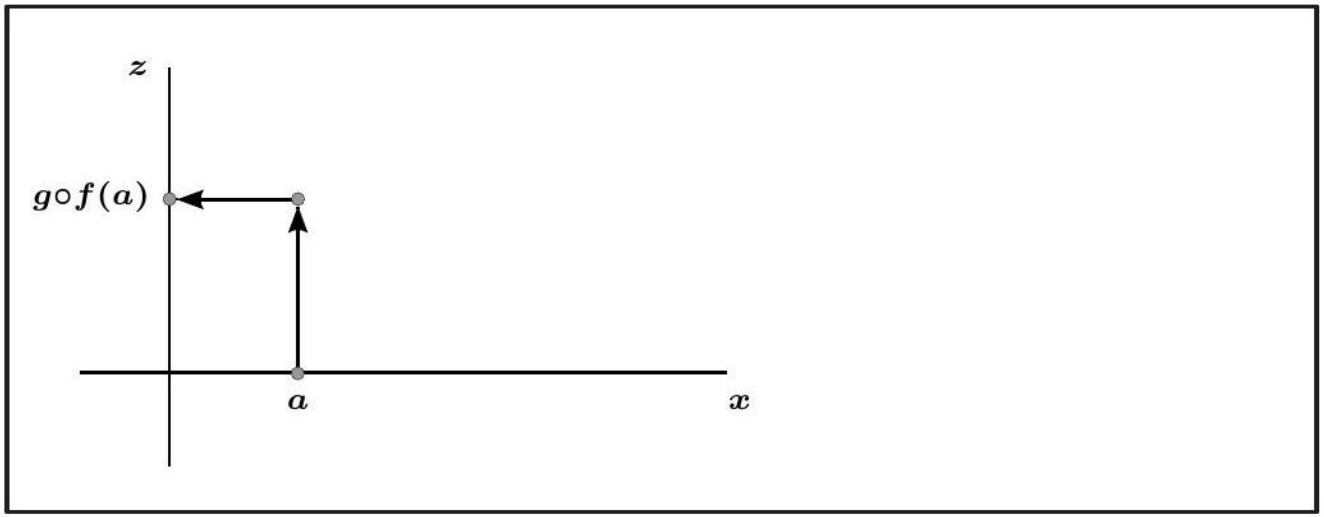




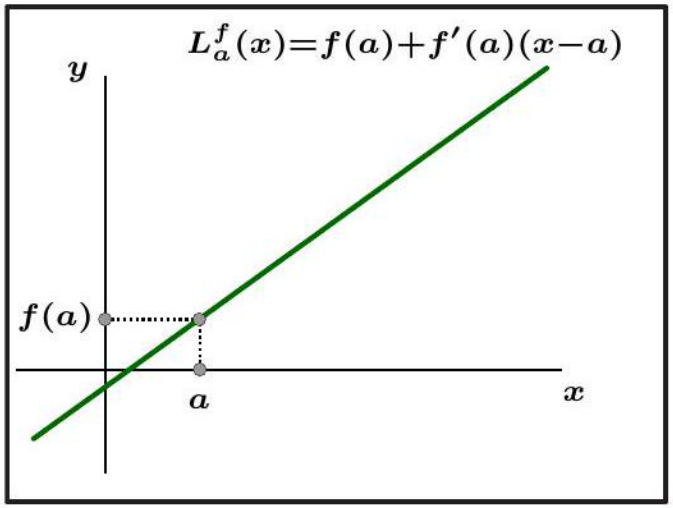
Chain Rule

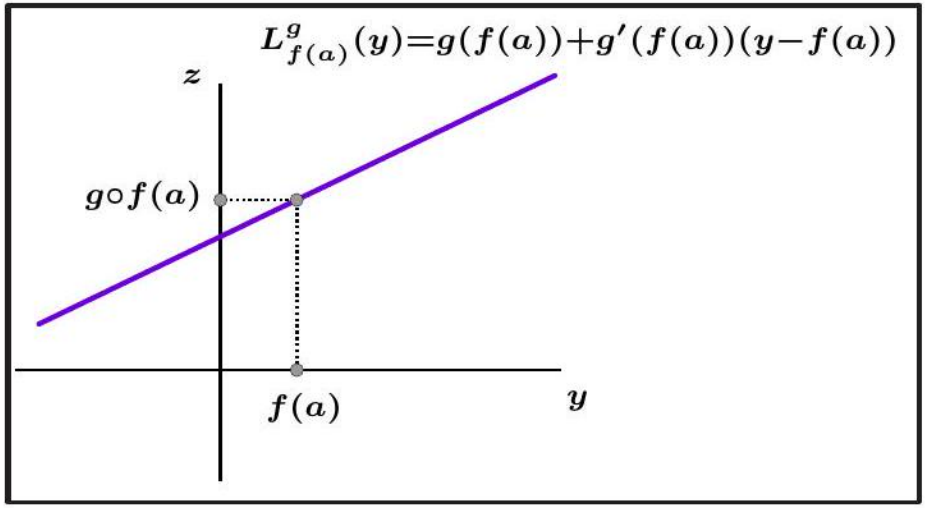

$L_a^f(x)=f(a)+f'(a)(x-a)$
y
$f(a)$
a
x

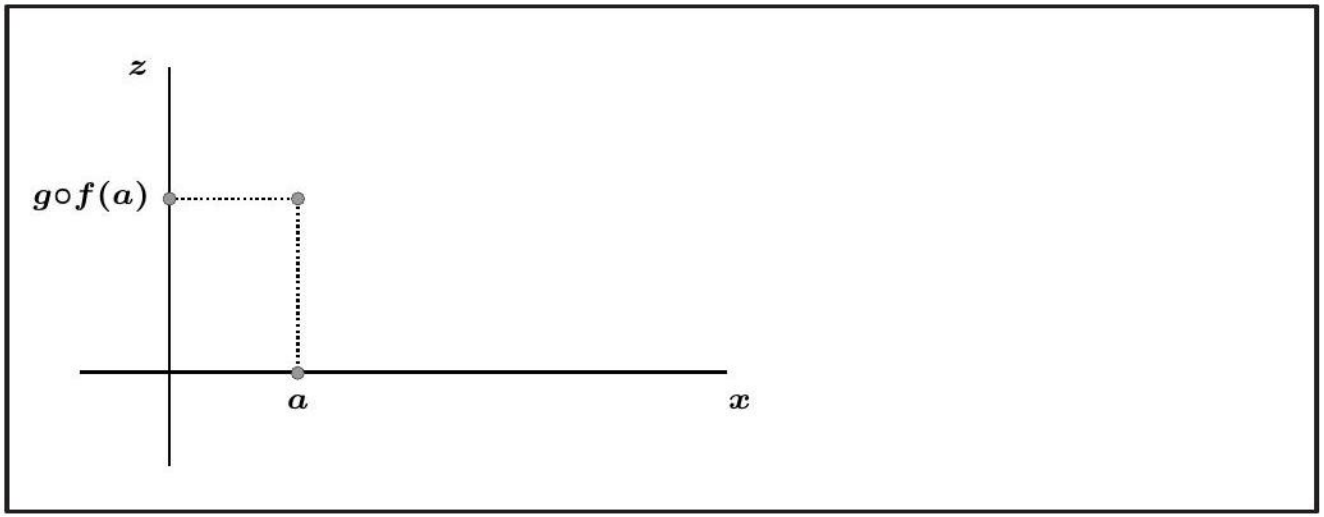

$L_{f(a)}^g(y)=g(f(a))+g'(f(a))(y-f(a))$
z
$g\circ f(a)$
$f(a)$
y


z
$g\circ f(a)$
a
x

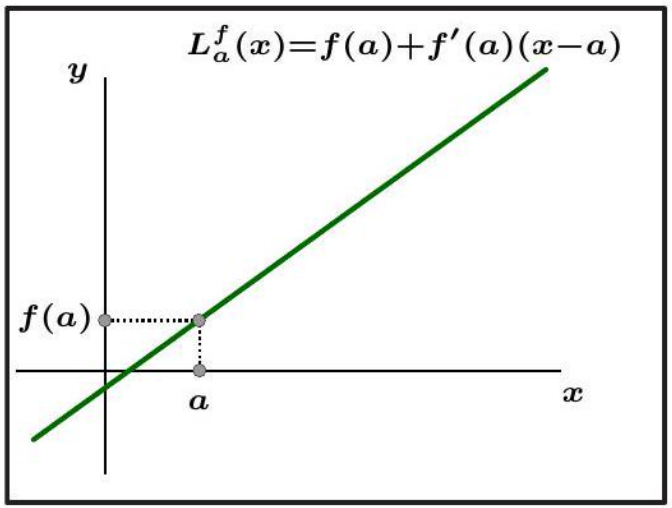
Chain Rule

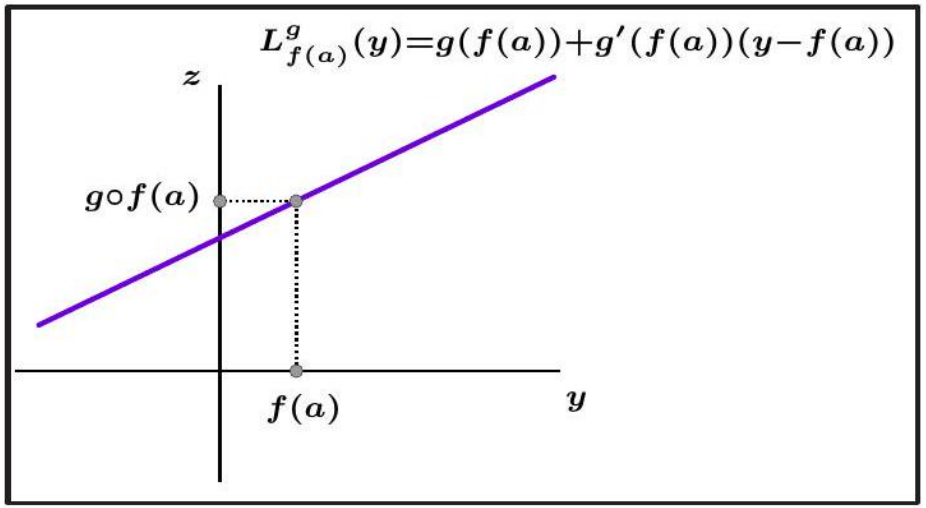

$L_a^f(x)=f(a)+f'(a)(x-a)$
y
$f(a)$
a
x

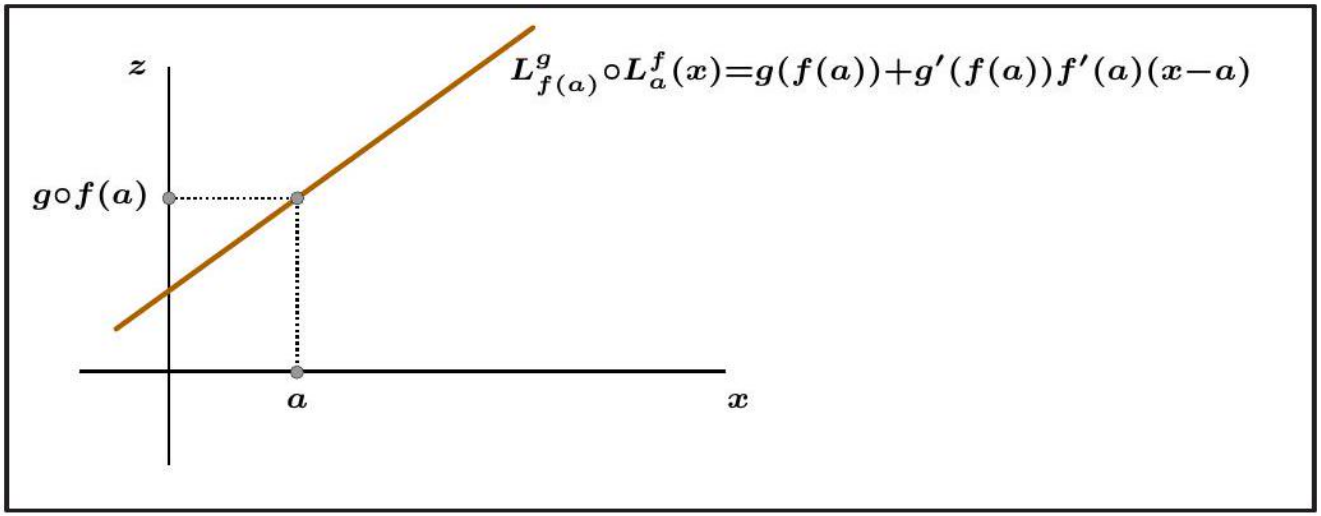

$L_{f(a)}^g(y)=g(f(a))+g'(f(a))(y-f(a))$
z
$g\circ f(a)$
$f(a)$
y


z
$g\circ f(a)$
a
x

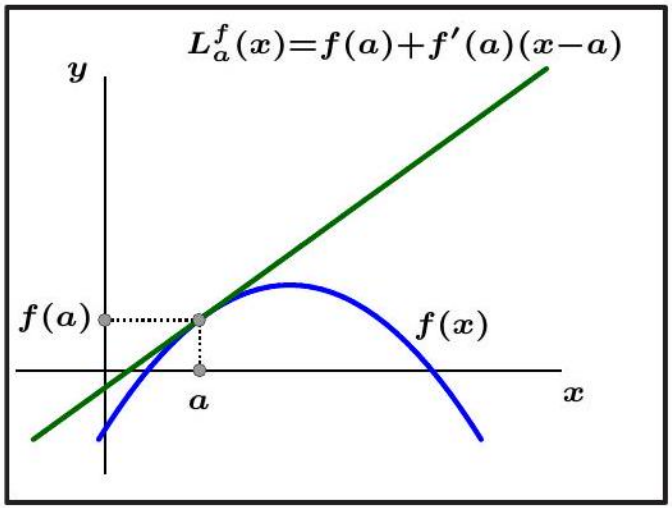
Chain Rule

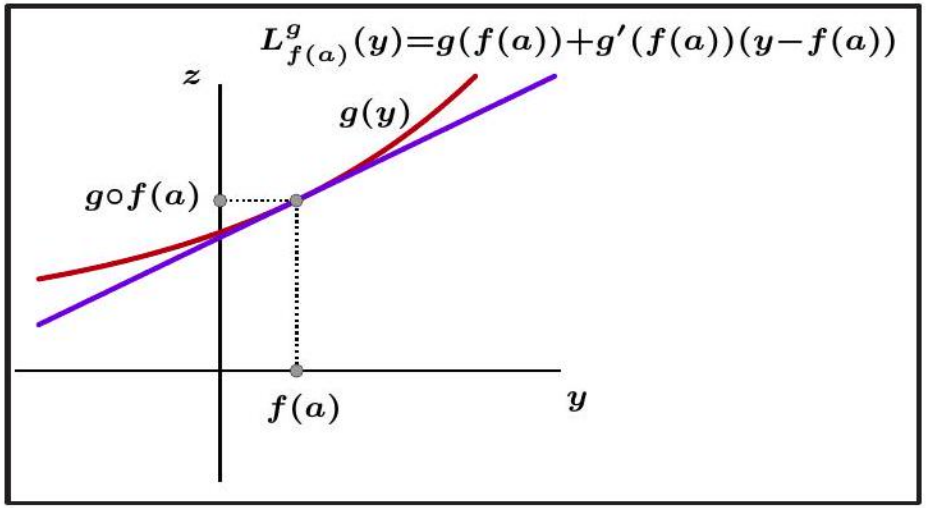

$L_a^f(x)=f(a)+f'(a)(x-a)$
y
$f(a)$
a
x

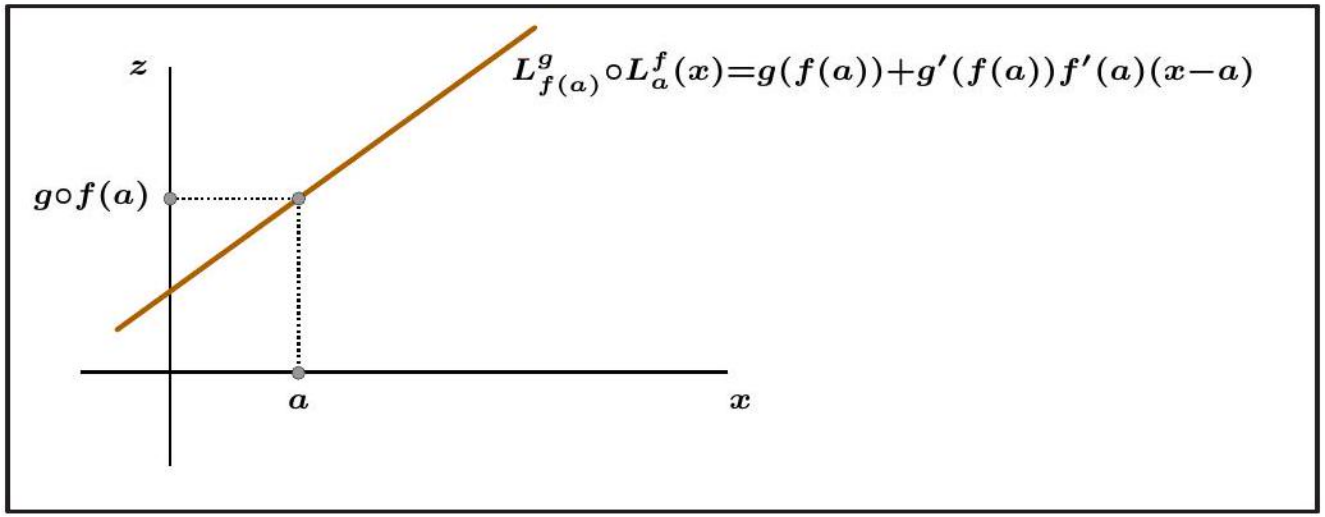

$L_{f(a)}^g(y)=g(f(a))+g'(f(a))(y-f(a))$
z
$g\circ f(a)$
$f(a)$
y


$L_{f(a)}^g \circ L_a^f(x)=g(f(a))+g'(f(a))f'(a)(x-a)$
z
$g\circ f(a)$
a
x

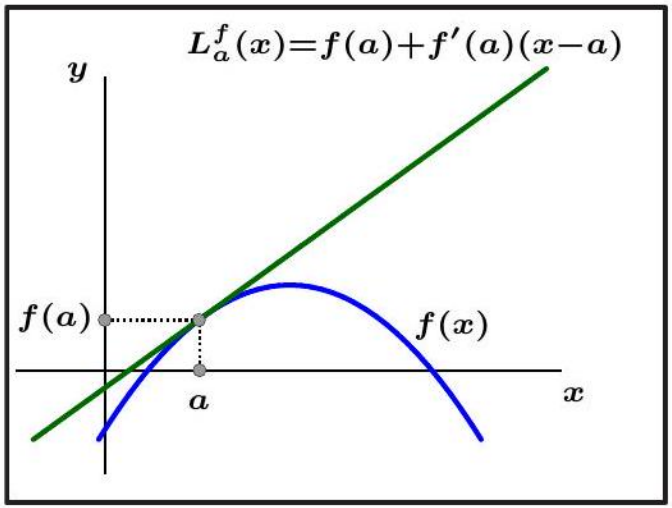
Chain Rule

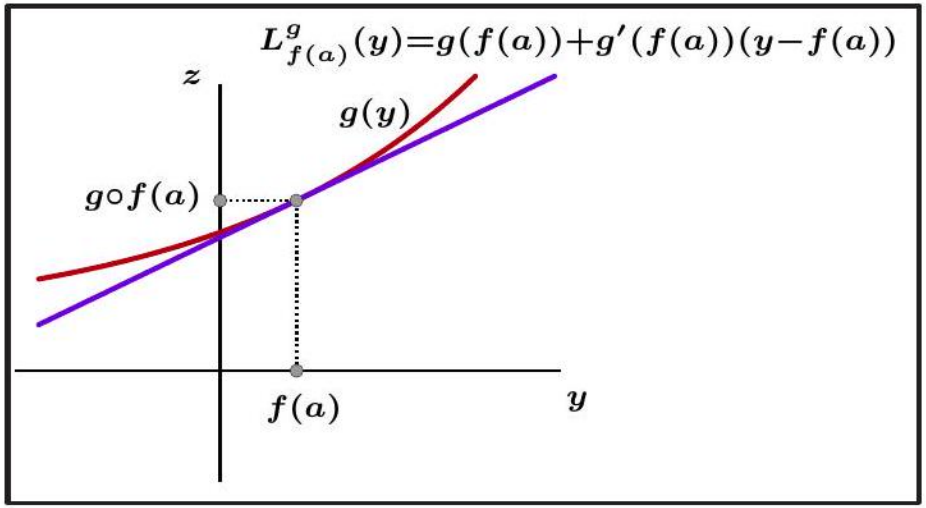

$L_a^f(x)=f(a)+f'(a)(x-a)$
y
$f(a)$
$f(x)$
a
x

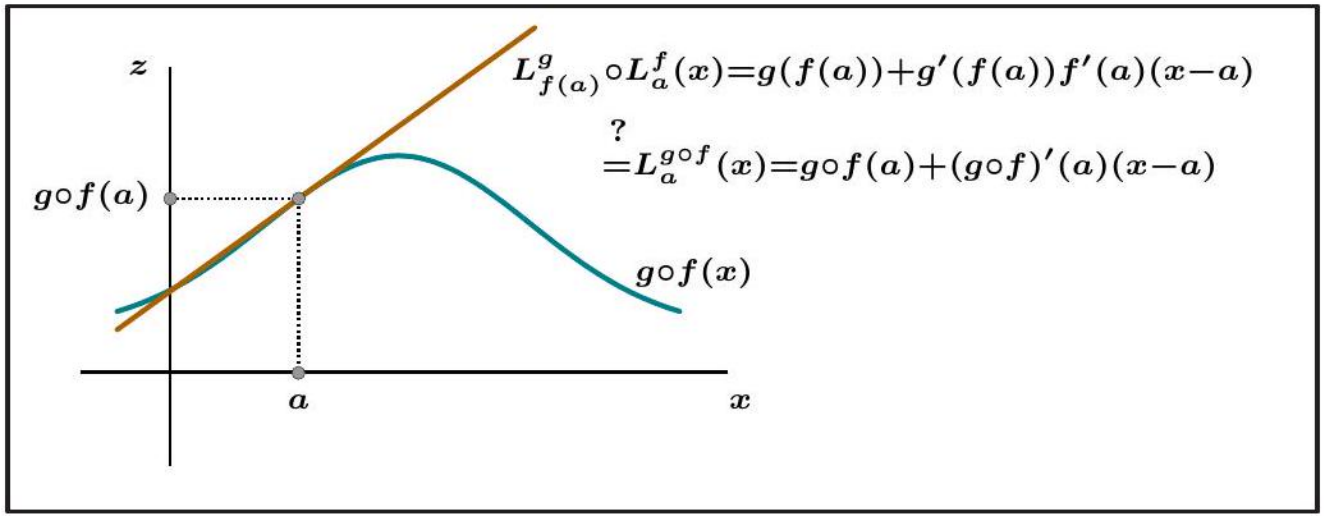

$L_{f(a)}^g(y)=g(f(a))+g'(f(a))(y-f(a))$
z
$g(y)$
$g\circ f(a)$
$f(a)$
y


$L_{f(a)}^g \circ L_a^f(x)=g(f(a))+g'(f(a))f'(a)(x-a)$
z
$g\circ f(a)$
a
x

Chain Rule


$L_a^f(x)=f(a)+f'(a)(x-a)$
y
$f(a)$
$f(x)$
a
x


$L_{f(a)}^g(y)=g(f(a))+g'(f(a))(y-f(a))$
z
$g(y)$
$g\circ f(a)$
$f(a)$
y


$L_{f(a)}^g \circ L_a^f(x)=g(f(a))+g'(f(a))f'(a)(x-a)$
$\stackrel{?}{=} L_a^{g\circ f}(x)=g\circ f(a)+(g\circ f)'(a)(x-a)$
z
$g\circ f(a)$
$g\circ f(x)$
a
x

Chain Rule

Theorem 3.2.5 (Chain Rule)

Assume that $f(x)$ *is differentiable at* $x = a$ *and* $g(y)$ *is differentiable at* $y = f(a)$. *Then* $h(x) = g \circ f(x) = g(f(x))$ *is differentiable at* $x = a$ *and*

$$h'(a) = g'(f(a))f'(a)$$

In particular,

$$L_a^h(x) = L_{f(a)}^g \circ L_a^f(x)$$

Chain Rule

Problem: Assume that $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are both differentiable. Let $h(x) = g \circ f(x)$.

The equation of the tangent line to $f(x)$ through the point $(1, 3)$ is

$$y = 3 + 5(x - 1)$$

and the equation of the tangent line to the graph of $h(x)$ through $(1, 4)$ is

$$z = 4 - 2(x - 1)$$

Find the equation of the tangent line to the graph of $z = g(y)$ through the point $(3, 4)$.

Chain Rule

Solution: The fact that the tangent line to $f(x)$ through the point $(1, 3)$ is

$$y = 3 + 5(x - 1) = L_1^f(x) = f(1) + f'(1)(x - 1)$$

tells us that $f(1) = 3$ and that $f'(1) = 5$.

Similarly, the fact that the tangent line to the graph of $h(x)$ through $(1, 4)$ is given by

$$z = 4 - 2(x - 1) = L_1^h(x) = h(1) + h'(1)(x - 1)$$

tells us that $h(1) = 4$ and $h'(1) = -2$.

Now $4 = h(1) = g(f(1)) = g(3)$ and from the Chain Rule, we get that

$$-2 = h'(1) = g'(f(1))f'(1) = g'(3)(5) \Rightarrow g'(3) = -\frac{2}{5}$$

It follows that the equation of the tangent line to the graph of $z = g(y)$ through the point $(3, 4)$ is given by

$$z = 4 - \frac{2}{5}(y - 3)$$

The Derivative of a^x

Example 3.2.6

If $h(x) = a^x$, find $h'(x)$.

Solution: Note that

$$a^x = e^{\ln(a)\cdot x}$$

If $g(u) = e^u$ and $f(x) = \ln(a) \cdot x$, then

$$a^x = h(x) = g \circ f(x)$$

By the Chain Rule

$$\begin{aligned} h'(x) &= g'(f(x)) \cdot f'(x) \\ &= e^{\ln(a)\cdot x} \cdot \ln(a) \\ &= \ln(a) \cdot a^x \end{aligned}$$

Derivative of $\sin(x)$

Theorem 3.2.7 (The Derivative of $\sin(x)$)

Assume that $f(x) = \sin(x)$. *Then*

$$f'(x) = \cos(x)$$

for all $x \in \mathbb{R}$.

Question: Can we use the properties of derivatives to find the derivative of other trigonometric functions?

Derivative of $\cos(x)$

Example: Find $\frac{d}{dx}(\cos(x))$. Solution: We know that

$$\cos(x) = \sin\left(x + \frac{\pi}{2}\right)$$

Let $y = \sin(u)$ and $u = x + \frac{\pi}{2}$. Substituting for u gives us that

$$y = y(x) = \sin\left(x + \frac{\pi}{2}\right) = \cos(x)$$

Therefore,

$$\begin{aligned}
\frac{d}{dx}(\cos(x)) &= \frac{dy}{dx} \\
&= \frac{dy}{du}\frac{du}{dx} \\
&= \cos(u) \cdot (1) \\
&= \cos\left(x + \frac{\pi}{2}\right) \\
&= \cos(x)\cos\left(\frac{\pi}{2}\right) - \sin(x)\sin\left(\frac{\pi}{2}\right) \\
&= -\sin(x)
\end{aligned}$$

Derivative of $\tan(x)$

Example: Find $\frac{d}{dx}(\tan(x))$.

Solution: Since

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

we get

$$\begin{aligned}
\frac{d}{dx}(\tan(x)) &= \frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right) \\
&= \frac{\left(\frac{d}{dx}\sin(x)\right)\cos(x) - (\sin(x))\left(\frac{d}{dx}\cos(x)\right)}{\cos^2(x)} \\
&= \frac{\cos(x)\cos(x) - (\sin(x))(-\sin(x))}{\cos^2(x)} \\
&= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \\
&= \frac{1}{\cos^2(x)} \\
&= \sec^2(x)
\end{aligned}$$

Derivative of $\cot(x)$

Note: We have just seen that

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

whenever $\cos(x) \neq 0$. A similar calculation shows that

$$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

whenever $\sin(x) \neq 0$.

Derivative of $\sec(x)$ and $\csc(x)$

Example: Find $\frac{d}{dx}(\sec(x))$.

Solution: Since $\sec(x) = \frac{1}{\cos(x)}$, we get

$$\begin{aligned}
\frac{d}{dx}\left(\frac{1}{\cos(x)}\right) &= \frac{\frac{d}{dx}(\cos(x))}{\cos^2(x)} \\
&= \frac{-(-\sin(x))}{\cos^2(x)} \\
&= \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{\cos(x)} \\
&= \tan(x)\sec(x)
\end{aligned}$$

A similar calculation shows that

$$\frac{d}{dx}(\csc(x)) = -\cot(x)\csc(x)$$

Section 3

Derivatives of Inverse Functions

Invertibility and Differentiability

Problem: If $f(x)$ is invertible with inverse $g(y)$ and if $f(x)$ is differentiable at $\boldsymbol{x = a}$, what can we say about the differentiablity of $\boldsymbol{g(y)}$ at $b = f(a)$?

Answer: We will see using the idea of linear approximations that

$$g'(b) = \frac{1}{f'(a)}$$

provided that $\boldsymbol{f'(a) \neq 0}$.

Invertibility and Differentiability

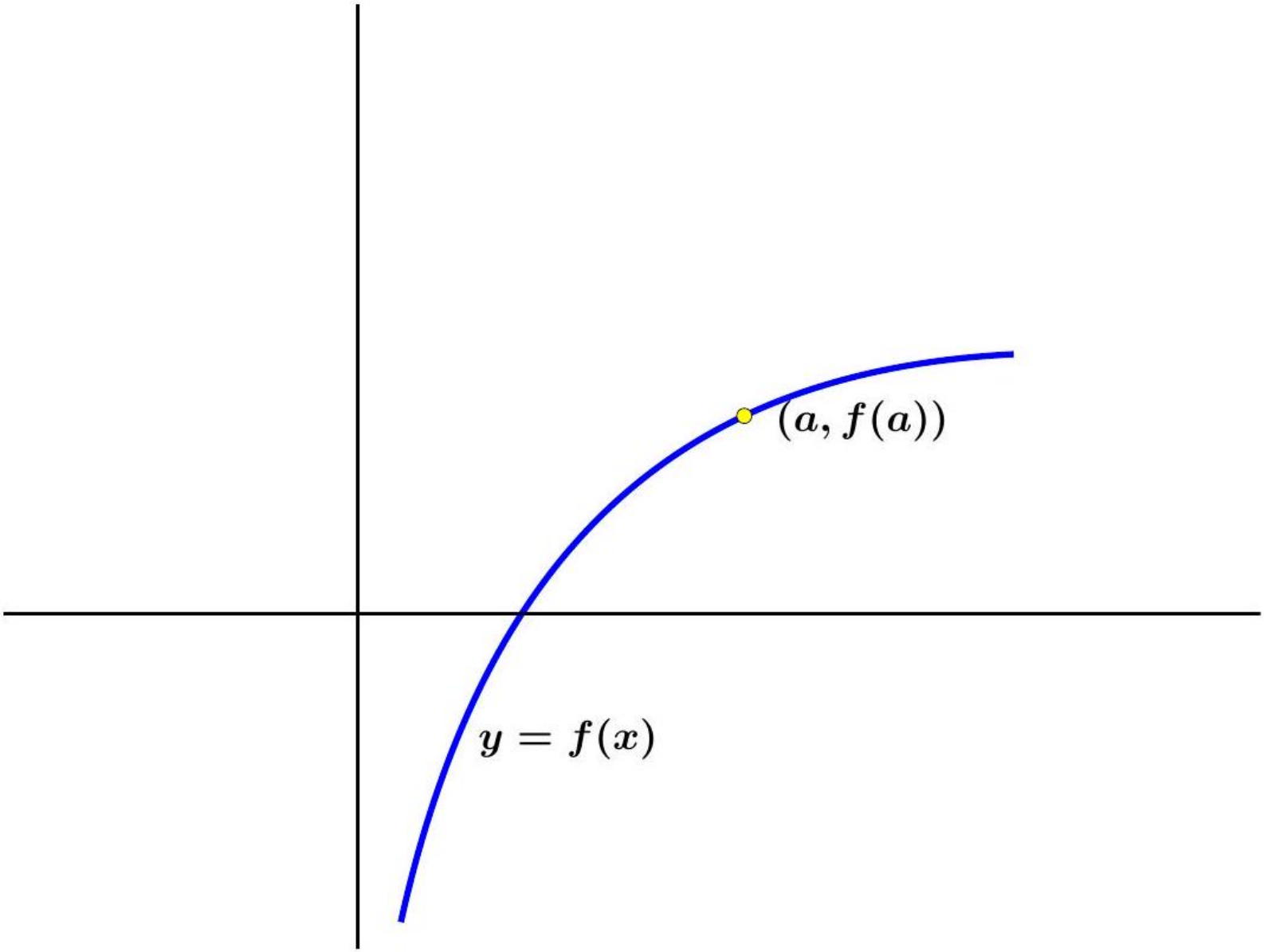
Observe: Given that $f(x)$ is differentiable at $x = a$ we have

$$y = L_a^f(x) = f(a) + f'(a)(x - a)$$

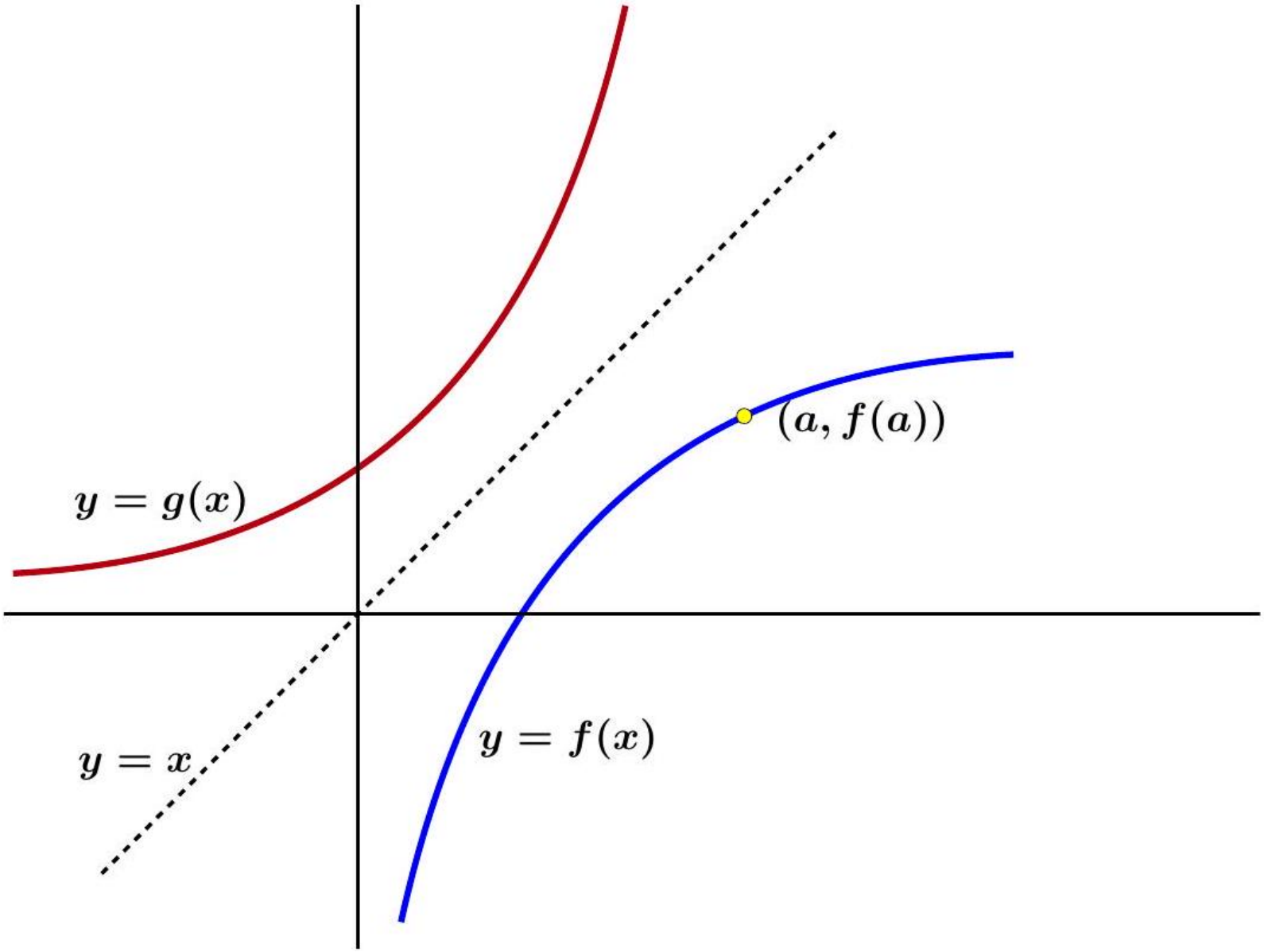
If $f'(a) \neq 0$, then $L_a^f(x)$ is invertible with

$$\begin{aligned}\left(L_a^f\right)^{-1}(x) &= a + \frac{1}{f'(a)}(x - f(a)) \\ &= g(f(a)) + \frac{1}{f'(a)}(x - f(a))\end{aligned}$$

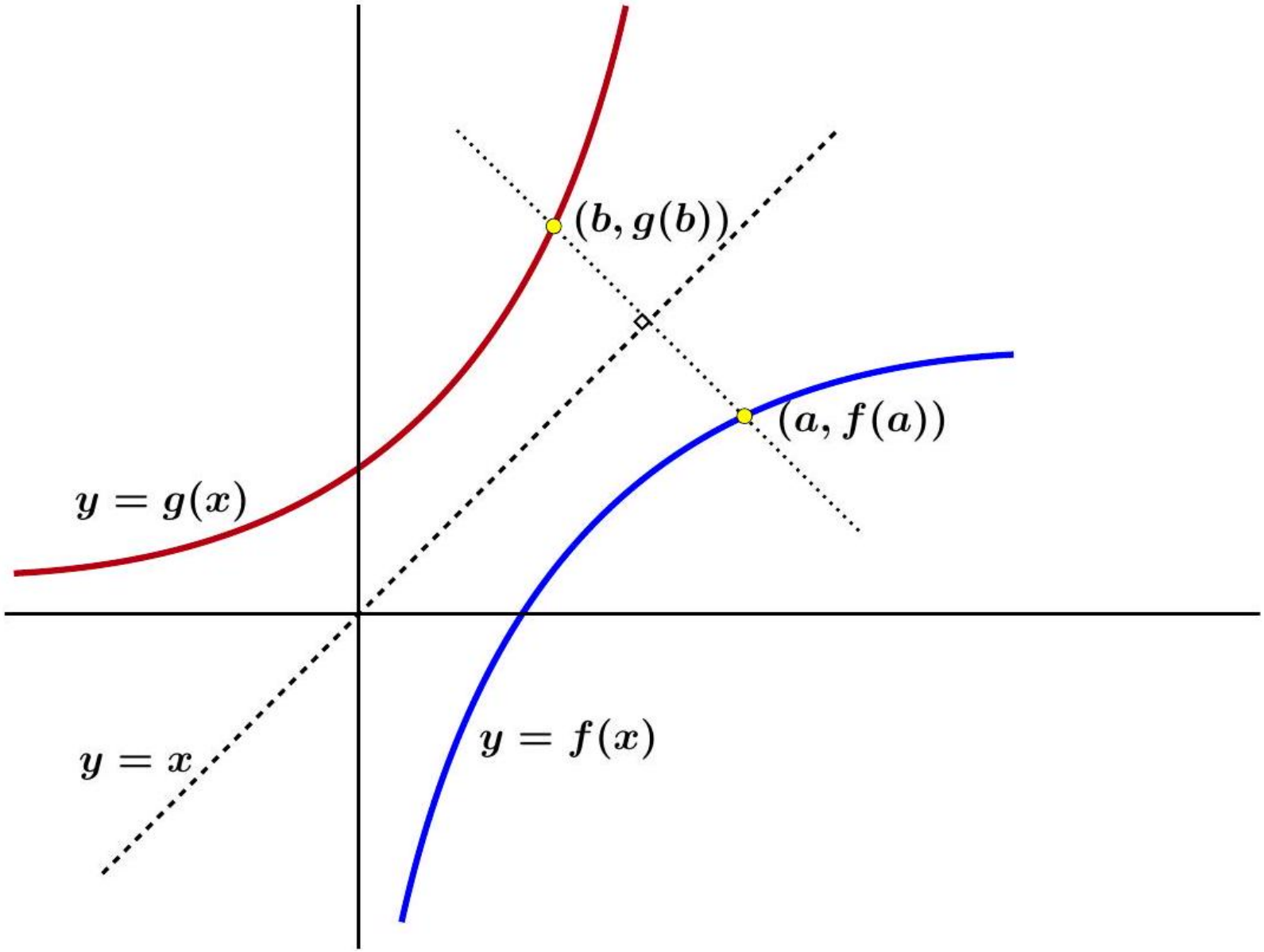
Inverse Function Theorem



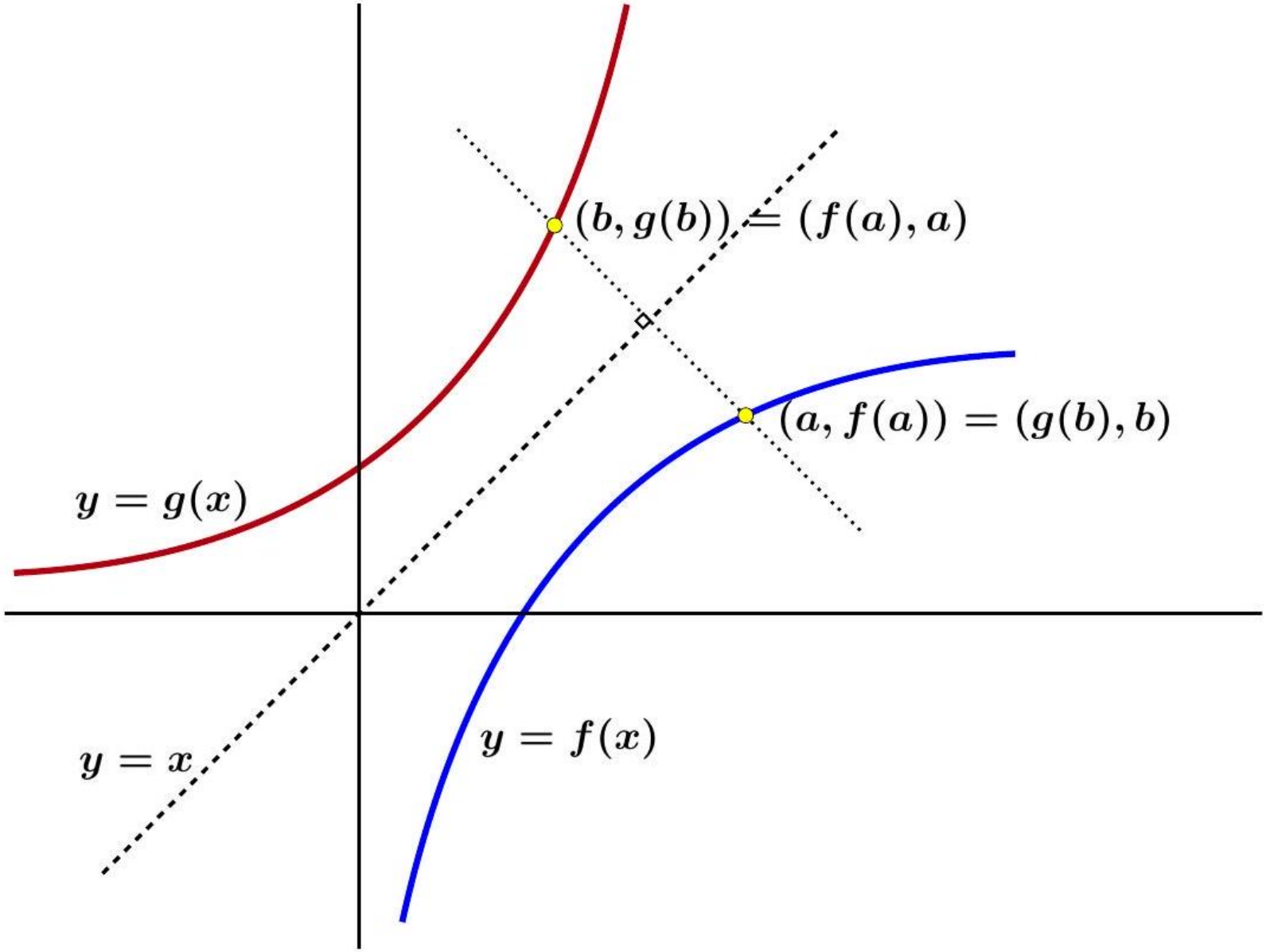
Inverse Function Theorem



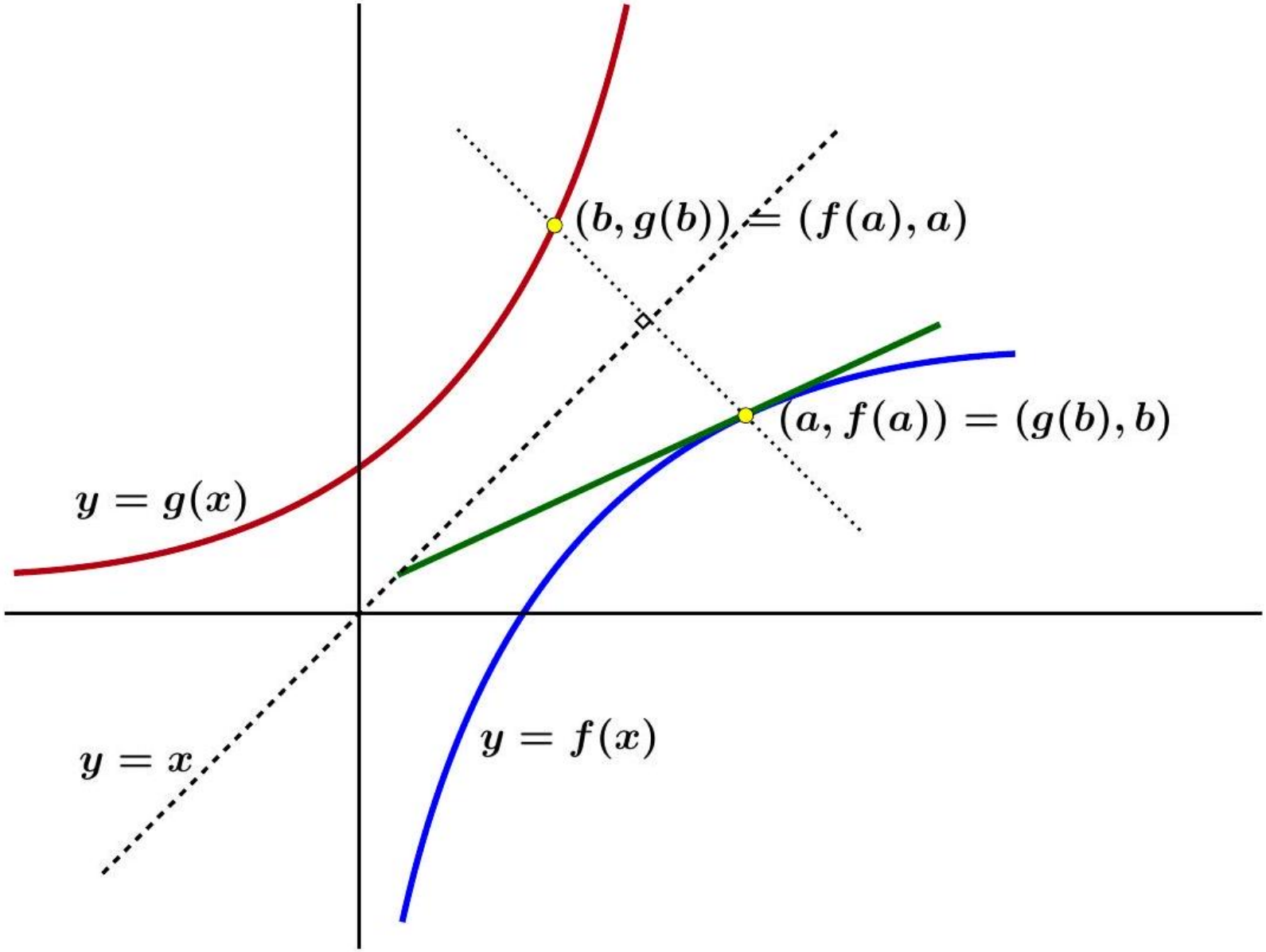
Inverse Function Theorem



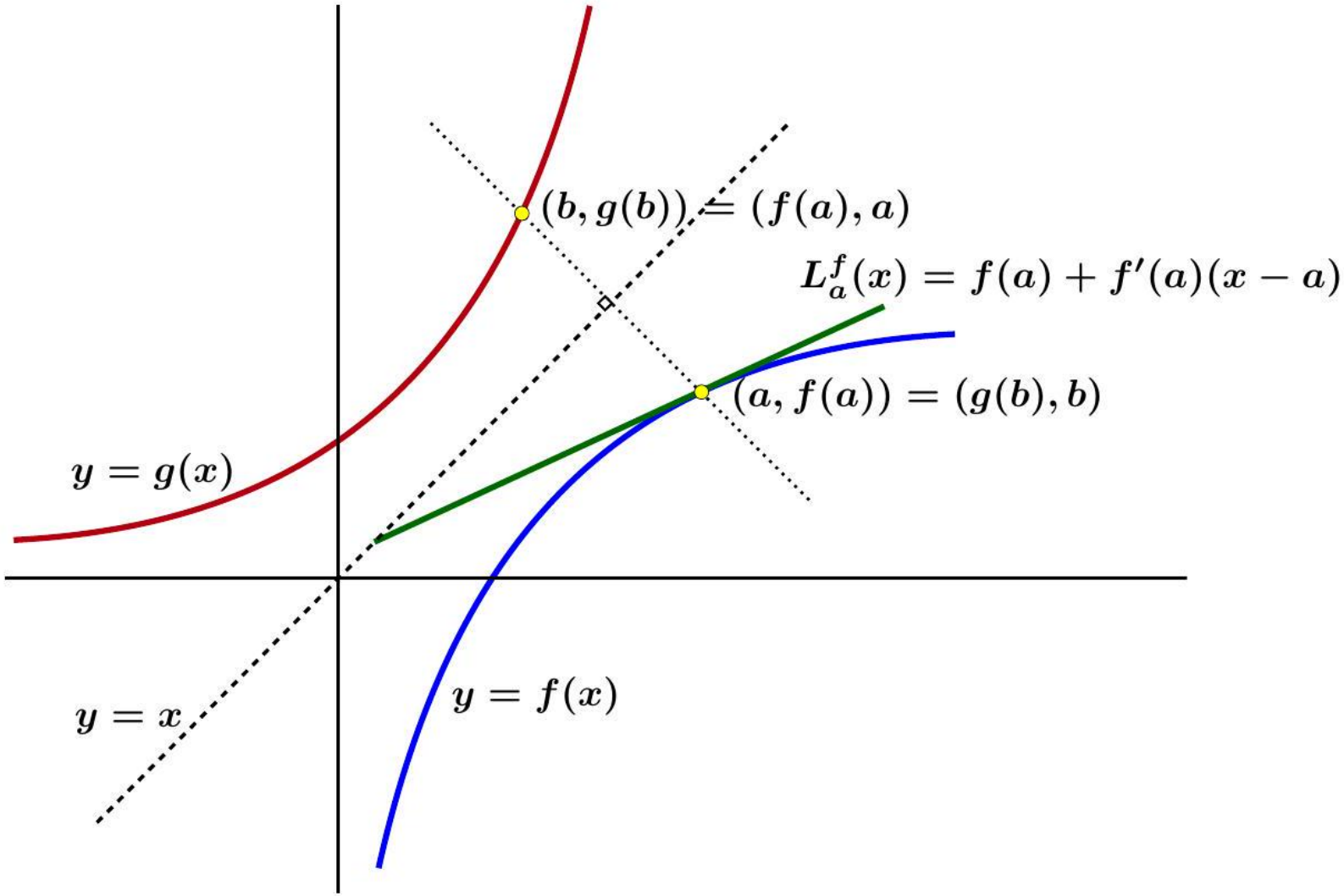
Inverse Function Theorem



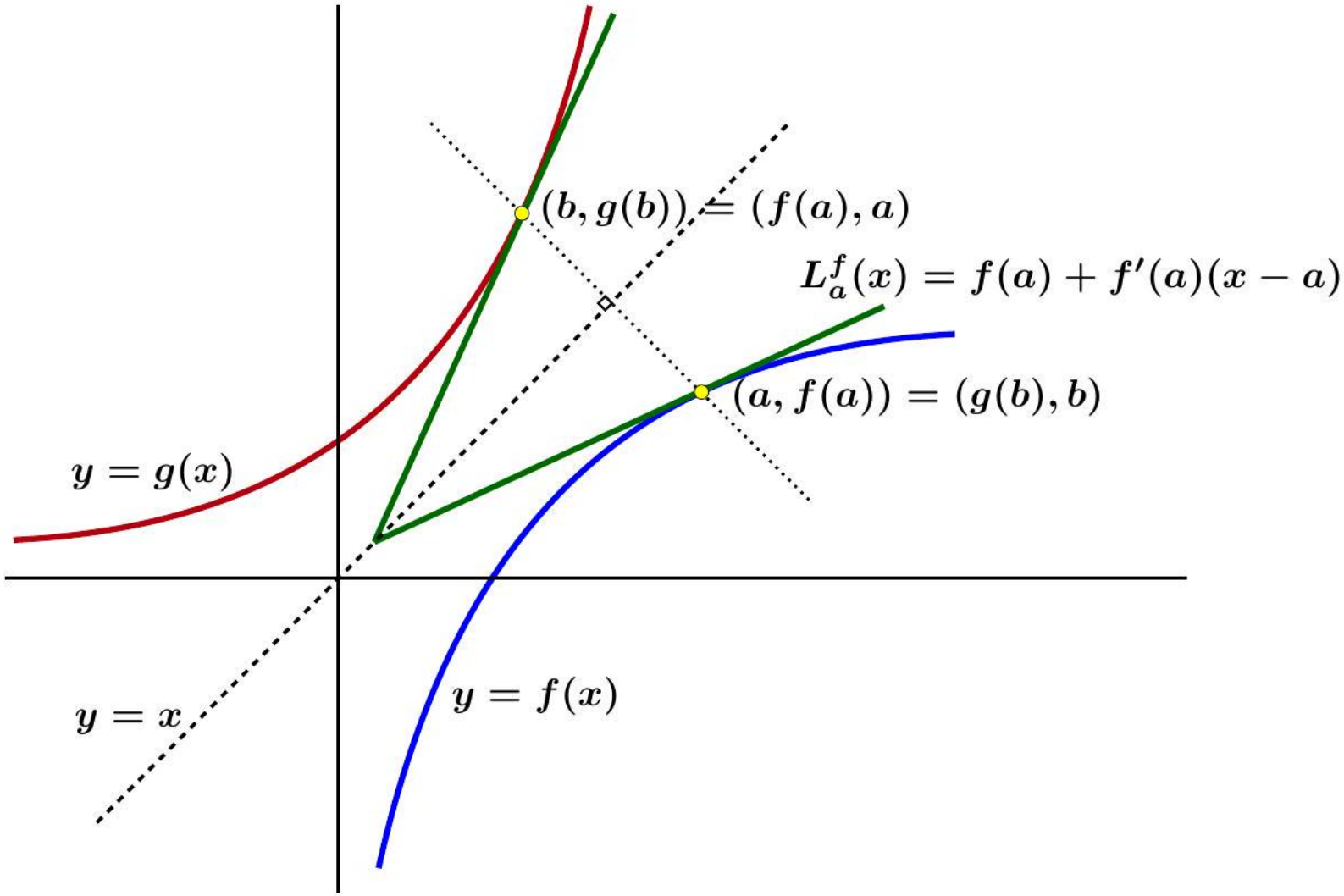
Inverse Function Theorem



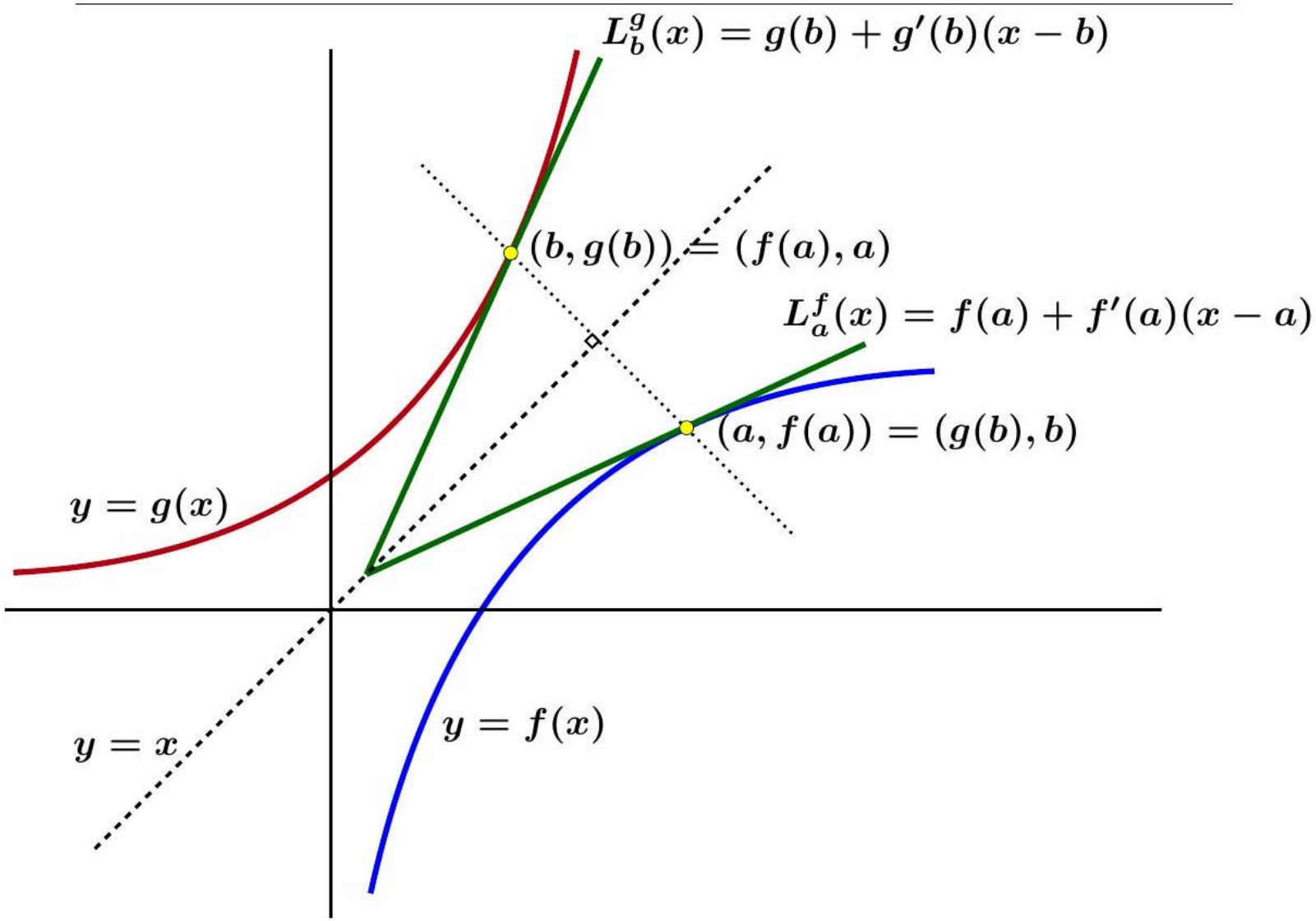
Inverse Function Theorem



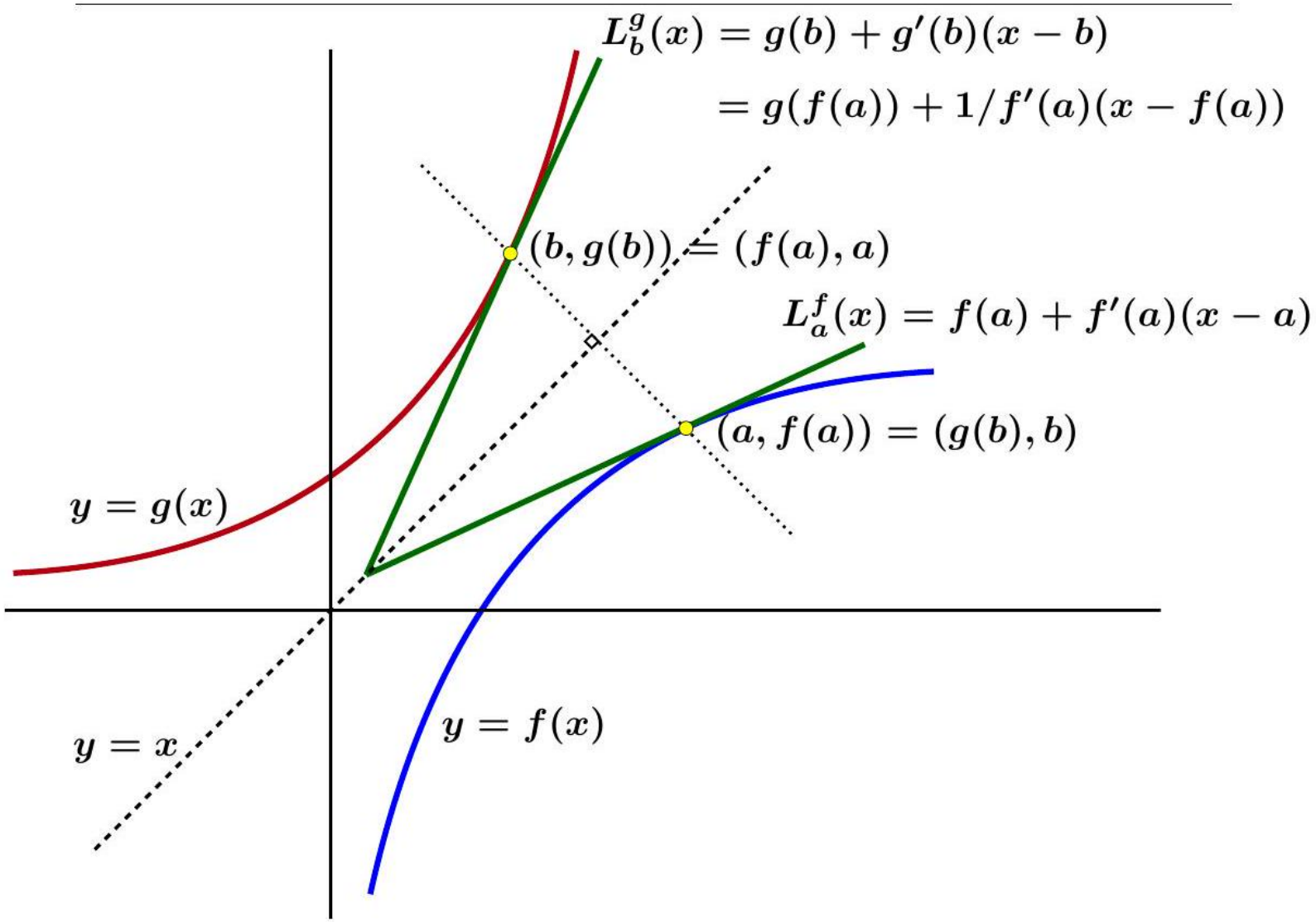
Inverse Function Theorem



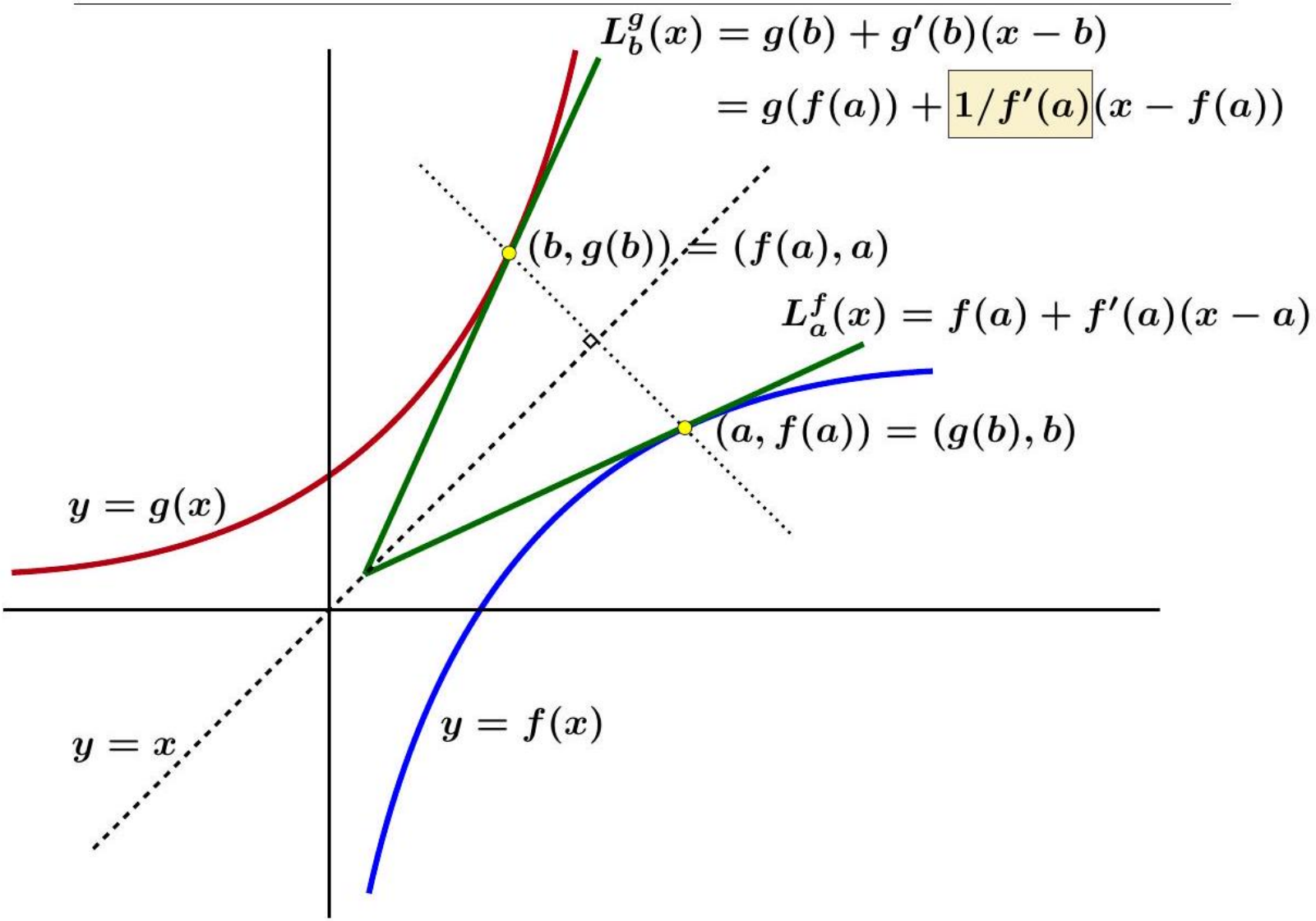
Inverse Function Theorem



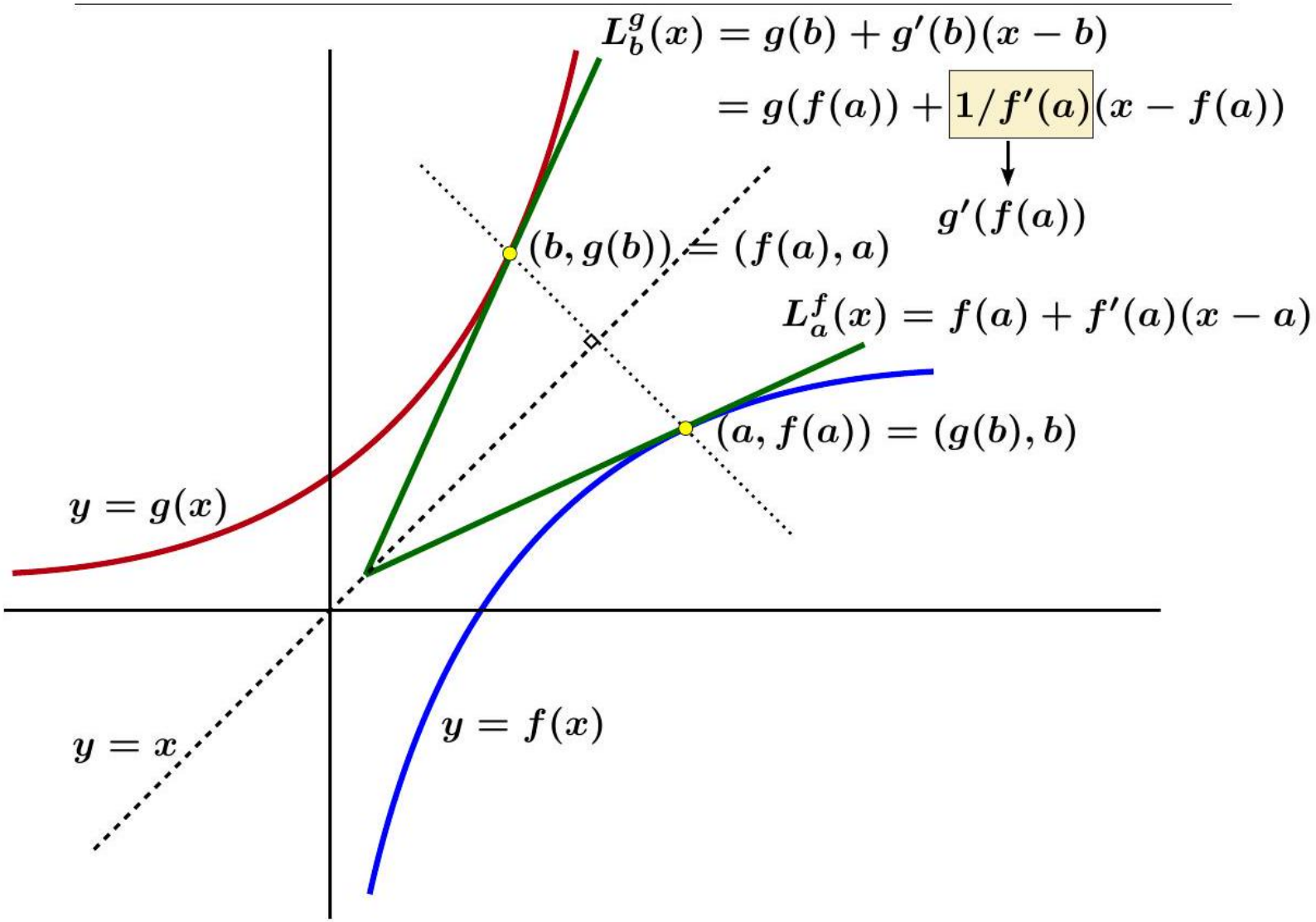
Inverse Function Theorem



Inverse Function Theorem



Inverse Function Theorem



Inverse Function Theorem

Theorem 3.3.1 (Inverse Function Theorem (IFT))

Assume that $f(x)$ is continuous and invertible on $[c, d]$ with inverse $g(y)$, and $f(x)$ is differentiable at $a \in (c, d)$. If $f'(a) \neq 0$, then $g(y)$ is differentiable at $b = f(a)$, and

$$g'(b) = \frac{1}{f'(a)} = \frac{1}{f'(g(b))}$$

Moreover, $L_a^f(x)$ is also invertible and

$$\left(L_a^f\right)^{-1}(x) = L_b^g(x) = L_{f(a)}^g(x)$$

Inverse Function Theorem

Example 3.3.2

Let $f(x) = x^3$ with $f^{-1}(y) = g(y) = y^{\frac{1}{3}}$. Let $a = 2$. Find

$$g'(f(a)) = g'(8).$$

Solution: We know that $f'(x) = 3x^2$, so by the Inverse Function Theorem:

$$\begin{aligned} g'(8) &= \frac{1}{f'(2)} \\ &= \frac{1}{12} \end{aligned}$$

We also know that $g'(y) = \frac{1}{3}y^{-\frac{2}{3}}$, so

$$\begin{aligned} g'(8) &= \frac{1}{3} \cdot 8^{-\frac{2}{3}} \\ &= \frac{1}{12} \end{aligned}$$

Inverse Function Theorem

Note:

Let $f(x) = x^3$ with $f^{-1}(y) = g(y) = y^{\frac{1}{3}}$.

Let $a = 0$, so $b = f(0) = 0$. We have

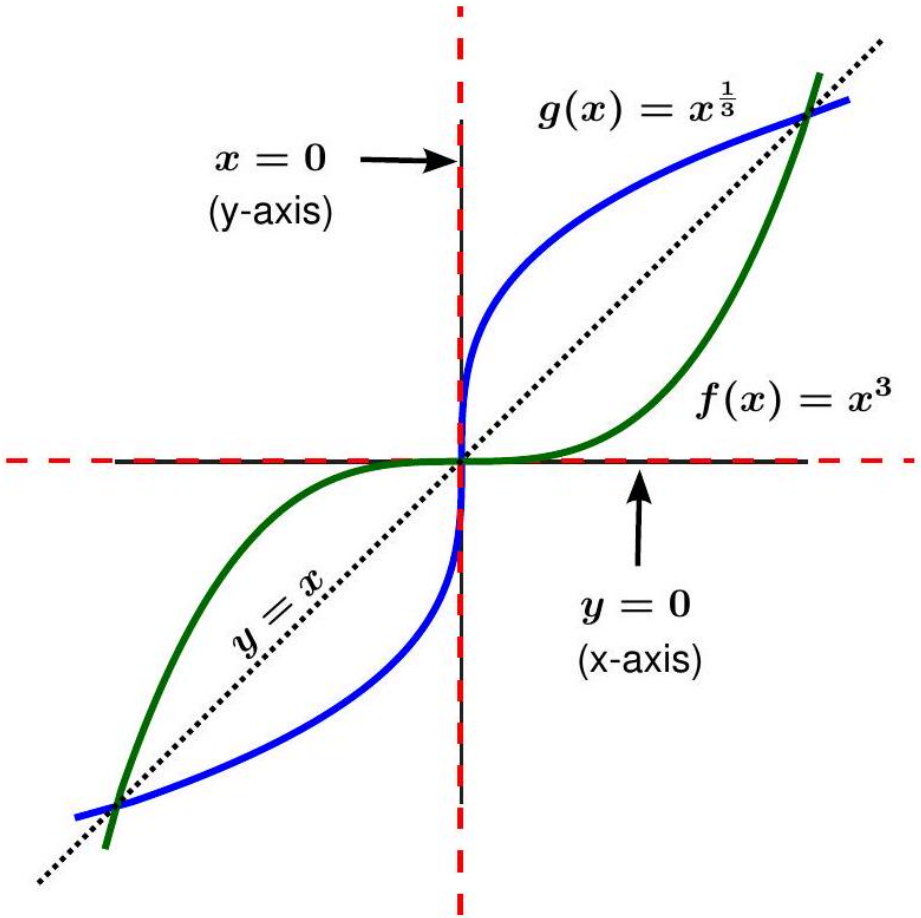
$$f'(0) = 3 \cdot 0^2 = 0$$

but

$$g'(y) = \frac{1}{3} y^{-\frac{2}{3}}$$

so $g(y)$ is not differentiable at $b = 0$.

Inverse Function Theorem



Derivative of $\ln(x)$

Example: We know that $f(x) = \ln(x)$ is invertible with inverse $g(y) = e^y$. Since e^y is differentiable for every $y \in \mathbb{R}$ the Inverse Function Theorem tells us that $f(x) = \ln(x)$ is differentiable for all $x > 0$ and that

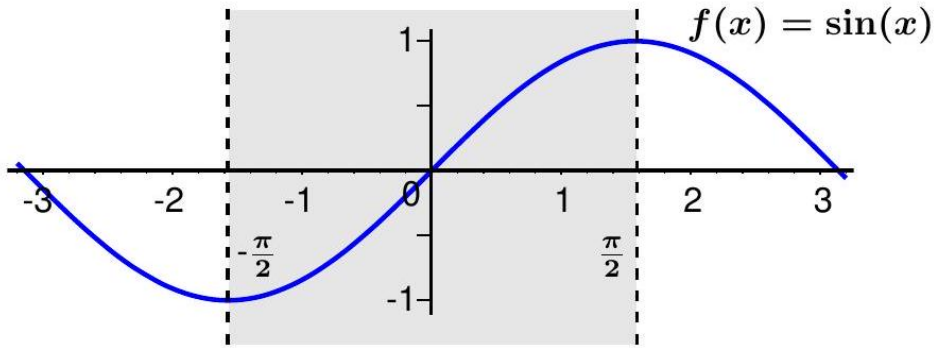
$$\begin{aligned} f'(x) &= \frac{1}{g'(f(x))} \\ &= \frac{1}{e^{f(x)}} \\ &= \frac{1}{e^{\ln(x)}} \\ &= \frac{1}{x} \end{aligned}$$

Theorem 3.3.3 (Derivative of $\ln(x)$)

The function $f(x) = \ln(x)$ is differentiable at $x > 0$, and

$$f'(x) = \frac{1}{x}$$

$\arcsin(x)$



Example: $\arcsin(x)$

Observe that $\sin(x)$ is not 1-1. However, if we restrict the domain to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then $\sin(x)$ is $1-1$ on this interval.

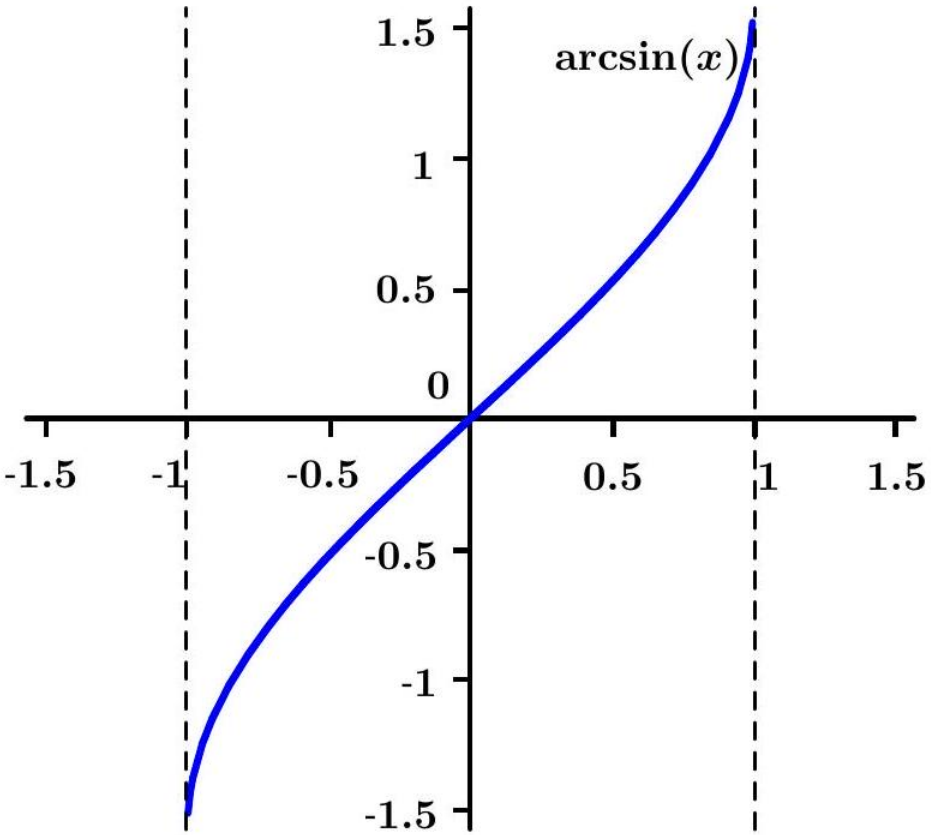
Definition 3.3.4 (arcsin)

Define $\arcsin(y) : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ by

$$\arcsin(y) = x \text{ if and only if } \sin(x) = y$$

for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$\arcsin(x)$



$\arcsin(x)$

Problem: Find $\frac{d}{dx}\arcsin(x)$.

By the IFT

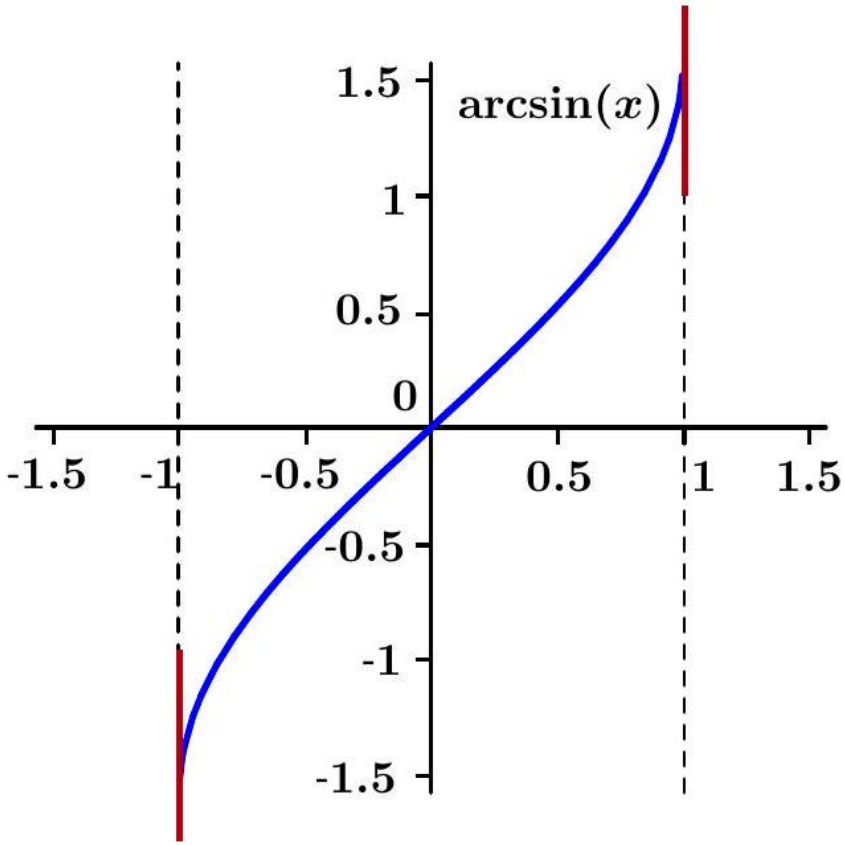
$$\begin{aligned}
\frac{d}{dx}\arcsin(x) &= \frac{1}{\frac{d}{dy}\sin(y)} \\
&= \frac{1}{\cos(y)} \\
&= \frac{1}{\sqrt{1-\sin^2(y)}} \\
&= \frac{1}{\sqrt{1-\sin^2(\arcsin(x))}} \\
&= \frac{1}{\sqrt{1-x^2}}
\end{aligned}$$

$\arcsin(x)$

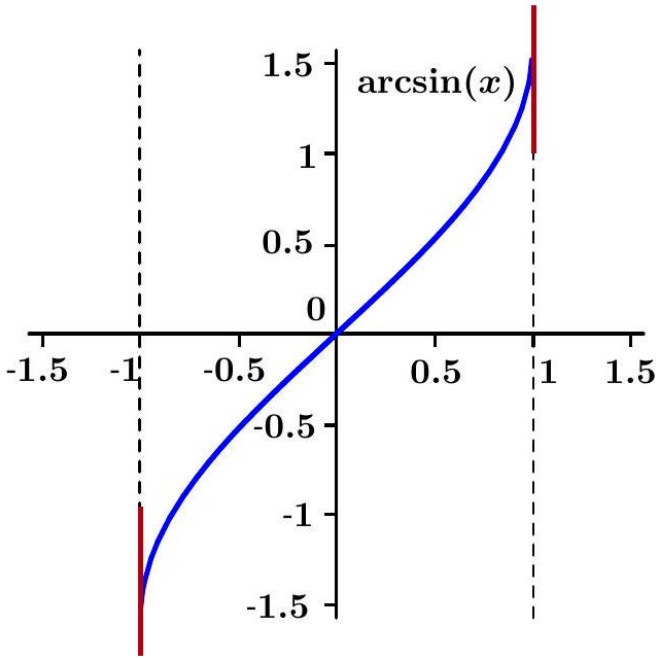
We have

$$h(x) = \frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

for $x \in (-1, 1)$.



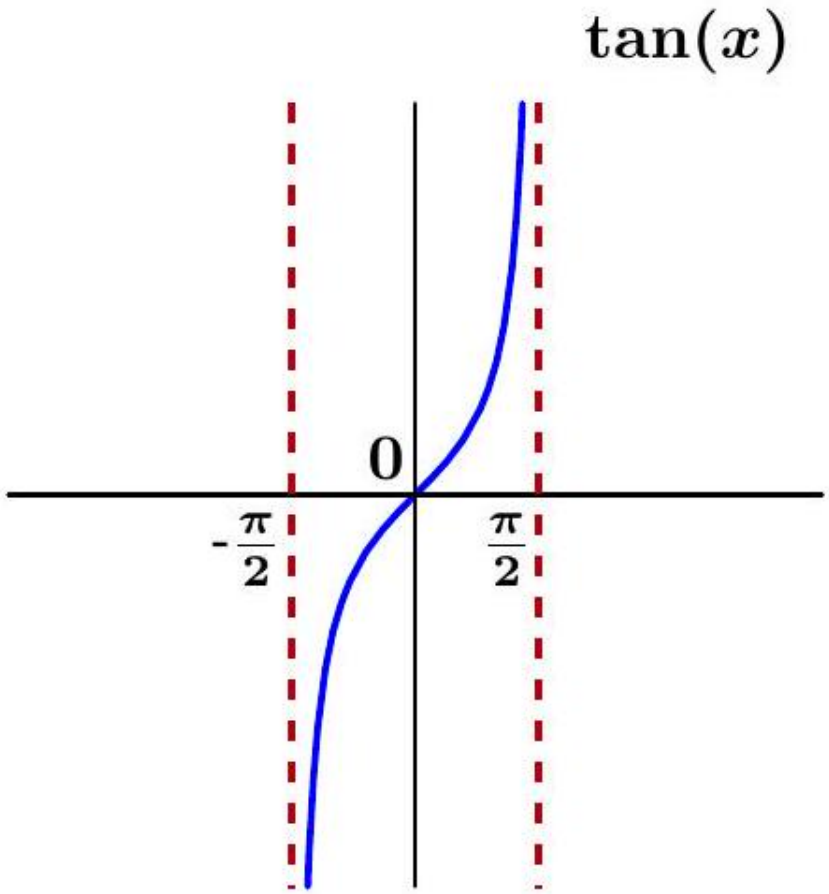
$\arcsin(x)$



Note:

1. $h(x) = \frac{1}{\sqrt{1-x^2}} > 0.$
2. $h(0) = \frac{1}{\sqrt{1-0^2}} = 1.$
3. $\lim_{x \to -1^+} \frac{1}{\sqrt{1-x^2}} = \infty.$
4. $\lim_{x \to 1^-} \frac{1}{\sqrt{1-x^2}} = \infty.$

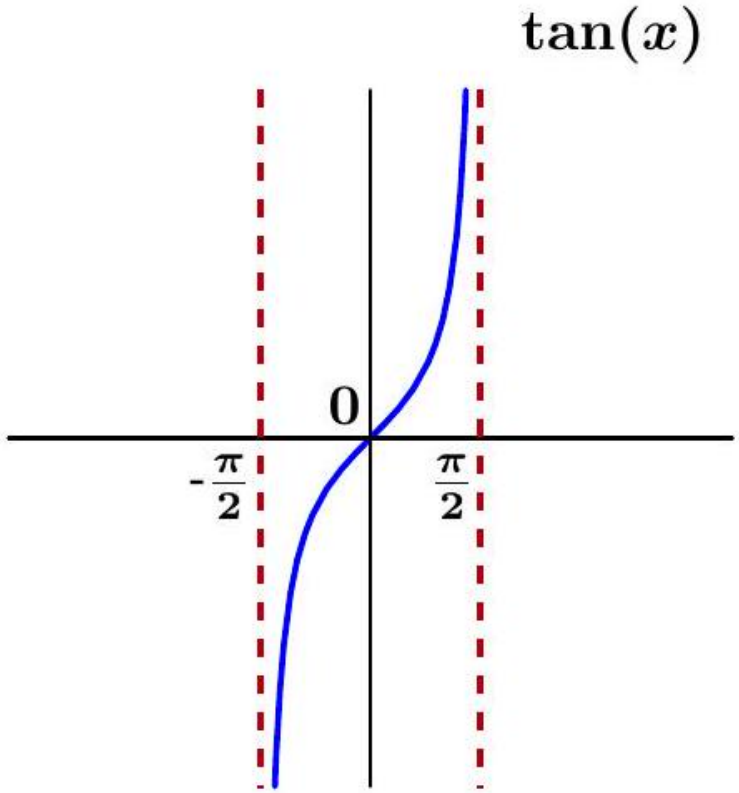
arctan(x)



Example: $\arctan(x)$
Observe that $\tan(x)$ is not 1-1.

However, if we restrict the domain to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then $\tan(x)$ is $1-1$ on this interval.

arctan(x)



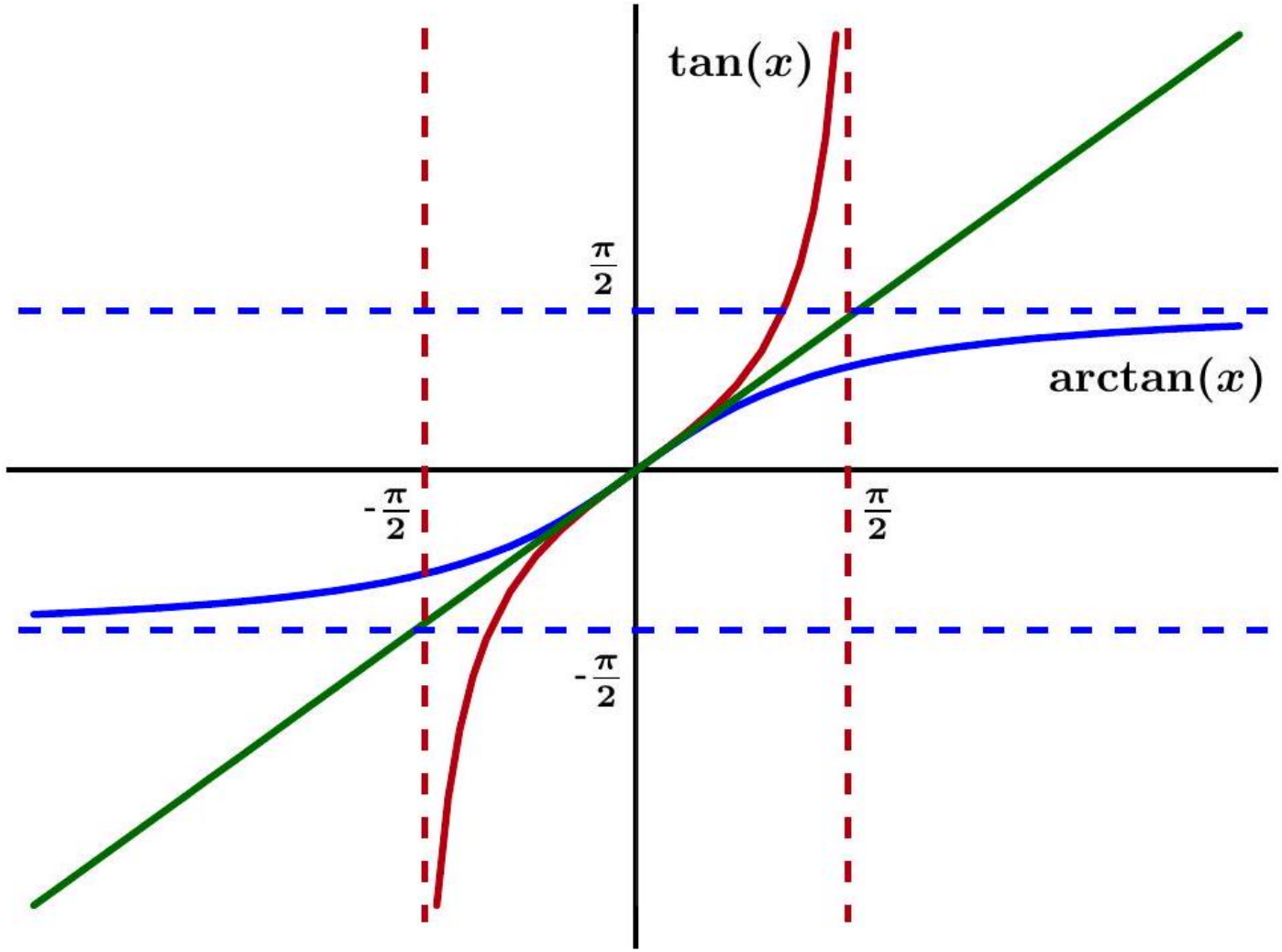
Definition 3.3.5 (arctan)

Define $\arctan(y) : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by

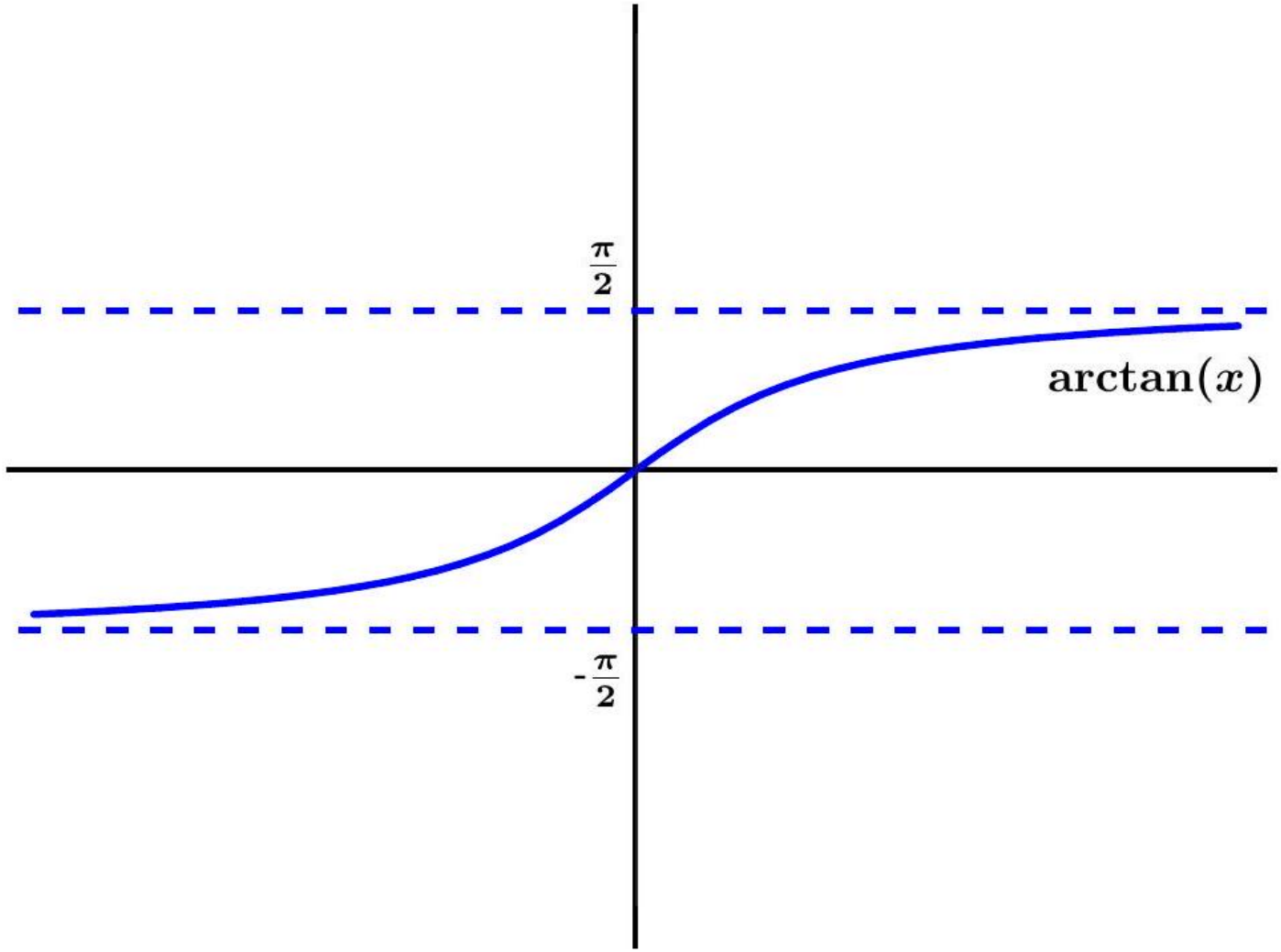
$$\arctan(y) = x \text{ if and only if } \tan(x) = y$$

for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

$\arctan(x)$



$\arctan(x)$



$\arctan(x)$

Problem: Find $\frac{d}{dx}\arctan(x)$.

By the IFT

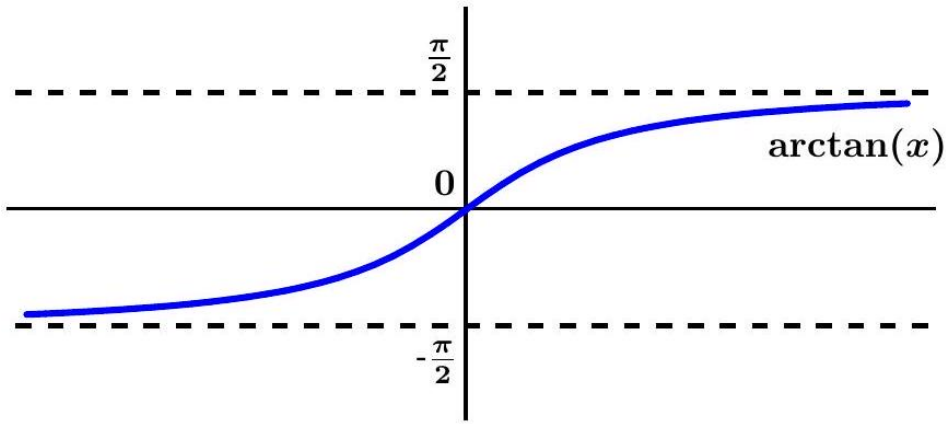
$$\begin{aligned}\frac{d}{dx}\arctan(x) &= \frac{1}{\frac{d}{dy}\tan(y)} \\ &= \frac{1}{\sec^2(y)} \\ &= \frac{1}{1+\tan^2(y)} \\ &= \frac{1}{1+\tan^2(\arctan(x))} \\ &= \frac{1}{1+x^2}\end{aligned}$$

$\arctan(x)$

We have

$$h(x) = \frac{d}{dx}\arctan(x) = \frac{1}{1+x^2}$$

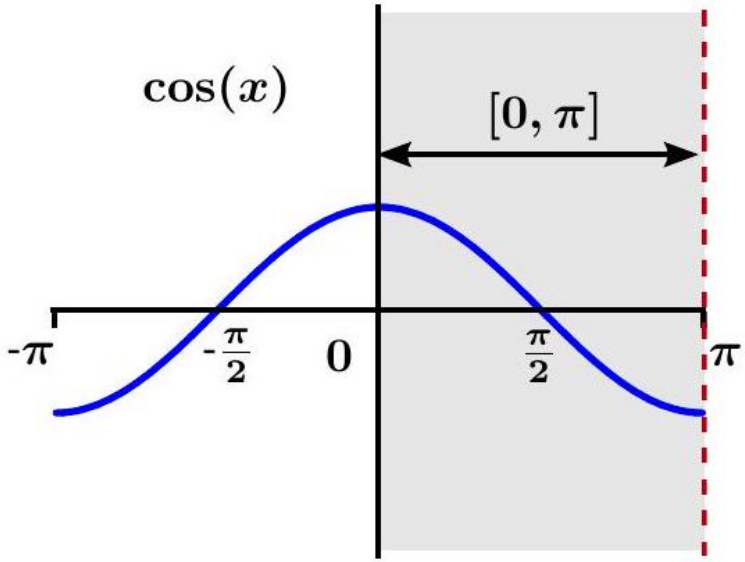
for $x \in \mathbb{R}$.



Note:

1. $h(x) = \frac{1}{1+x^2} > 0.$
2. $h(0) = \frac{1}{1+0^2} = 1.$
3. $\lim_{x\to-\infty} \frac{1}{1+x^2} = 0 = \lim_{x\to\infty} \frac{1}{1+x^2}.$

$\arccos(x)$



Example: $\arccos(x)$ Observe that $\cos(x)$ is not 1-1.

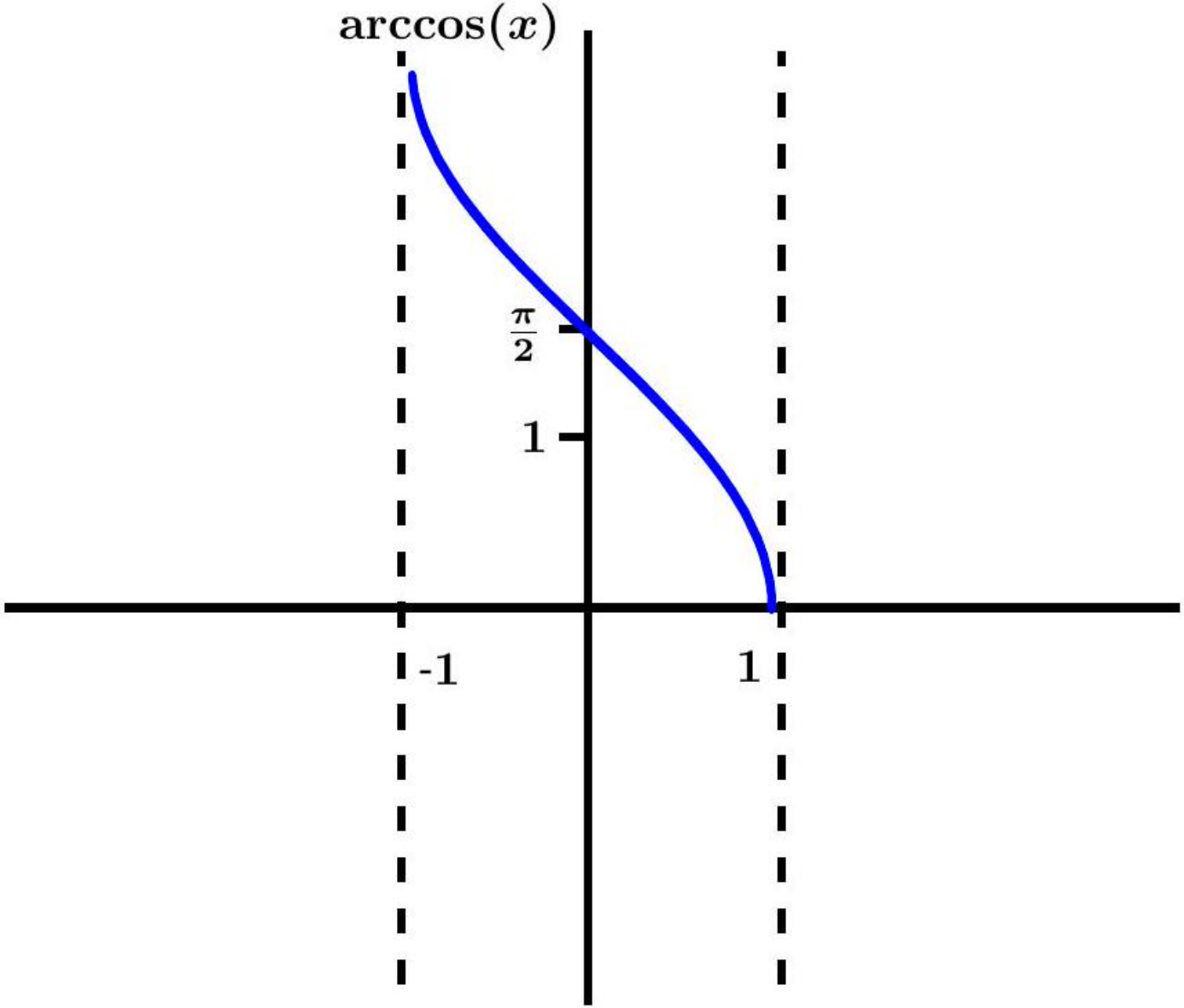
However, if we restrict the domain to $[0, \pi]$, then $\cos(x)$ is 1-1 on this interval.

Definition 3.3.6 (arccos)

Define $\arccos(y) : [-1, 1] \to [0, \pi]$ by

$$\arccos(y) = x \text{ if and only if } \cos(x) = y$$

for $x \in [0, \pi]$.


arccos(x)
π/2
1
-1
1

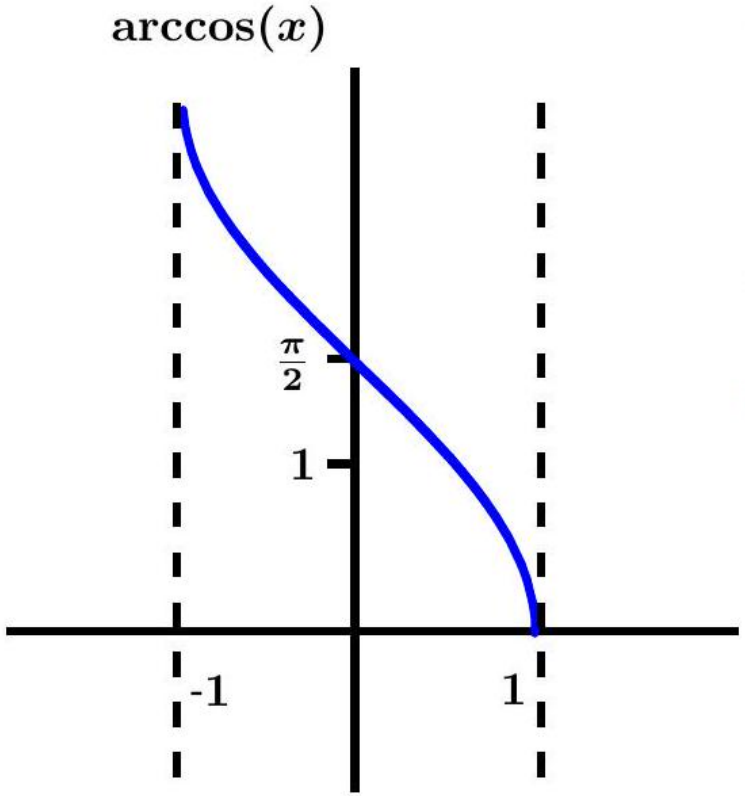
$\arccos(x)$

Problem: Find $\frac{d}{dx}\arccos(x)$.

By the IFT

$$\begin{aligned}\frac{d}{dx}\arccos(x) &= \frac{1}{\frac{d}{dy}\cos(y)} \\ &= \frac{1}{-\sin(y)} \\ &= \frac{-1}{\sqrt{1-\cos^2(y)}} \\ &= \frac{-1}{\sqrt{1-\cos^2(\arccos(x))}} \\ &= \frac{-1}{\sqrt{1-x^2}}\end{aligned}$$

$\arccos(x)$

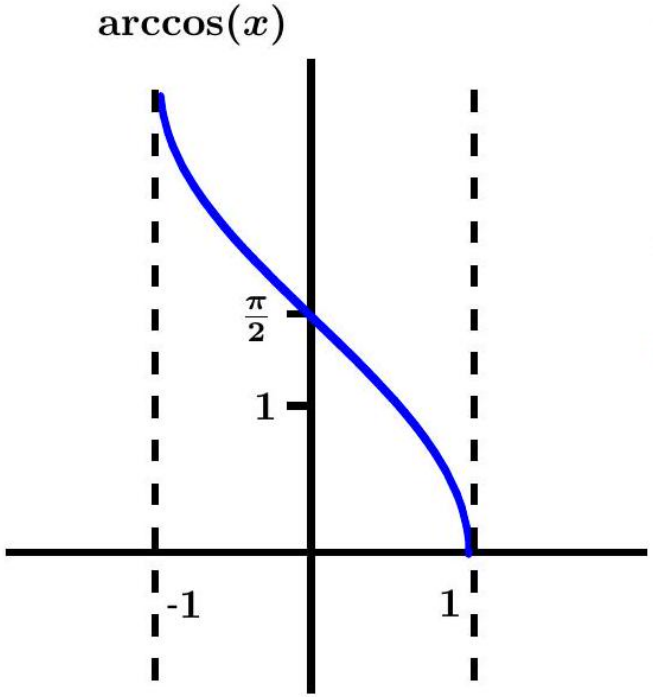


We have

$$h(x) = \frac{d}{dx}\arccos(x) = \frac{-1}{\sqrt{1-x^2}}$$

for $x \in (-1, 1)$

$\arccos(x)$



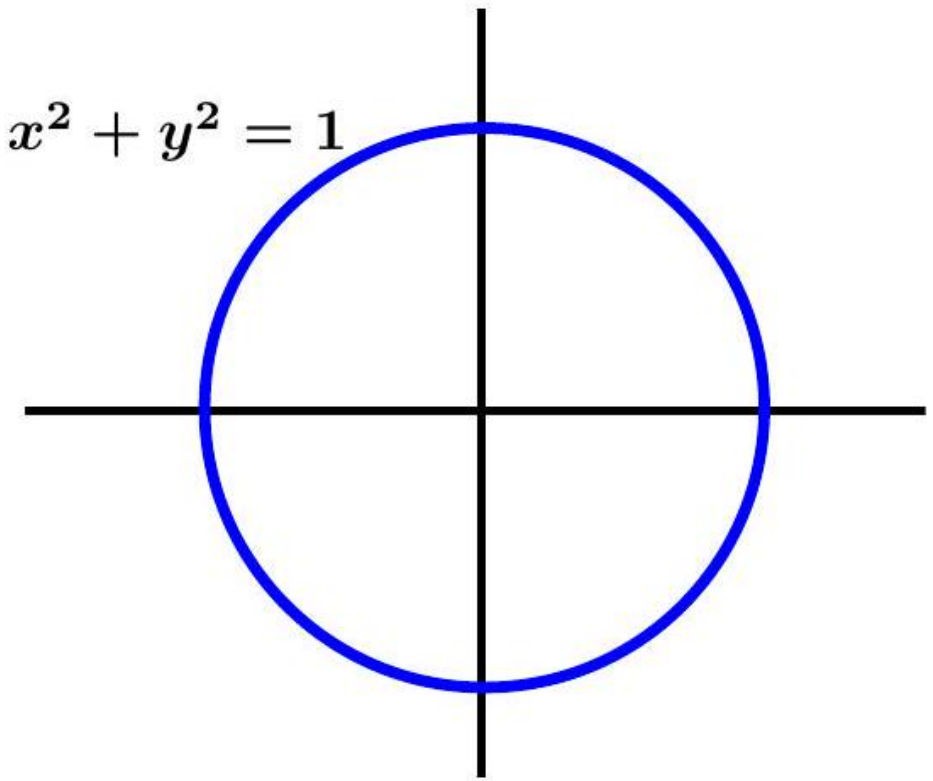
Note:

1. $h(x) = \frac{-1}{\sqrt{1-x^2}} < 0.$
2. $h(0) = \frac{-1}{\sqrt{1-0^2}} = -1.$
3. $\lim_{x \to -1^+} \frac{-1}{\sqrt{1-x^2}} = -\infty.$
4. $\lim_{x \to 1^-} \frac{-1}{\sqrt{1-x^2}} = -\infty.$

Section 4

Implicit Differentiation

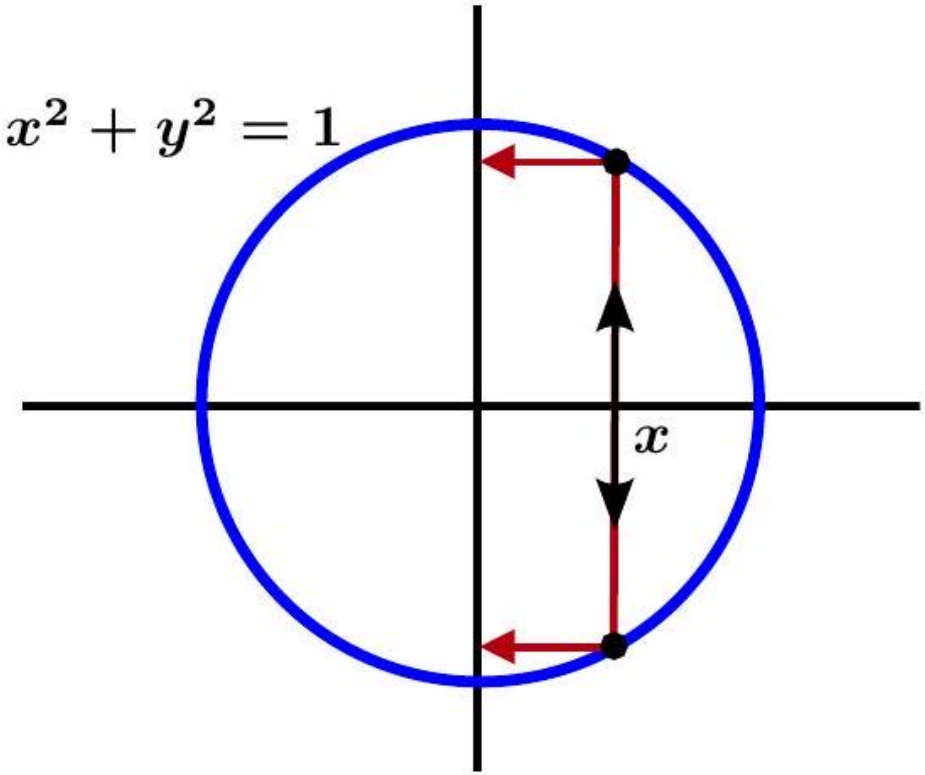
Relations and Implicit Functions



Example: Consider the relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

Relations and Implicit Functions

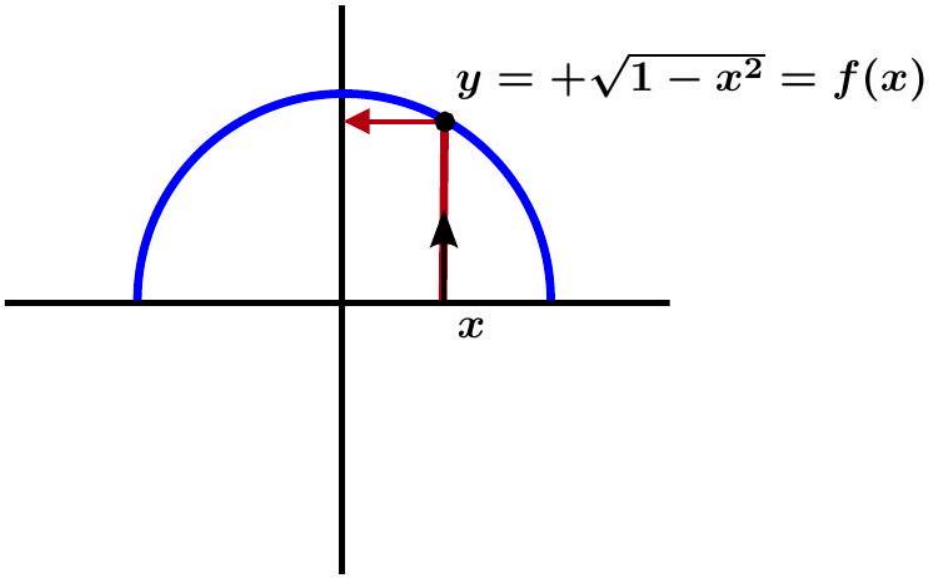


Example: Consider the relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

Note: This relation is not a function.

Relations and Implicit Functions

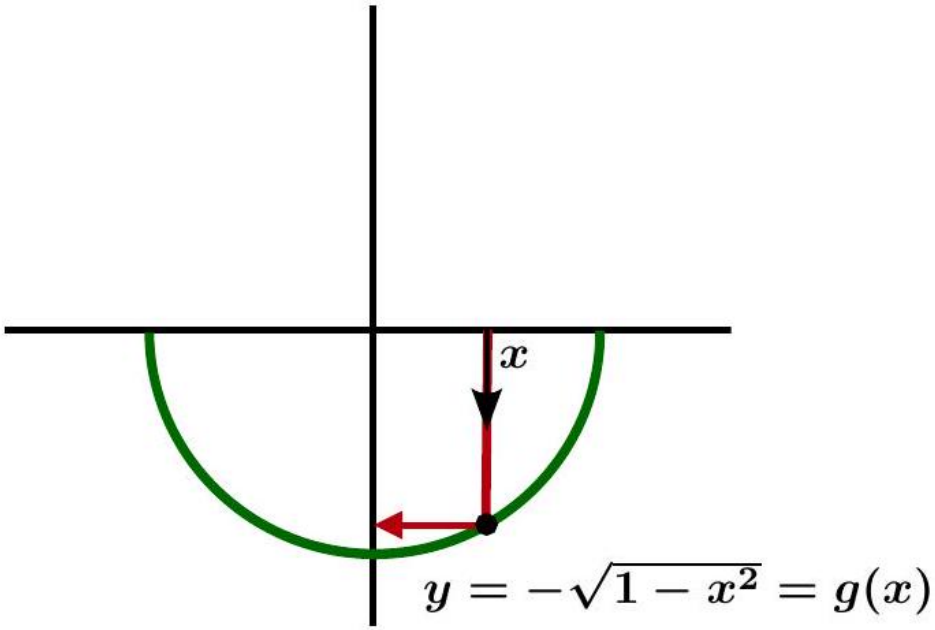


Example: Consider the relation

$$\left\{(x, y) \mid x^2 + y^2 = 1\right\}$$

Note: This relation is not a function. However, by restricting the range of the relation we can generate many functions.

Relations and Implicit Functions

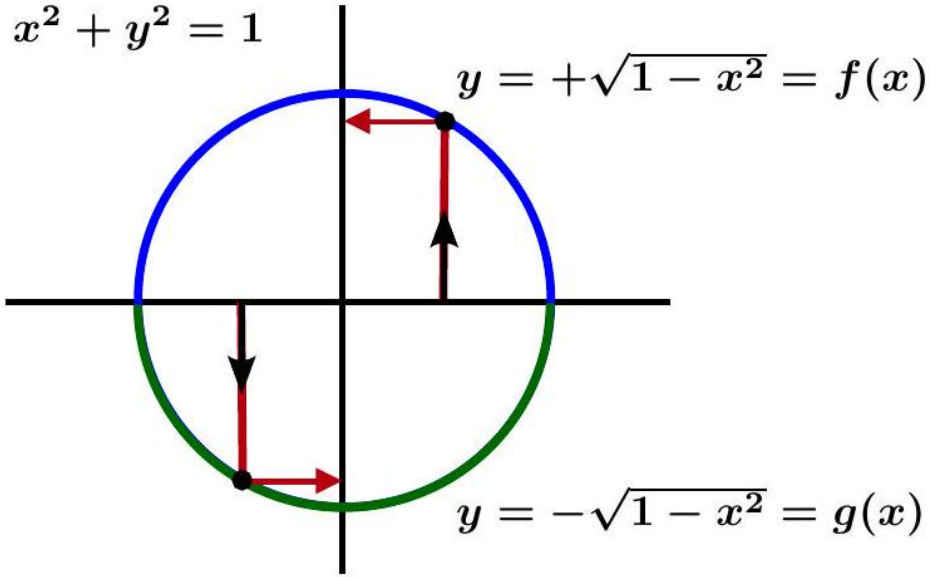


Example: Consider the relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

Note: This relation is not a function. However, by restricting the range of the relation we can generate many functions. We say these functions are implicitly defined by the relation.

Relations and Implicit Functions



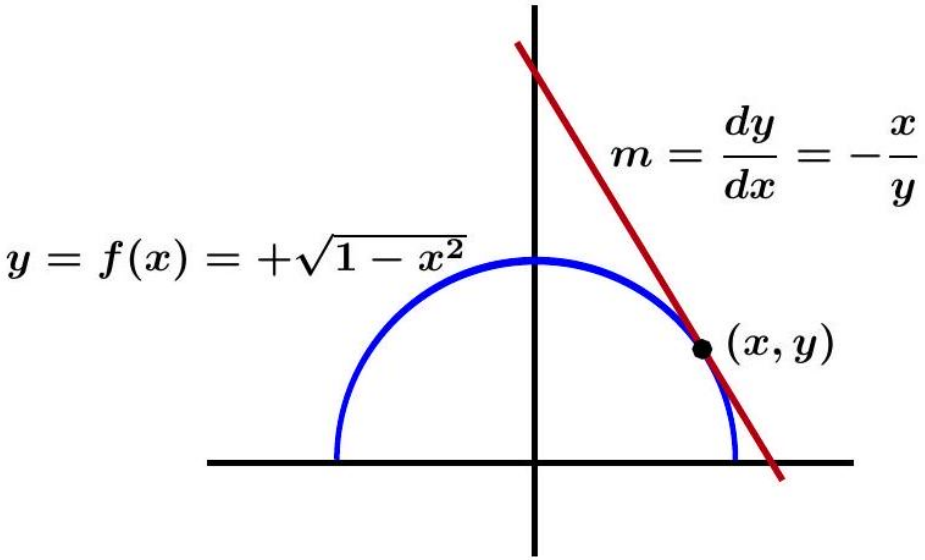
Example (continued): The relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

implies two functions

$$y = f(x) = +\sqrt{1 - x^2} \quad \text{and} \quad y = g(x) = -\sqrt{1 - x^2}$$

Relations and Implicit Functions



Example (continued): The relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

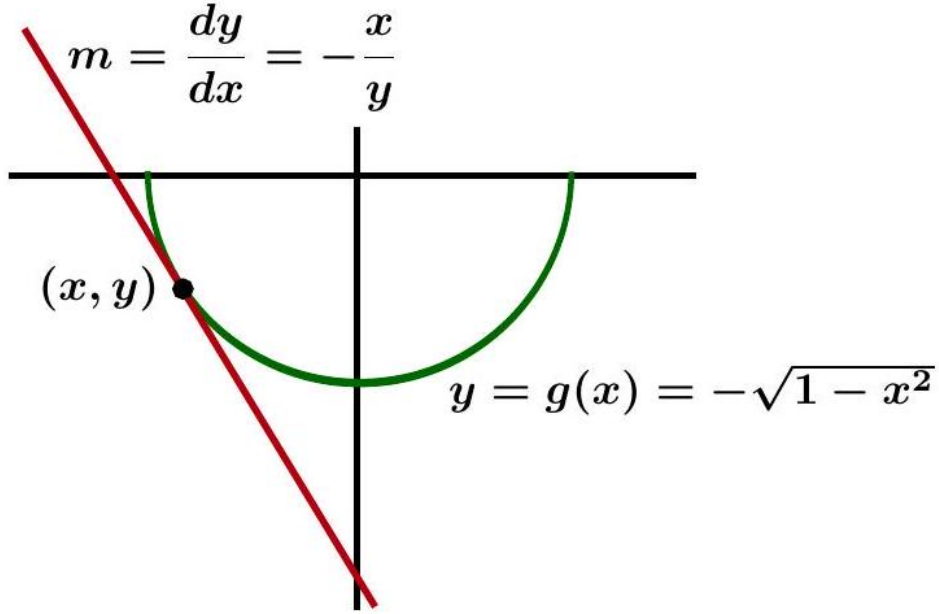
implies two functions

$$y = f(x) = +\sqrt{1-x^2} \quad \text{and} \quad y = g(x) = -\sqrt{1-x^2}$$

with derivatives

$$f'(x) = -\frac{x}{\sqrt{1-x^2}} = -\frac{x}{y}$$

Relations and Implicit Functions



Example (continued): The relation

$$\{(x, y) \mid x^2 + y^2 = 1\}$$

implies two functions

$$y = f(x) = +\sqrt{1 - x^2} \quad \text{and} \quad y = g(x) = -\sqrt{1 - x^2}$$

with derivatives

$$f'(x) = -\frac{x}{\sqrt{1 - x^2}} = -\frac{x}{y} \quad \text{and} \quad g'(x) = \frac{x}{\sqrt{1 - x^2}} = -\frac{x}{y}$$

Implicit Differentiation

Example (continued): If

$$y = h(x)$$

is any differentiable function which satisfies

$$x^2 + y^2 = 1$$

then

$$\frac{d}{dx}\left(x^2 + y^2\right) = \frac{d}{dx}(1)$$

Hence

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

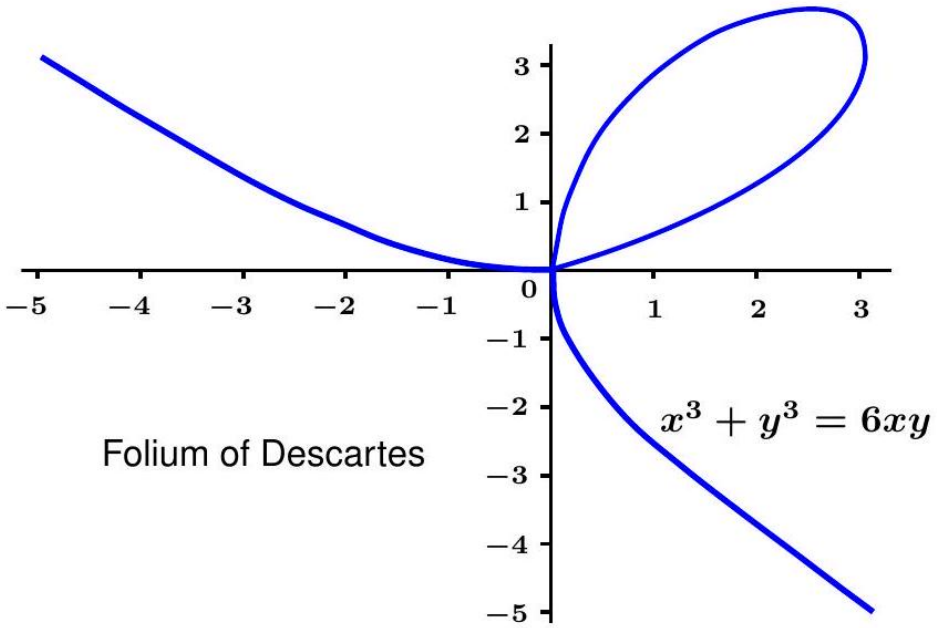
Finally

$$2y \cdot \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

provided that $\boldsymbol{y \neq 0}$.

Note: The process of finding the derivative without knowing the explicit formula for the function is called implicit differentiation.

Folium of Descartes

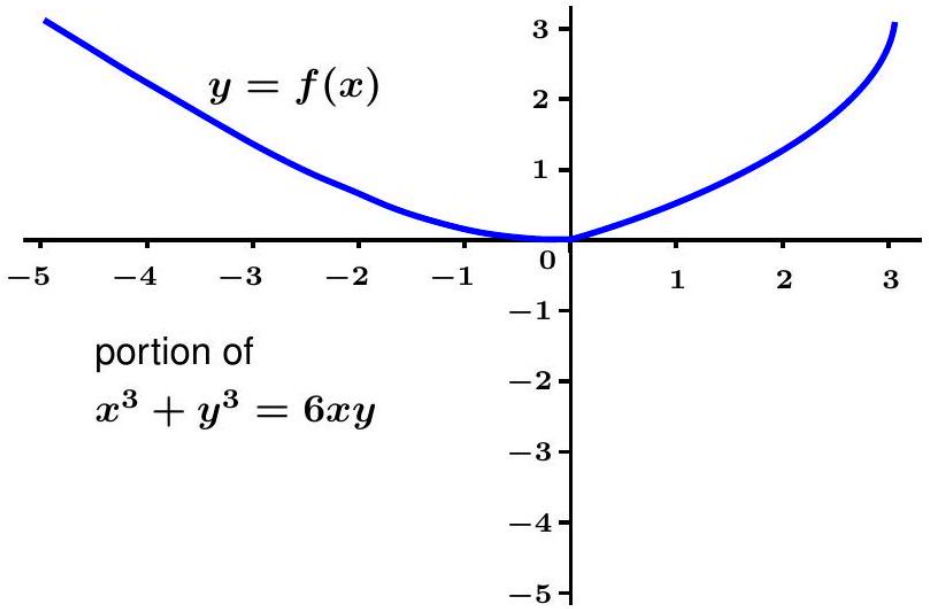


Example: The graph of the relation

$$\{(x, y) \mid x^3 + y^3 = 6xy\}$$

is called the Folium of Descartes.

Folium of Descartes

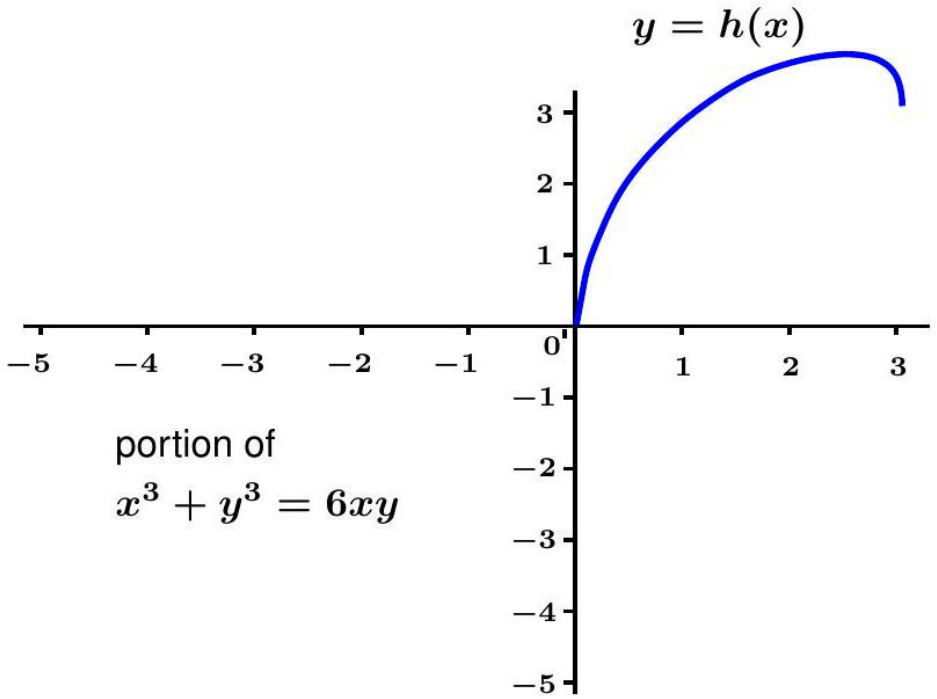


Example: The graph of the relation

$$\{(x, y) \mid x^3 + y^3 = 6xy\}$$

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Folium of Descartes

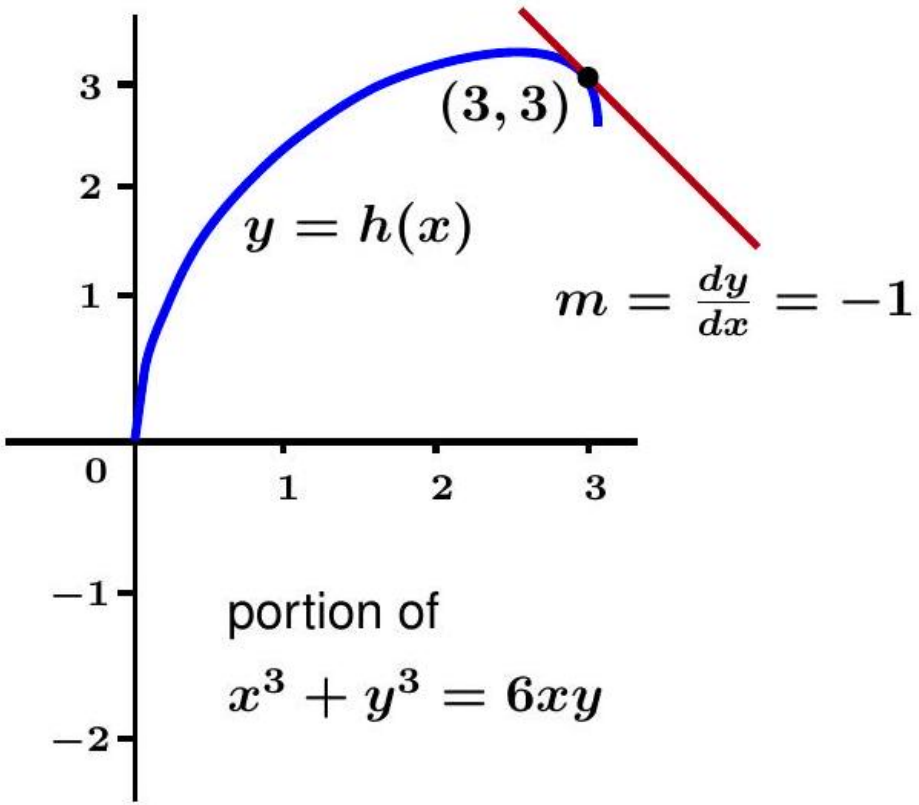


Example: The graph of the relation

$$\{(x, y) \mid x^3 + y^3 = 6xy\}$$

is called the Folium of Descartes.

Folium of Descartes



Example 3.4.1

Example: Find $h'(3)$ if $y = h(x)$ is a differentiable function with $h(3) = 3$ satisfying

$$m = \frac{dy}{dx} = -1$$

Folium of Descartes

Example 1.4.1 (continue)

$$x^3 + y^3 = 6xy$$

Then

$$\frac{d}{dx}\left(x^3 + y^3\right) = \frac{d}{dx}(6xy)$$

Hence

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$$

and

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x}$$

Letting $x = 3$ and $y = 3$ gives

$$h'(3) = \left.\frac{dy}{dx}\right|_{(3,3)} = \frac{6(3) - 3\left(3^2\right)}{3\left(3^2\right) - 6(3)} = -1$$

Caution!

Example 3.4.2

Suppose that

$$x^4 + y^4 = -1 - x^2y^2$$

Find $\frac{dy}{dx}$.

Solution: We get

$$\frac{d}{dx}\left(x^4 + y^4\right) = \frac{d}{dx}\left(-1 - x^2y^2\right)$$

so

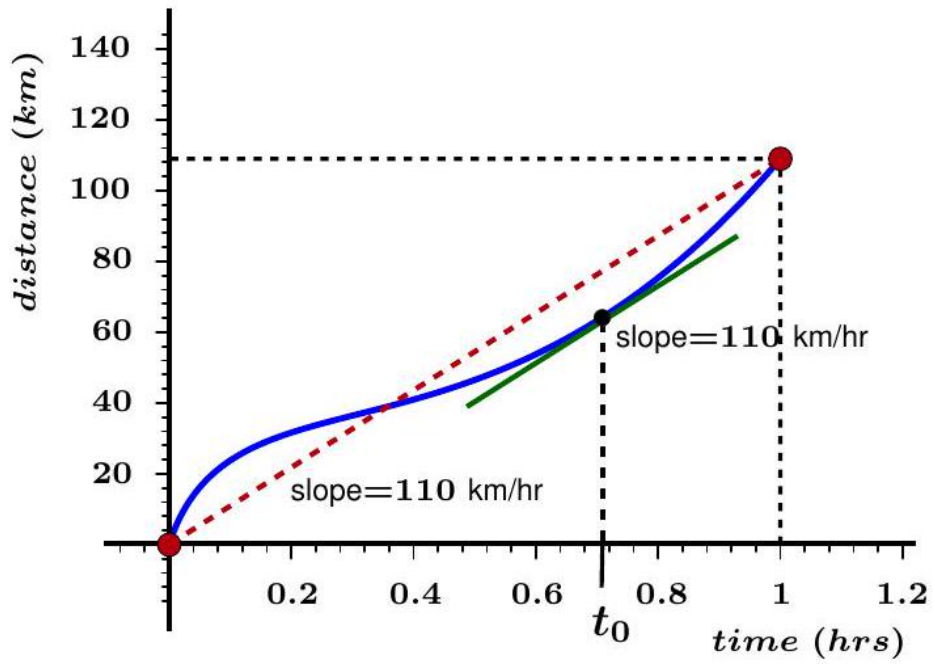
$$\frac{dy}{dx} = \frac{-2xy^2 - 4x^3}{4y^3 + 2x^2y} \quad (**)$$

Important Observation: The equation $(*)$ has no solutions, so $(**)$ is meaningless!

Section 5

Mean Value Theorem

A Problem

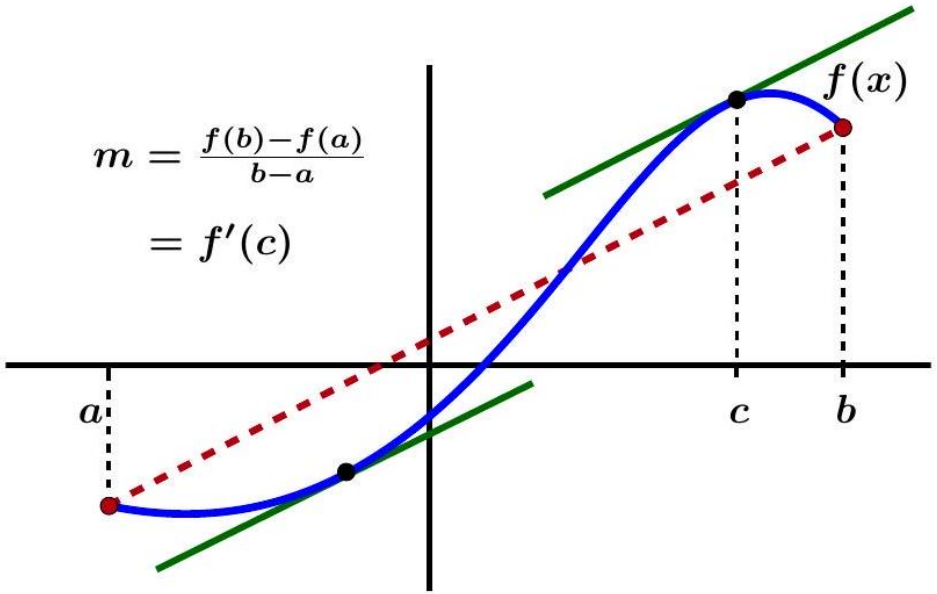


Problem: A car travels forward a distance of 110 km in one hour along a road with a posted speed limit of 100 km/hr. Prove that at some point in the journey the car was speeding.

Average velocity $= 110$ km/hr.

At t_0 the tangent line is parallel to the secant line $\Rightarrow v(t_o) = 110$ km/hr.

Generic Case

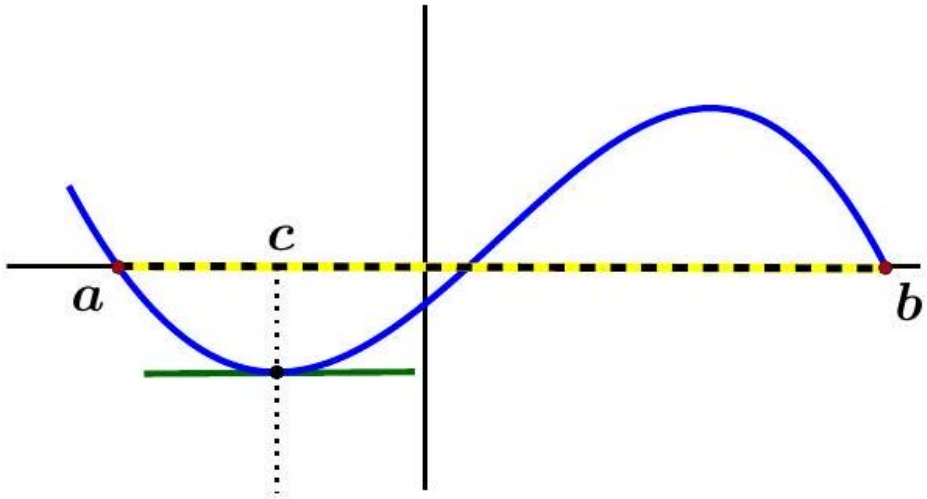


Question: Is there always a point at which the instantaneous rate of change equals the average rate of change?

Average rate of change $= \frac{f(b)-f(a)}{b-a}$

$$= f'(c)?$$

Rolle's Theorem



Theorem 3.5.1 (Rolle's Theorem)

Assume that $f(x)$ is continuous on $[a,b]$, differentiable on (a,b), and that $f(a) = 0 = f(b)$. Then there exists a $c \in (a,b)$ with

$$f'(c) = 0$$

Proof: Case 1: $f(x) = 0$ for all $x \in [a,b] \Rightarrow f'(c) = 0$ for any $c \in (a,b)$.
Case 2: $f(x) \neq 0$ for some $x \in [a,b]$.
Then $f(x)$ either attains its maximum at some point $c \in (a,b)$ and $f'(c) = 0$, or $f(x)$ attains its minimum at some point $c \in (a,b)$ and $f'(c) = 0$.

Mean Value Theorem

Theorem 3.5.2 (Mean Value Theorem)

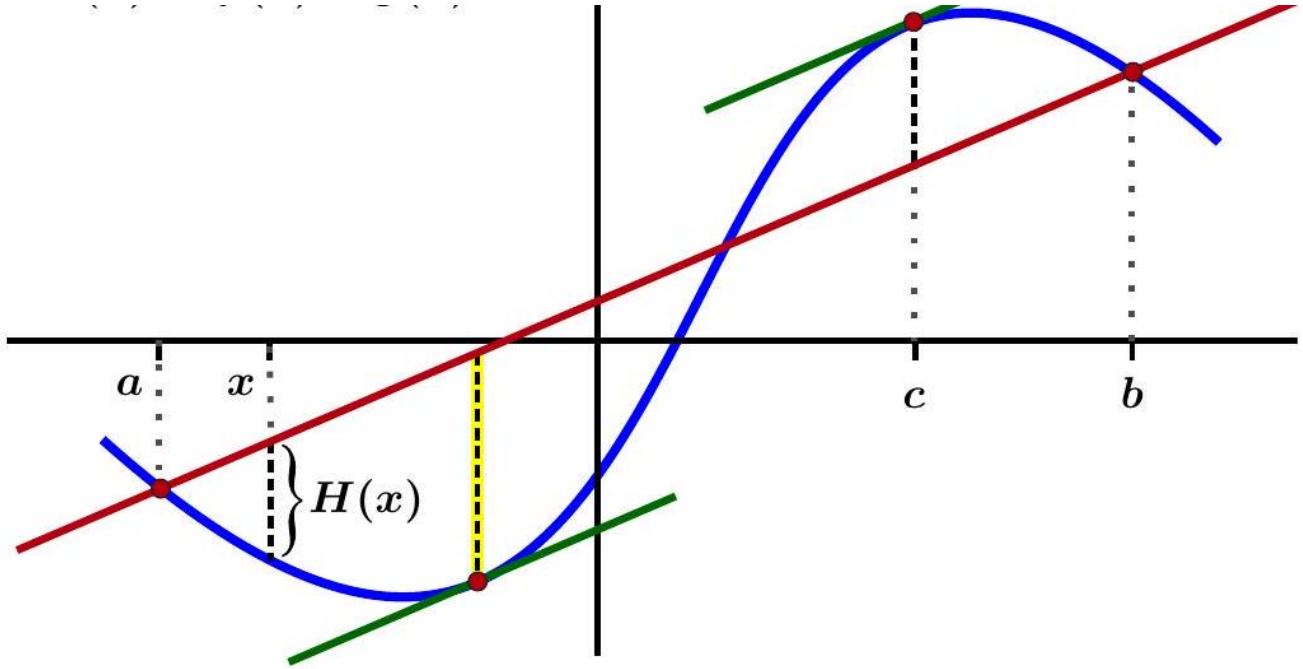
Assume that $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists a $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometric Proof

- $f(x)$
- $g(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$

  $H(x) = f(x) - g(x)$



Analytic Proof

Let

$$g(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \Rightarrow g'(x) = \frac{f(b) - f(a)}{b - a}$$

for all $x \in (a, b)$. Let $H(x) = f(x) - g(x)$. Then

$$H(a) = f(a) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(a - a) \right) = f(a) - f(a) = 0$$

$$\begin{aligned} H(b) &= f(b) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(b - a) \right) \\ &= f(b) - (f(a) + f(b) - f(a)) \\ &= 0 \end{aligned}$$

Rolle's Theorem $\Rightarrow$ there exists a $c \in (a, b)$ such that

$$0 = H'(c) = f'(c) - g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$$

MVT Problem

Problem: Two runners compete in a $100\ \mathrm{m}$ race. If they finish the race in exactly the same amount of time $\boldsymbol{t_f}$, show that there exists some time $c \in (0, t_f)$ such that at time c the two runners were traveling exactly the same speed.